

Mathematical supplement to A minimal local covariant action for holographic dark energy: constraints, perturbations, and nonlinear dynamics

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Version V52 — 27 September 2026. This document contains the complete mathematical catalogue accompanying the article cited in Ref. [1]. The catalogue, expanded coefficient tables, available verification programs, and saved results form a separate supplementary package prepared for GitHub. They are not included in the main article’s PDF or arXiv source archive. Repository and release metadata have not yet been supplied; the local package is technically prepared but requires completion of those metadata before publication.

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1 Mathematical catalogue

1.1 Scope and indexing

This standalone mathematical supplement accompanies version V52 of the main article [1]. References explicitly identified as belonging to the main article, including Appendices A–E, use the numbers in that version. All other section and equation numbers refer to this supplement. The two documents compile independently; the external numbers have been checked against the V52 build. The definitions and domains of validity are stated below.

Subsections 1.2–1.5 and 1.7, together with the kinematics and normalizations given below, apply to the radiation-like branch $U = 0$ in the absence of an independent matter sector. The independent parameters used are the ratio of the magnitudes of the comoving momenta $r = k'/k > 0$, the cosine of the angle between them $-1 < z < 1$, and the variable $x = k\eta$. Subsection 1.6 contains a separate local result for a general potential $U \in C^6$ at $\chi < 0$ and $\Delta_\Phi \neq 0$. The condition $U = 0$ and the kinematics of the radiation-like branch do not apply to that section. The catalogue does not give the general gravitational amplitude on an arbitrary background. The formulas in the article retain the external frequency factors, the overall scale, and the separate volume normalization of exactly homogeneous lines.

The position indices 0, 1, 2, 3 correspond to positions 1, 2, 3, 4 in the article. The field indices 0, 1, 2 denote the scalar and the first and second linear tensor polarizations. The inward-directed momenta in units of k are

$$\begin{aligned} \mathbf{p}_0 &= (0, 0, 1), & \mathbf{p}_1 &= (0, 0, -1), \\ \mathbf{p}_2 &= (-r\sqrt{1-z^2}, 0, -rz), & \mathbf{p}_3 &= (r\sqrt{1-z^2}, 0, rz). \end{aligned} \quad (1)$$

The signed frequencies are $(c_{A_0}, c_{A_1}, -rc_{A_2}, -rc_{A_3})$, where $c_0 = 1/\sqrt{3}$, $c_1 = c_2 = 1$. For each external momentum $\mathbf{p}$, set $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$. We use the vectors $\mathbf{e} = (0, 1, 0)$, $\mathbf{b} = (-n_z, 0, n_x)$ and the tensors $E_1 = \mathbf{e}\mathbf{e} - \mathbf{b}\mathbf{b}$, $E_2 = \mathbf{e}\mathbf{b} + \mathbf{b}\mathbf{e}$. Both tensors have squared norm two. The parameter t_θ of the original algebraic calculation is related to z by $t_\theta^2 = (1-z)/(1+z)$.

For an internal momentum $\mathbf{p}$ of nonzero norm, we use the unnormalized vector $\mathbf{b} = (-p_z, 0, p_x)$: $E_1 = \mathbf{e}\mathbf{e} - \mathbf{b}\mathbf{b}/p^2$, $E_2 = \mathbf{e}\mathbf{b} + \mathbf{b}\mathbf{e}$. The squared norms of the scalar and the two tensors are 1, 2, $2p^2$. Reversing $\mathbf{p}$ to $-\mathbf{p}$ between the ends of the internal line gives the factors $\tau_I = (1, 1, -1)$.

At $\mathbf{p} = 0$, we use five symmetric trace-free matrices. The two diagonal matrices are $\text{diag}(1, -1, 0)$ and $\text{diag}(1, 1, -2)$. The other three have unit entries at the pairs of positions xy, yx , xz, zx , and yz, zy , respectively. Their squared norms are 2, 6, 2, 2, 2. The kinetic sign of the homogeneous scalar coordinate is -1 . The covariances of the geometric coordinates are divided by the positive squared geometric norm. The kinetic sign of the scalar coordinate is accounted for in the free equation of motion without introducing a negative state probability.

1.2 Thirteen representative contact components and reconstruction of 81 components

For each contact component, four sets of coefficients of powers of x are given below: h_p for the full Hamiltonian contact term; b_p for the original Lagrangian contact term taken with the opposite sign; u_p for the sum of contact terms from eliminating auxiliary variables; and v_p for the sum of Legendre-transform contact terms. In all cases,

$$\frac{H_4}{g_R^2 k^4} = \sum_{p=-4}^0 h_p x^p, \quad h_p = b_p + u_p + v_p. \quad (2)$$

The individual partitions needed to reconstruct the angular dependence are recovered from the sums u_p, v_p as specified in Subsection 1.5.2. Coefficients of powers not listed are zero. The symbols R_s in the following formulas denote the explicit algebraic functions in Subsection 1.7, not external electronic records. When expressed in terms of z , some sources for the global Cartesian tensors contain $\sqrt{1-z^2}$; after changing to t_θ , all functions are rational.

Let T exchange the incoming and outgoing pairs, and let S exchange the two positions within one pair. The exact rules are

$$h_p^{BACD}(r, z) = h_p^{ABDC}(r, z) = h_p^{ABCD}(r, -z), \quad h_p^{CDAB}(r, z) = r^{4+p} \overline{h_p^{ABCD}(1/r, z)}. \quad (3)$$

The indices within each pair are ordered first, and the pairs are then ordered lexicographically. An odd number of within-pair exchanges gives $z \mapsto -z$. Exchanging the pairs corresponds to the substitution $r \mapsto 1/r$, complex conjugation, and multiplication by r^{4+p} . These identities have been checked separately for each of the 81 components. The same identities also hold separately for b_p, u_p, v_p defined in (2). The starting point is the total Lagrangian density $\mathcal{L}_\psi + \mathcal{L}_g$ on the radiation-like background, given in Eqs. (342)–(343) of the main article [1]. Substituting the linear constraints into its quadrilinear part gives b_p with the opposite sign. Eliminating the second-order auxiliary variables gives u_p , and the Legendre transform in Eq. (191) of the main article gives v_p . The corresponding decomposition of the full contact term is given in Eq. (351) of the main article [1].

In the quadrilinear coefficients of these parts of the contact term, all external-wave data are permuted: field types, momenta, signed frequencies, and polarizations. For u_p and v_p , each partition is first transformed according to (504), including the replacement $+\leftrightarrow -$ when the two positions within one external pair are exchanged. The three partitions $0, +, -$ are then summed according to (498). The resulting sums satisfy (3). For a fixed partition label, the rules (504) must be used. The original part b_p requires no such summation.

Components containing an odd number of fields with the second linear tensor polarization vanish identically by reflection symmetry under $y \mapsto -y$. There are 40 such components; the remaining 41 are reconstructed from 13 representatives.

1.2.1 Component 0000

$$\begin{aligned} h_{-4}^{0000} &= R_1 \\ h_{-3}^{0000} &= R_2 \\ h_{-2}^{0000} &= R_3 \\ h_{-1}^{0000} &= R_4 \\ h_0^{0000} &= R_5 \end{aligned} \quad (4)$$

$$\begin{aligned} b_{-4}^{0000} &= R_6 \\ b_{-3}^{0000} &= R_7 \\ b_{-2}^{0000} &= R_8 \\ b_{-1}^{0000} &= R_9 \\ b_0^{0000} &= R_{10} \end{aligned} \quad (5)$$

$$\begin{aligned} u_{-4}^{0000} &= R_{11} \\ u_{-3}^{0000} &= R_{12} \\ u_{-2}^{0000} &= R_{13} \\ u_{-1}^{0000} &= R_{14} \\ u_0^{0000} &= R_{15} \end{aligned} \quad (6)$$

$$\begin{aligned} v_{-4}^{0000} &= R_{16} \\ v_{-3}^{0000} &= R_{17} \\ v_{-2}^{0000} &= R_{18} \\ v_{-1}^{0000} &= R_{19} \\ v_0^{0000} &= R_{20} \end{aligned} \quad (7)$$

1.2.2 Component 0001

$$\begin{aligned}
h_{-4}^{0001} &= R_{21} \\
h_{-3}^{0001} &= R_{22} \\
h_{-2}^{0001} &= R_{23} \\
h_{-1}^{0001} &= R_{24}
\end{aligned} \tag{8}$$

$$\begin{aligned}
b_{-4}^{0001} &= R_{25} \\
b_{-3}^{0001} &= R_{26} \\
b_{-2}^{0001} &= R_{27} \\
b_{-1}^{0001} &= R_{28}
\end{aligned} \tag{9}$$

$$\begin{aligned}
u_{-4}^{0001} &= R_{29} \\
u_{-3}^{0001} &= R_{30} \\
u_{-2}^{0001} &= R_{31} \\
u_{-1}^{0001} &= R_{32}
\end{aligned} \tag{10}$$

$$\begin{aligned}
v_{-4}^{0001} &= R_{33} \\
v_{-3}^{0001} &= R_{34} \\
v_{-2}^{0001} &= R_{35} \\
v_{-1}^{0001} &= R_{36}
\end{aligned} \tag{11}$$

1.2.3 Component 0011

$$\begin{aligned}
h_{-4}^{0011} &= R_{37} \\
h_{-3}^{0011} &= R_{38} \\
h_{-2}^{0011} &= R_{39} \\
h_{-1}^{0011} &= R_{40}
\end{aligned} \tag{12}$$

$$\begin{aligned}
b_{-4}^{0011} &= R_{41} \\
b_{-3}^{0011} &= R_{42} \\
b_{-2}^{0011} &= R_{43}
\end{aligned} \tag{13}$$

$$\begin{aligned}
u_{-4}^{0011} &= R_{44} \\
u_{-3}^{0011} &= R_{45} \\
u_{-2}^{0011} &= R_{46} \\
u_{-1}^{0011} &= R_{47}
\end{aligned} \tag{14}$$

$$\begin{aligned}
v_{-4}^{0011} &= R_{48} \\
v_{-3}^{0011} &= R_{49} \\
v_{-2}^{0011} &= R_{50} \\
v_{-1}^{0011} &= R_{51}
\end{aligned} \tag{15}$$

1.2.4 Component 0022

$$\begin{aligned}
h_{-4}^{0022} &= R_{52} \\
h_{-3}^{0022} &= R_{53} \\
h_{-2}^{0022} &= R_{54}
\end{aligned} \tag{16}$$

$$\begin{aligned}
b_{-4}^{0022} &= R_{55} \\
b_{-3}^{0022} &= R_{56} \\
b_{-2}^{0022} &= R_{57}
\end{aligned} \tag{17}$$

$$\begin{aligned}
u_{-4}^{0022} &= R_{58} \\
u_{-3}^{0022} &= R_{59} \\
u_{-2}^{0022} &= R_{60} \\
u_{-1}^{0022} &= R_{51}
\end{aligned} \tag{18}$$

$$\begin{aligned}
v_{-4}^{0022} &= R_{61} \\
v_{-3}^{0022} &= R_{62} \\
v_{-2}^{0022} &= R_{63} \\
v_{-1}^{0022} &= R_{64}
\end{aligned} \tag{19}$$

1.2.5 Component 0101

$$\begin{aligned}
h_{-4}^{0101} &= R_{65} \\
h_{-3}^{0101} &= R_{66} \\
h_{-2}^{0101} &= R_{67} \\
h_{-1}^{0101} &= R_{68}
\end{aligned} \tag{20}$$

$$\begin{aligned}
b_{-4}^{0101} &= R_{69} \\
b_{-3}^{0101} &= R_{70} \\
b_{-2}^{0101} &= R_{71}
\end{aligned} \tag{21}$$

$$\begin{aligned}
u_{-4}^{0101} &= R_{72} \\
u_{-3}^{0101} &= R_{73} \\
u_{-2}^{0101} &= R_{74} \\
u_{-1}^{0101} &= R_{75}
\end{aligned} \tag{22}$$

$$\begin{aligned}
v_{-4}^{0101} &= R_{76} \\
v_{-3}^{0101} &= R_{77} \\
v_{-2}^{0101} &= R_{78} \\
v_{-1}^{0101} &= R_{79}
\end{aligned} \tag{23}$$

1.2.6 Component 0111

$$\begin{aligned}
h_{-4}^{0111} &= R_{80} \\
h_{-3}^{0111} &= R_{81} \\
h_{-2}^{0111} &= R_{82} \\
h_{-1}^{0111} &= R_{83}
\end{aligned} \tag{24}$$

$$\begin{aligned}
b_{-3}^{0111} &= R_{84} \\
b_{-2}^{0111} &= R_{85}
\end{aligned} \tag{25}$$

$$\begin{aligned}
u_{-4}^{0111} &= R_{86} \\
u_{-3}^{0111} &= R_{87} \\
u_{-2}^{0111} &= R_{88} \\
u_{-1}^{0111} &= R_{83}
\end{aligned} \tag{26}$$

$$\begin{aligned}
v_{-4}^{0111} &= R_{89} \\
v_{-3}^{0111} &= R_{90} \\
v_{-2}^{0111} &= R_{91}
\end{aligned} \tag{27}$$

1.2.7 Component 0122

$$\begin{aligned} h_{-3}^{0122} &= R_{92} \\ h_{-2}^{0122} &= R_{93} \end{aligned} \tag{28}$$

$$\begin{aligned} b_{-3}^{0122} &= R_{94} \\ b_{-2}^{0122} &= R_{95} \end{aligned} \tag{29}$$

$$\begin{aligned} u_{-3}^{0122} &= R_{96} \\ u_{-2}^{0122} &= R_{97} \end{aligned} \tag{30}$$

$$v^{0122} = 0. \tag{31}$$

1.2.8 Component 0202

$$\begin{aligned} h_{-4}^{0202} &= R_{98} \\ h_{-3}^{0202} &= R_{99} \\ h_{-2}^{0202} &= R_{100} \\ h_{-1}^{0202} &= R_{101} \end{aligned} \tag{32}$$

$$\begin{aligned} b_{-4}^{0202} &= R_{102} \\ b_{-3}^{0202} &= R_{103} \\ b_{-2}^{0202} &= R_{104} \end{aligned} \tag{33}$$

$$\begin{aligned} u_{-4}^{0202} &= R_{105} \\ u_{-3}^{0202} &= R_{106} \\ u_{-2}^{0202} &= R_{107} \\ u_{-1}^{0202} &= R_{108} \end{aligned} \tag{34}$$

$$\begin{aligned} v_{-4}^{0202} &= R_{109} \\ v_{-3}^{0202} &= R_{110} \\ v_{-2}^{0202} &= R_{111} \\ v_{-1}^{0202} &= R_{112} \end{aligned} \tag{35}$$

1.2.9 Component 0212

$$\begin{aligned} h_{-4}^{0212} &= R_{113} \\ h_{-3}^{0212} &= R_{114} \\ h_{-2}^{0212} &= R_{115} \\ h_{-1}^{0212} &= R_{116} \end{aligned} \tag{36}$$

$$\begin{aligned} b_{-3}^{0212} &= R_{117} \\ b_{-2}^{0212} &= R_{118} \end{aligned} \tag{37}$$

$$\begin{aligned} u_{-4}^{0212} &= R_{119} \\ u_{-3}^{0212} &= R_{120} \\ u_{-2}^{0212} &= R_{121} \\ u_{-1}^{0212} &= R_{116} \end{aligned} \tag{38}$$

$$\begin{aligned} v_{-4}^{0212} &= R_{122} \\ v_{-3}^{0212} &= R_{123} \\ v_{-2}^{0212} &= R_{124} \end{aligned} \tag{39}$$

1.2.10 Component 1111

$$\begin{aligned} h_{-4}^{1111} &= R_{125} \\ h_{-3}^{1111} &= R_{126} \\ h_{-2}^{1111} &= R_{127} \\ h_{-1}^{1111} &= R_{128} \end{aligned} \tag{40}$$

$$b_{-2}^{1111} = R_{129} \tag{41}$$

$$\begin{aligned} u_{-4}^{1111} &= R_{130} \\ u_{-3}^{1111} &= R_{131} \\ u_{-2}^{1111} &= R_{132} \\ u_{-1}^{1111} &= R_{128} \end{aligned} \tag{42}$$

$$\begin{aligned} v_{-4}^{1111} &= R_{133} \\ v_{-3}^{1111} &= R_{134} \\ v_{-2}^{1111} &= R_{135} \end{aligned} \tag{43}$$

1.2.11 Component 1122

$$\begin{aligned} h_{-4}^{1122} &= R_{136} \\ h_{-3}^{1122} &= R_{137} \\ h_{-2}^{1122} &= R_{138} \end{aligned} \tag{44}$$

$$b_{-2}^{1122} = R_{139} \tag{45}$$

$$\begin{aligned} u_{-4}^{1122} &= R_{140} \\ u_{-3}^{1122} &= R_{141} \\ u_{-2}^{1122} &= R_{142} \end{aligned} \tag{46}$$

$$\begin{aligned} v_{-4}^{1122} &= R_{143} \\ v_{-3}^{1122} &= R_{144} \\ v_{-2}^{1122} &= R_{145} \end{aligned} \tag{47}$$

1.2.12 Component 1212

$$\begin{aligned} h_{-4}^{1212} &= R_{146} \\ h_{-3}^{1212} &= R_{147} \\ h_{-2}^{1212} &= R_{148} \\ h_{-1}^{1212} &= R_{149} \end{aligned} \tag{48}$$

$$b_{-2}^{1212} = R_{150} \tag{49}$$

$$\begin{aligned} u_{-4}^{1212} &= R_{151} \\ u_{-3}^{1212} &= R_{152} \\ u_{-2}^{1212} &= R_{153} \\ u_{-1}^{1212} &= R_{149} \end{aligned} \tag{50}$$

$$\begin{aligned} v_{-4}^{1212} &= R_{154} \\ v_{-3}^{1212} &= R_{155} \\ v_{-2}^{1212} &= R_{156} \end{aligned} \tag{51}$$

1.2.13 Component 2222

$$\begin{aligned}
h_{-4}^{2222} &= R_{157} \\
h_{-3}^{2222} &= R_{158} \\
h_{-2}^{2222} &= R_{159} \\
h_{-1}^{2222} &= R_{160}
\end{aligned} \tag{52}$$

$$b_{-2}^{2222} = R_{161} \tag{53}$$

$$\begin{aligned}
u_{-4}^{2222} &= R_{162} \\
u_{-3}^{2222} &= R_{163} \\
u_{-2}^{2222} &= R_{164} \\
u_{-1}^{2222} &= R_{160}
\end{aligned} \tag{54}$$

$$\begin{aligned}
v_{-4}^{2222} &= R_{165} \\
v_{-3}^{2222} &= R_{166} \\
v_{-2}^{2222} &= R_{167}
\end{aligned} \tag{55}$$

1.2.14 Complete table

A value of one in column S denotes $z \mapsto -z$; in column T , it denotes the exchange of the incoming and outgoing pairs defined in (3). No transformations are required for zero rows.

Component	Representative	S	T	Nonzero
0000	0000	0	0	1
0001	0001	0	0	1
0002	0002	0	0	0
0010	0001	1	0	1
0011	0011	0	0	1
0012	0012	0	0	0
0020	0002	1	0	0
0021	0012	1	0	0
0022	0022	0	0	1
0100	0001	0	1	1
0101	0101	0	0	1
0102	0102	0	0	0
0110	0101	1	0	1
0111	0111	0	0	1
0112	0112	0	0	0
0120	0102	1	0	0
0121	0112	1	0	0
0122	0122	0	0	1
0200	0002	0	1	0
0201	0102	0	1	0
0202	0202	0	0	1
0210	0102	1	1	0
0211	0211	0	0	0
0212	0212	0	0	1
0220	0202	1	0	1
0221	0212	1	0	1
0222	0222	0	0	0
1000	0001	1	1	1
1001	0101	1	0	1
1002	0102	1	0	0

1010	0101	0	0	1
1011	0111	1	0	1
1012	0112	1	0	0
1020	0102	0	0	0
1021	0112	0	0	0
1022	0122	1	0	1
1100	0011	0	1	1
1101	0111	0	1	1
1102	0211	0	1	0
1110	0111	1	1	1
1111	1111	0	0	1
1112	1112	0	0	0
1120	0211	1	1	0
1121	1112	1	0	0
1122	1122	0	0	1
1200	0012	0	1	0
1201	0112	0	1	0
1202	0212	0	1	1
1210	0112	1	1	0
1211	1112	0	1	0
1212	1212	0	0	1
1220	0212	1	1	1
1221	1212	1	0	1
1222	1222	0	0	0
2000	0002	1	1	0
2001	0102	1	1	0
2002	0202	1	0	1
2010	0102	0	1	0
2011	0211	1	0	0
2012	0212	1	0	1
2020	0202	0	0	1
2021	0212	0	0	1
2022	0222	1	0	0
2100	0012	1	1	0
2101	0112	1	1	0
2102	0212	1	1	1
2110	0112	0	1	0
2111	1112	1	1	0
2112	1212	1	0	1
2120	0212	0	1	1
2121	1212	0	0	1
2122	1222	1	0	0
2200	0022	0	1	1
2201	0122	0	1	1
2202	0222	0	1	0
2210	0122	1	1	1
2211	1122	0	1	1
2212	1222	0	1	0
2220	0222	1	1	0
2221	1222	1	1	0
2222	2222	0	0	1

1.3 Coefficients of the internal field and its x-derivative for all 54 pairs

In the heading $ij : ab$, the first two indices denote the momentum positions, and the second two denote the types of the two external fields. The coefficients are defined by

$$\frac{[L_3]_{ij,I}}{g_R k^3} = e^{-i\Omega_{ij}x} [A_{ij,I}(x)X_I + B_{ij,I}(x)\partial_x X_I], \quad A_{ij,I} = \sum_{p=-3}^0 A_{ij,I,p}x^p, \quad B_{ij,I} = \sum_{p=-2}^0 B_{ij,I,p}x^p. \quad (56)$$

The overall factor $g_R k^3$ has been factored out in accordance with the normalization in the article. The derivative ∂_x is taken with respect to $x = k\eta$. The spatial momentum of the trial internal line is $-(\mathbf{p}_i + \mathbf{p}_j)$, but its polarization basis is constructed from $\mathbf{p}_i + \mathbf{p}_j$. The index $I = 0, 1, 2$ refers to a line with nonzero momentum; for the pairs 01 and 23, the index $I = 0, 1, \dots, 5$ refers to six homogeneous directions. All coefficients not listed are zero.

1.3.1 Pair 01:00

$$A^{0100;0} = 0. \quad (57)$$

$$A^{0100;1} = 0. \quad (58)$$

$$\begin{aligned} A_{-3}^{0100;2} &= R_{168} \\ A_{-2}^{0100;2} &= R_{169} \\ A_{-1}^{0100;2} &= R_{170} \end{aligned} \quad (59)$$

$$A^{0100;3} = 0. \quad (60)$$

$$A^{0100;4} = 0. \quad (61)$$

$$A^{0100;5} = 0. \quad (62)$$

$$\begin{aligned} B_{-2}^{0100;0} &= R_{140} \\ B_{-1}^{0100;0} &= R_{171} \\ B_0^{0100;0} &= R_{172} \end{aligned} \quad (63)$$

$$B^{0100;1} = 0. \quad (64)$$

$$\begin{aligned} B_{-2}^{0100;2} &= R_{173} \\ B_{-1}^{0100;2} &= R_{174} \end{aligned} \quad (65)$$

$$B^{0100;3} = 0. \quad (66)$$

$$B^{0100;4} = 0. \quad (67)$$

$$B^{0100;5} = 0. \quad (68)$$

1.3.2 Pair 01:01

$$A^{0101;0} = 0. \quad (69)$$

$$\begin{aligned} A_{-3}^{0101;1} &= R_{143} \\ A_{-2}^{0101;1} &= R_{175} \end{aligned} \quad (70)$$

$$A^{0101;2} = 0. \quad (71)$$

$$A^{0101;3} = 0. \quad (72)$$

$$A^{0101;4} = 0. \quad (73)$$

$$A^{0101;5} = 0. \quad (74)$$

$$B^{0101;0} = 0. \quad (75)$$

$$B_{-2}^{0101;1} = R_{176}$$

$$B_{-1}^{0101;1} = R_{177} \quad (76)$$

$$B^{0101;2} = 0. \quad (77)$$

$$B^{0101;3} = 0. \quad (78)$$

$$B^{0101;4} = 0. \quad (79)$$

$$B^{0101;5} = 0. \quad (80)$$

1.3.3 Pair 01:02

$$A^{0102;0} = 0. \quad (81)$$

$$A^{0102;1} = 0. \quad (82)$$

$$A^{0102;2} = 0. \quad (83)$$

$$A_{-3}^{0102;3} = R_{176}$$

$$A_{-2}^{0102;3} = R_{177} \quad (84)$$

$$A^{0102;4} = 0. \quad (85)$$

$$A^{0102;5} = 0. \quad (86)$$

$$B^{0102;0} = 0. \quad (87)$$

$$B^{0102;1} = 0. \quad (88)$$

$$B^{0102;2} = 0. \quad (89)$$

$$B_{-2}^{0102;3} = R_{143}$$

$$B_{-1}^{0102;3} = R_{175} \quad (90)$$

$$B^{0102;4} = 0. \quad (91)$$

$$B^{0102;5} = 0. \quad (92)$$

1.3.4 Pair 01:10

$$A^{0110;0} = 0. \quad (93)$$

$$A_{-3}^{0110;1} = R_{143}$$

$$A_{-2}^{0110;1} = R_{175} \quad (94)$$

$$A^{0110;2} = 0. \quad (95)$$

$$A^{0110;3} = 0. \quad (96)$$

$$A^{0110;4} = 0. \quad (97)$$

$$A^{0110;5} = 0. \quad (98)$$

$$B^{0110;0} = 0. \quad (99)$$

$$B_{-2}^{0110;1} = R_{176}$$

$$B_{-1}^{0110;1} = R_{177} \quad (100)$$

$$B^{0110;2} = 0. \quad (101)$$

$$B^{0110;3} = 0. \quad (102)$$

$$B^{0110;4} = 0. \quad (103)$$

$$B^{0110;5} = 0. \quad (104)$$

1.3.5 Pair 01:11

$$A^{0111;0} = 0. \quad (105)$$

$$A^{0111;1} = 0. \quad (106)$$

$$A_{-1}^{0111;2} = R_{178} \quad (107)$$

$$A^{0111;3} = 0. \quad (108)$$

$$A^{0111;4} = 0. \quad (109)$$

$$A^{0111;5} = 0. \quad (110)$$

$$B_{-2}^{0111;0} = R_{179}$$

$$B_{-1}^{0111;0} = R_{174} \quad (111)$$

$$B^{0111;1} = 0. \quad (112)$$

$$B^{0111;2} = 0. \quad (113)$$

$$B^{0111;3} = 0. \quad (114)$$

$$B^{0111;4} = 0. \quad (115)$$

$$B^{0111;5} = 0. \quad (116)$$

1.3.6 Pair 01:12

$$A^{0112;0} = 0. \quad (117)$$

$$A^{0112;1} = 0. \quad (118)$$

$$A^{0112;2} = 0. \quad (119)$$

$$A^{0112;3} = 0. \quad (120)$$

$$A^{0112;4} = 0. \quad (121)$$

$$A^{0112;5} = 0. \quad (122)$$

$$B^{0112;0} = 0. \quad (123)$$

$$B^{0112;1} = 0. \quad (124)$$

$$B^{0112;2} = 0. \quad (125)$$

$$B^{0112;3} = 0. \quad (126)$$

$$B^{0112;4} = 0. \quad (127)$$

$$B^{0112;5} = 0. \quad (128)$$

1.3.7 Pair 01:20

$$A^{0120;0} = 0. \quad (129)$$

$$A^{0120;1} = 0. \quad (130)$$

$$A^{0120;2} = 0. \quad (131)$$

$$A_{-3}^{0120;3} = R_{143}$$

$$A_{-2}^{0120;3} = R_{175} \quad (132)$$

$$A^{0120;4} = 0. \quad (133)$$

$$A^{0120;5} = 0. \quad (134)$$

$$B^{0120;0} = 0. \quad (135)$$

$$B^{0120;1} = 0. \quad (136)$$

$$B^{0120;2} = 0. \quad (137)$$

$$B_{-2}^{0120;3} = R_{176}$$

$$B_{-1}^{0120;3} = R_{177} \quad (138)$$

$$B^{0120;4} = 0. \quad (139)$$

$$B^{0120;5} = 0. \quad (140)$$

1.3.8 Pair 01:21

$$A^{0121;0} = 0. \quad (141)$$

$$A^{0121;1} = 0. \quad (142)$$

$$A^{0121;2} = 0. \quad (143)$$

$$A^{0121;3} = 0. \quad (144)$$

$$A^{0121;4} = 0. \quad (145)$$

$$A^{0121;5} = 0. \quad (146)$$

$$B^{0121;0} = 0. \quad (147)$$

$$B^{0121;1} = 0. \quad (148)$$

$$B^{0121;2} = 0. \quad (149)$$

$$B^{0121;3} = 0. \quad (150)$$

$$B^{0121;4} = 0. \quad (151)$$

$$B^{0121;5} = 0. \quad (152)$$

1.3.9 Pair 01:22

$$A^{0122;0} = 0. \quad (153)$$

$$A^{0122;1} = 0. \quad (154)$$

$$A_{-1}^{0122;2} = R_{180} \quad (155)$$

$$A^{0122;3} = 0. \quad (156)$$

$$A^{0122;4} = 0. \quad (157)$$

$$A^{0122;5} = 0. \quad (158)$$

$$B_{-2}^{0122;0} = R_{181}$$

$$B_{-1}^{0122;0} = R_{169} \quad (159)$$

$$B^{0122;1} = 0. \quad (160)$$

$$B^{0122;2} = 0. \quad (161)$$

$$B^{0122;3} = 0. \quad (162)$$

$$B^{0122;4} = 0. \quad (163)$$

$$B^{0122;5} = 0. \quad (164)$$

1.3.10 Pair 02:00

$$A_{-3}^{0200;0} = R_{182}$$

$$A_{-2}^{0200;0} = R_{183}$$

$$A_{-1}^{0200;0} = R_{184}$$

$$A_0^{0200;0} = R_{185} \quad (165)$$

$$A_{-3}^{0200;1} = R_{186}$$

$$A_{-2}^{0200;1} = R_{187}$$

$$A_{-1}^{0200;1} = R_{188} \quad (166)$$

$$A^{0200;2} = 0. \quad (167)$$

$$B_{-2}^{0200;0} = R_{189}$$

$$B_{-1}^{0200;0} = R_{190}$$

$$B_0^{0200;0} = R_{191} \quad (168)$$

$$B_{-2}^{0200;1} = R_{192}$$

$$B_{-1}^{0200;1} = R_{193} \quad (169)$$

$$B^{0200;2} = 0. \quad (170)$$

1.3.11 Pair 02:01

$$\begin{aligned} A_{-3}^{0201;0} &= R_{194} \\ A_{-2}^{0201;0} &= R_{195} \\ A_{-1}^{0201;0} &= R_{196} \end{aligned} \tag{171}$$

$$\begin{aligned} A_{-3}^{0201;1} &= R_{197} \\ A_{-2}^{0201;1} &= R_{198} \\ A_{-1}^{0201;1} &= R_{199} \end{aligned} \tag{172}$$

$$A^{0201;2} = 0. \tag{173}$$

$$\begin{aligned} B_{-2}^{0201;0} &= R_{200} \\ B_{-1}^{0201;0} &= R_{201} \end{aligned} \tag{174}$$

$$\begin{aligned} B_{-2}^{0201;1} &= R_{202} \\ B_{-1}^{0201;1} &= R_{203} \end{aligned} \tag{175}$$

$$B^{0201;2} = 0. \tag{176}$$

1.3.12 Pair 02:02

$$A^{0202;0} = 0. \tag{177}$$

$$A^{0202;1} = 0. \tag{178}$$

$$\begin{aligned} A_{-3}^{0202;2} &= R_{204} \\ A_{-2}^{0202;2} &= R_{205} \\ A_{-1}^{0202;2} &= R_{206} \end{aligned} \tag{179}$$

$$B^{0202;0} = 0. \tag{180}$$

$$B^{0202;1} = 0. \tag{181}$$

$$\begin{aligned} B_{-2}^{0202;2} &= R_{207} \\ B_{-1}^{0202;2} &= R_{208} \end{aligned} \tag{182}$$

1.3.13 Pair 02:10

$$\begin{aligned} A_{-3}^{0210;0} &= R_{209} \\ A_{-2}^{0210;0} &= R_{210} \\ A_{-1}^{0210;0} &= R_{211} \end{aligned} \tag{183}$$

$$\begin{aligned} A_{-3}^{0210;1} &= R_{212} \\ A_{-2}^{0210;1} &= R_{213} \\ A_{-1}^{0210;1} &= R_{214} \end{aligned} \tag{184}$$

$$A^{0210;2} = 0. \tag{185}$$

$$\begin{aligned} B_{-2}^{0210;0} &= R_{215} \\ B_{-1}^{0210;0} &= R_{216} \end{aligned} \tag{186}$$

$$\begin{aligned} B_{-2}^{0210;1} &= R_{217} \\ B_{-1}^{0210;1} &= R_{218} \end{aligned} \tag{187}$$

$$B^{0210;2} = 0. \tag{188}$$

1.3.14 Pair 02:11

$$\begin{aligned} A_{-3}^{0211;0} &= R_{219} \\ A_{-2}^{0211;0} &= R_{220} \\ A_{-1}^{0211;0} &= R_{221} \end{aligned} \tag{189}$$

$$A_{-1}^{0211;1} = R_{222} \tag{190}$$

$$A^{0211;2} = 0. \tag{191}$$

$$\begin{aligned} B_{-2}^{0211;0} &= R_{223} \\ B_{-1}^{0211;0} &= R_{224} \end{aligned} \tag{192}$$

$$B^{0211;1} = 0. \tag{193}$$

$$B^{0211;2} = 0. \tag{194}$$

1.3.15 Pair 02:12

$$A^{0212;0} = 0. \tag{195}$$

$$A^{0212;1} = 0. \tag{196}$$

$$A_{-1}^{0212;2} = R_{225} \tag{197}$$

$$B^{0212;0} = 0. \tag{198}$$

$$B^{0212;1} = 0. \tag{199}$$

$$B^{0212;2} = 0. \tag{200}$$

1.3.16 Pair 02:20

$$A^{0220;0} = 0. \tag{201}$$

$$A^{0220;1} = 0. \tag{202}$$

$$\begin{aligned} A_{-3}^{0220;2} &= R_{226} \\ A_{-2}^{0220;2} &= R_{227} \\ A_{-1}^{0220;2} &= R_{228} \end{aligned} \tag{203}$$

$$B^{0220;0} = 0. \tag{204}$$

$$B^{0220;1} = 0. \tag{205}$$

$$\begin{aligned} B_{-2}^{0220;2} &= R_{229} \\ B_{-1}^{0220;2} &= R_{230} \end{aligned} \tag{206}$$

1.3.17 Pair 02:21

$$A^{0221;0} = 0. \tag{207}$$

$$A^{0221;1} = 0. \tag{208}$$

$$A_{-1}^{0221;2} = R_{231} \tag{209}$$

$$B^{0221;0} = 0. \tag{210}$$

$$B^{0221;1} = 0. \tag{211}$$

$$B^{0221;2} = 0. \tag{212}$$

1.3.18 Pair 02:22

$$\begin{aligned} A_{-3}^{0222;0} &= R_{232} \\ A_{-2}^{0222;0} &= R_{233} \\ A_{-1}^{0222;0} &= R_{234} \end{aligned} \tag{213}$$

$$A_{-1}^{0222;1} = R_{235} \tag{214}$$

$$A^{0222;2} = 0. \tag{215}$$

$$\begin{aligned} B_{-2}^{0222;0} &= R_{236} \\ B_{-1}^{0222;0} &= R_{237} \end{aligned} \tag{216}$$

$$B^{0222;1} = 0. \tag{217}$$

$$B^{0222;2} = 0. \tag{218}$$

1.3.19 Pair 03:00

$$\begin{aligned} A_{-3}^{0300;0} &= R_{238} \\ A_{-2}^{0300;0} &= R_{239} \\ A_{-1}^{0300;0} &= R_{240} \\ A_0^{0300;0} &= R_{241} \end{aligned} \tag{219}$$

$$\begin{aligned} A_{-3}^{0300;1} &= R_{242} \\ A_{-2}^{0300;1} &= R_{243} \\ A_{-1}^{0300;1} &= R_{244} \end{aligned} \tag{220}$$

$$A^{0300;2} = 0. \tag{221}$$

$$\begin{aligned} B_{-2}^{0300;0} &= R_{245} \\ B_{-1}^{0300;0} &= R_{246} \\ B_0^{0300;0} &= R_{247} \end{aligned} \tag{222}$$

$$\begin{aligned} B_{-2}^{0300;1} &= R_{248} \\ B_{-1}^{0300;1} &= R_{249} \end{aligned} \tag{223}$$

$$B^{0300;2} = 0. \tag{224}$$

1.3.20 Pair 03:01

$$\begin{aligned} A_{-3}^{0301;0} &= R_{250} \\ A_{-2}^{0301;0} &= R_{251} \\ A_{-1}^{0301;0} &= R_{196} \end{aligned} \tag{225}$$

$$\begin{aligned} A_{-3}^{0301;1} &= R_{252} \\ A_{-2}^{0301;1} &= R_{253} \\ A_{-1}^{0301;1} &= R_{254} \end{aligned} \tag{226}$$

$$A^{0301;2} = 0. \tag{227}$$

$$\begin{aligned} B_{-2}^{0301;0} &= R_{255} \\ B_{-1}^{0301;0} &= R_{256} \end{aligned} \tag{228}$$

$$\begin{aligned} B_{-2}^{0301;1} &= R_{257} \\ B_{-1}^{0301;1} &= R_{258} \end{aligned} \tag{229}$$

$$B^{0301;2} = 0. \tag{230}$$

1.3.21 Pair 03:02

$$A^{0302;0} = 0. \quad (231)$$

$$A^{0302;1} = 0. \quad (232)$$

$$\begin{aligned} A_{-3}^{0302;2} &= R_{259} \\ A_{-2}^{0302;2} &= R_{260} \\ A_{-1}^{0302;2} &= R_{261} \end{aligned} \quad (233)$$

$$B^{0302;0} = 0. \quad (234)$$

$$B^{0302;1} = 0. \quad (235)$$

$$\begin{aligned} B_{-2}^{0302;2} &= R_{262} \\ B_{-1}^{0302;2} &= R_{263} \end{aligned} \quad (236)$$

1.3.22 Pair 03:10

$$\begin{aligned} A_{-3}^{0310;0} &= R_{264} \\ A_{-2}^{0310;0} &= R_{265} \\ A_{-1}^{0310;0} &= R_{211} \end{aligned} \quad (237)$$

$$\begin{aligned} A_{-3}^{0310;1} &= R_{266} \\ A_{-2}^{0310;1} &= R_{267} \\ A_{-1}^{0310;1} &= R_{268} \end{aligned} \quad (238)$$

$$A^{0310;2} = 0. \quad (239)$$

$$\begin{aligned} B_{-2}^{0310;0} &= R_{269} \\ B_{-1}^{0310;0} &= R_{270} \end{aligned} \quad (240)$$

$$\begin{aligned} B_{-2}^{0310;1} &= R_{271} \\ B_{-1}^{0310;1} &= R_{272} \end{aligned} \quad (241)$$

$$B^{0310;2} = 0. \quad (242)$$

1.3.23 Pair 03:11

$$\begin{aligned} A_{-3}^{0311;0} &= R_{219} \\ A_{-2}^{0311;0} &= R_{273} \\ A_{-1}^{0311;0} &= R_{274} \end{aligned} \quad (243)$$

$$A_{-1}^{0311;1} = R_{275} \quad (244)$$

$$A^{0311;2} = 0. \quad (245)$$

$$\begin{aligned} B_{-2}^{0311;0} &= R_{223} \\ B_{-1}^{0311;0} &= R_{276} \end{aligned} \quad (246)$$

$$B^{0311;1} = 0. \quad (247)$$

$$B^{0311;2} = 0. \quad (248)$$

1.3.24 Pair 03:12

$$A^{0312;0} = 0. \quad (249)$$

$$A^{0312;1} = 0. \quad (250)$$

$$A_{-1}^{0312;2} = R_{277} \quad (251)$$

$$B^{0312;0} = 0. \quad (252)$$

$$B^{0312;1} = 0. \quad (253)$$

$$B^{0312;2} = 0. \quad (254)$$

1.3.25 Pair 03:20

$$A^{0320;0} = 0. \quad (255)$$

$$A^{0320;1} = 0. \quad (256)$$

$$A_{-3}^{0320;2} = R_{278}$$

$$A_{-2}^{0320;2} = R_{279}$$

$$A_{-1}^{0320;2} = R_{280} \quad (257)$$

$$B^{0320;0} = 0. \quad (258)$$

$$B^{0320;1} = 0. \quad (259)$$

$$B_{-2}^{0320;2} = R_{281}$$

$$B_{-1}^{0320;2} = R_{282} \quad (260)$$

1.3.26 Pair 03:21

$$A^{0321;0} = 0. \quad (261)$$

$$A^{0321;1} = 0. \quad (262)$$

$$A_{-1}^{0321;2} = R_{283} \quad (263)$$

$$B^{0321;0} = 0. \quad (264)$$

$$B^{0321;1} = 0. \quad (265)$$

$$B^{0321;2} = 0. \quad (266)$$

1.3.27 Pair 03:22

$$A_{-3}^{0322;0} = R_{284}$$

$$A_{-2}^{0322;0} = R_{285}$$

$$A_{-1}^{0322;0} = R_{286} \quad (267)$$

$$A_{-1}^{0322;1} = R_{287} \quad (268)$$

$$A^{0322;2} = 0. \quad (269)$$

$$B_{-2}^{0322;0} = R_{288}$$

$$B_{-1}^{0322;0} = R_{289} \quad (270)$$

$$B^{0322;1} = 0. \quad (271)$$

$$B^{0322;2} = 0. \quad (272)$$

1.3.28 Pair 12:00

$$\begin{aligned}
A_{-3}^{1200;0} &= R_{238} \\
A_{-2}^{1200;0} &= R_{239} \\
A_{-1}^{1200;0} &= R_{240} \\
A_0^{1200;0} &= R_{241}
\end{aligned} \tag{273}$$

$$\begin{aligned}
A_{-3}^{1200;1} &= R_{242} \\
A_{-2}^{1200;1} &= R_{243} \\
A_{-1}^{1200;1} &= R_{244}
\end{aligned} \tag{274}$$

$$A^{1200;2} = 0. \tag{275}$$

$$\begin{aligned}
B_{-2}^{1200;0} &= R_{245} \\
B_{-1}^{1200;0} &= R_{246} \\
B_0^{1200;0} &= R_{247}
\end{aligned} \tag{276}$$

$$\begin{aligned}
B_{-2}^{1200;1} &= R_{248} \\
B_{-1}^{1200;1} &= R_{249}
\end{aligned} \tag{277}$$

$$B^{1200;2} = 0. \tag{278}$$

1.3.29 Pair 12:01

$$\begin{aligned}
A_{-3}^{1201;0} &= R_{250} \\
A_{-2}^{1201;0} &= R_{251} \\
A_{-1}^{1201;0} &= R_{196}
\end{aligned} \tag{279}$$

$$\begin{aligned}
A_{-3}^{1201;1} &= R_{252} \\
A_{-2}^{1201;1} &= R_{253} \\
A_{-1}^{1201;1} &= R_{254}
\end{aligned} \tag{280}$$

$$A^{1201;2} = 0. \tag{281}$$

$$\begin{aligned}
B_{-2}^{1201;0} &= R_{255} \\
B_{-1}^{1201;0} &= R_{256}
\end{aligned} \tag{282}$$

$$\begin{aligned}
B_{-2}^{1201;1} &= R_{257} \\
B_{-1}^{1201;1} &= R_{258}
\end{aligned} \tag{283}$$

$$B^{1201;2} = 0. \tag{284}$$

1.3.30 Pair 12:02

$$A^{1202;0} = 0. \tag{285}$$

$$A^{1202;1} = 0. \tag{286}$$

$$\begin{aligned}
A_{-3}^{1202;2} &= R_{259} \\
A_{-2}^{1202;2} &= R_{260} \\
A_{-1}^{1202;2} &= R_{261}
\end{aligned} \tag{287}$$

$$B^{1202;0} = 0. \tag{288}$$

$$B^{1202;1} = 0. \tag{289}$$

$$\begin{aligned}
B_{-2}^{1202;2} &= R_{262} \\
B_{-1}^{1202;2} &= R_{263}
\end{aligned} \tag{290}$$

1.3.31 Pair 12:10

$$\begin{aligned} A_{-3}^{1210;0} &= R_{264} \\ A_{-2}^{1210;0} &= R_{265} \\ A_{-1}^{1210;0} &= R_{211} \end{aligned} \tag{291}$$

$$\begin{aligned} A_{-3}^{1210;1} &= R_{266} \\ A_{-2}^{1210;1} &= R_{267} \\ A_{-1}^{1210;1} &= R_{268} \end{aligned} \tag{292}$$

$$A^{1210;2} = 0. \tag{293}$$

$$\begin{aligned} B_{-2}^{1210;0} &= R_{269} \\ B_{-1}^{1210;0} &= R_{270} \end{aligned} \tag{294}$$

$$\begin{aligned} B_{-2}^{1210;1} &= R_{271} \\ B_{-1}^{1210;1} &= R_{272} \end{aligned} \tag{295}$$

$$B^{1210;2} = 0. \tag{296}$$

1.3.32 Pair 12:11

$$\begin{aligned} A_{-3}^{1211;0} &= R_{219} \\ A_{-2}^{1211;0} &= R_{273} \\ A_{-1}^{1211;0} &= R_{274} \end{aligned} \tag{297}$$

$$A_{-1}^{1211;1} = R_{275} \tag{298}$$

$$A^{1211;2} = 0. \tag{299}$$

$$\begin{aligned} B_{-2}^{1211;0} &= R_{223} \\ B_{-1}^{1211;0} &= R_{276} \end{aligned} \tag{300}$$

$$B^{1211;1} = 0. \tag{301}$$

$$B^{1211;2} = 0. \tag{302}$$

1.3.33 Pair 12:12

$$A^{1212;0} = 0. \tag{303}$$

$$A^{1212;1} = 0. \tag{304}$$

$$A_{-1}^{1212;2} = R_{277} \tag{305}$$

$$B^{1212;0} = 0. \tag{306}$$

$$B^{1212;1} = 0. \tag{307}$$

$$B^{1212;2} = 0. \tag{308}$$

1.3.34 Pair 12:20

$$A^{1220;0} = 0. \tag{309}$$

$$A^{1220;1} = 0. \tag{310}$$

$$A_{-3}^{1220;2} = R_{278}$$

$$A_{-2}^{1220;2} = R_{279}$$

$$A_{-1}^{1220;2} = R_{280} \tag{311}$$

$$B^{1220;0} = 0. \tag{312}$$

$$B^{1220;1} = 0. \tag{313}$$

$$B_{-2}^{1220;2} = R_{281}$$

$$B_{-1}^{1220;2} = R_{282} \tag{314}$$

1.3.35 Pair 12:21

$$A^{1221;0} = 0. \quad (315)$$

$$A^{1221;1} = 0. \quad (316)$$

$$A_{-1}^{1221;2} = R_{283} \quad (317)$$

$$B^{1221;0} = 0. \quad (318)$$

$$B^{1221;1} = 0. \quad (319)$$

$$B^{1221;2} = 0. \quad (320)$$

1.3.36 Pair 12:22

$$\begin{aligned} A_{-3}^{1222;0} &= R_{284} \\ A_{-2}^{1222;0} &= R_{285} \\ A_{-1}^{1222;0} &= R_{286} \end{aligned} \quad (321)$$

$$A_{-1}^{1222;1} = R_{287} \quad (322)$$

$$A^{1222;2} = 0. \quad (323)$$

$$\begin{aligned} B_{-2}^{1222;0} &= R_{288} \\ B_{-1}^{1222;0} &= R_{289} \end{aligned} \quad (324)$$

$$B^{1222;1} = 0. \quad (325)$$

$$B^{1222;2} = 0. \quad (326)$$

1.3.37 Pair 13:00

$$\begin{aligned} A_{-3}^{1300;0} &= R_{182} \\ A_{-2}^{1300;0} &= R_{183} \\ A_{-1}^{1300;0} &= R_{184} \\ A_0^{1300;0} &= R_{185} \end{aligned} \quad (327)$$

$$\begin{aligned} A_{-3}^{1300;1} &= R_{186} \\ A_{-2}^{1300;1} &= R_{187} \\ A_{-1}^{1300;1} &= R_{188} \end{aligned} \quad (328)$$

$$A^{1300;2} = 0. \quad (329)$$

$$\begin{aligned} B_{-2}^{1300;0} &= R_{189} \\ B_{-1}^{1300;0} &= R_{190} \\ B_0^{1300;0} &= R_{191} \end{aligned} \quad (330)$$

$$\begin{aligned} B_{-2}^{1300;1} &= R_{192} \\ B_{-1}^{1300;1} &= R_{193} \end{aligned} \quad (331)$$

$$B^{1300;2} = 0. \quad (332)$$

1.3.38 Pair 13:01

$$\begin{aligned} A_{-3}^{1301;0} &= R_{194} \\ A_{-2}^{1301;0} &= R_{195} \\ A_{-1}^{1301;0} &= R_{196} \end{aligned} \tag{333}$$

$$\begin{aligned} A_{-3}^{1301;1} &= R_{197} \\ A_{-2}^{1301;1} &= R_{198} \\ A_{-1}^{1301;1} &= R_{199} \end{aligned} \tag{334}$$

$$A^{1301;2} = 0. \tag{335}$$

$$\begin{aligned} B_{-2}^{1301;0} &= R_{200} \\ B_{-1}^{1301;0} &= R_{201} \end{aligned} \tag{336}$$

$$\begin{aligned} B_{-2}^{1301;1} &= R_{202} \\ B_{-1}^{1301;1} &= R_{203} \end{aligned} \tag{337}$$

$$B^{1301;2} = 0. \tag{338}$$

1.3.39 Pair 13:02

$$A^{1302;0} = 0. \tag{339}$$

$$A^{1302;1} = 0. \tag{340}$$

$$\begin{aligned} A_{-3}^{1302;2} &= R_{204} \\ A_{-2}^{1302;2} &= R_{205} \\ A_{-1}^{1302;2} &= R_{206} \end{aligned} \tag{341}$$

$$B^{1302;0} = 0. \tag{342}$$

$$B^{1302;1} = 0. \tag{343}$$

$$\begin{aligned} B_{-2}^{1302;2} &= R_{207} \\ B_{-1}^{1302;2} &= R_{208} \end{aligned} \tag{344}$$

1.3.40 Pair 13:10

$$\begin{aligned} A_{-3}^{1310;0} &= R_{209} \\ A_{-2}^{1310;0} &= R_{210} \\ A_{-1}^{1310;0} &= R_{211} \end{aligned} \tag{345}$$

$$\begin{aligned} A_{-3}^{1310;1} &= R_{212} \\ A_{-2}^{1310;1} &= R_{213} \\ A_{-1}^{1310;1} &= R_{214} \end{aligned} \tag{346}$$

$$A^{1310;2} = 0. \tag{347}$$

$$\begin{aligned} B_{-2}^{1310;0} &= R_{215} \\ B_{-1}^{1310;0} &= R_{216} \end{aligned} \tag{348}$$

$$\begin{aligned} B_{-2}^{1310;1} &= R_{217} \\ B_{-1}^{1310;1} &= R_{218} \end{aligned} \tag{349}$$

$$B^{1310;2} = 0. \tag{350}$$

1.3.41 Pair 13:11

$$\begin{aligned} A_{-3}^{1311;0} &= R_{219} \\ A_{-2}^{1311;0} &= R_{220} \\ A_{-1}^{1311;0} &= R_{221} \end{aligned} \tag{351}$$

$$A_{-1}^{1311;1} = R_{222} \tag{352}$$

$$A^{1311;2} = 0. \tag{353}$$

$$\begin{aligned} B_{-2}^{1311;0} &= R_{223} \\ B_{-1}^{1311;0} &= R_{224} \end{aligned} \tag{354}$$

$$B^{1311;1} = 0. \tag{355}$$

$$B^{1311;2} = 0. \tag{356}$$

1.3.42 Pair 13:12

$$A^{1312;0} = 0. \tag{357}$$

$$A^{1312;1} = 0. \tag{358}$$

$$A_{-1}^{1312;2} = R_{225} \tag{359}$$

$$B^{1312;0} = 0. \tag{360}$$

$$B^{1312;1} = 0. \tag{361}$$

$$B^{1312;2} = 0. \tag{362}$$

1.3.43 Pair 13:20

$$A^{1320;0} = 0. \tag{363}$$

$$A^{1320;1} = 0. \tag{364}$$

$$\begin{aligned} A_{-3}^{1320;2} &= R_{226} \\ A_{-2}^{1320;2} &= R_{227} \\ A_{-1}^{1320;2} &= R_{228} \end{aligned} \tag{365}$$

$$B^{1320;0} = 0. \tag{366}$$

$$B^{1320;1} = 0. \tag{367}$$

$$\begin{aligned} B_{-2}^{1320;2} &= R_{229} \\ B_{-1}^{1320;2} &= R_{230} \end{aligned} \tag{368}$$

1.3.44 Pair 13:21

$$A^{1321;0} = 0. \tag{369}$$

$$A^{1321;1} = 0. \tag{370}$$

$$A_{-1}^{1321;2} = R_{231} \tag{371}$$

$$B^{1321;0} = 0. \tag{372}$$

$$B^{1321;1} = 0. \tag{373}$$

$$B^{1321;2} = 0. \tag{374}$$

1.3.45 Pair 13:22

$$\begin{aligned} A_{-3}^{1322;0} &= R_{232} \\ A_{-2}^{1322;0} &= R_{233} \\ A_{-1}^{1322;0} &= R_{234} \end{aligned} \tag{375}$$

$$A_{-1}^{1322;1} = R_{235} \tag{376}$$

$$A^{1322;2} = 0. \tag{377}$$

$$\begin{aligned} B_{-2}^{1322;0} &= R_{236} \\ B_{-1}^{1322;0} &= R_{237} \end{aligned} \tag{378}$$

$$B^{1322;1} = 0. \tag{379}$$

$$B^{1322;2} = 0. \tag{380}$$

1.3.46 Pair 23:00

$$A^{2300;0} = 0. \tag{381}$$

$$\begin{aligned} A_{-3}^{2300;1} &= R_{290} \\ A_{-2}^{2300;1} &= R_{291} \\ A_{-1}^{2300;1} &= R_{292} \end{aligned} \tag{382}$$

$$\begin{aligned} A_{-3}^{2300;2} &= R_{293} \\ A_{-2}^{2300;2} &= R_{294} \\ A_{-1}^{2300;2} &= R_{295} \end{aligned} \tag{383}$$

$$A^{2300;3} = 0. \tag{384}$$

$$\begin{aligned} A_{-3}^{2300;4} &= R_{296} \\ A_{-2}^{2300;4} &= R_{297} \\ A_{-1}^{2300;4} &= R_{298} \end{aligned} \tag{385}$$

$$A^{2300;5} = 0. \tag{386}$$

$$\begin{aligned} B_{-2}^{2300;0} &= R_{140} \\ B_{-1}^{2300;0} &= R_{299} \\ B_0^{2300;0} &= R_{300} \end{aligned} \tag{387}$$

$$\begin{aligned} B_{-2}^{2300;1} &= R_{301} \\ B_{-1}^{2300;1} &= R_{302} \end{aligned} \tag{388}$$

$$\begin{aligned} B_{-2}^{2300;2} &= R_{303} \\ B_{-1}^{2300;2} &= R_{304} \end{aligned} \tag{389}$$

$$B^{2300;3} = 0. \tag{390}$$

$$\begin{aligned} B_{-2}^{2300;4} &= R_{305} \\ B_{-1}^{2300;4} &= R_{306} \end{aligned} \tag{391}$$

$$B^{2300;5} = 0. \tag{392}$$

1.3.47 Pair 23:01

$$A^{2301;0} = 0. \quad (393)$$

$$\begin{aligned} A_{-3}^{2301;1} &= R_{307} \\ A_{-2}^{2301;1} &= R_{308} \end{aligned} \quad (394)$$

$$\begin{aligned} A_{-3}^{2301;2} &= R_{309} \\ A_{-2}^{2301;2} &= R_{310} \end{aligned} \quad (395)$$

$$A^{2301;3} = 0. \quad (396)$$

$$\begin{aligned} A_{-3}^{2301;4} &= R_{311} \\ A_{-2}^{2301;4} &= R_{312} \end{aligned} \quad (397)$$

$$A^{2301;5} = 0. \quad (398)$$

$$B^{2301;0} = 0. \quad (399)$$

$$\begin{aligned} B_{-2}^{2301;1} &= R_{219} \\ B_{-1}^{2301;1} &= R_{313} \end{aligned} \quad (400)$$

$$\begin{aligned} B_{-2}^{2301;2} &= R_{314} \\ B_{-1}^{2301;2} &= R_{315} \end{aligned} \quad (401)$$

$$B^{2301;3} = 0. \quad (402)$$

$$\begin{aligned} B_{-2}^{2301;4} &= R_{316} \\ B_{-1}^{2301;4} &= R_{317} \end{aligned} \quad (403)$$

$$B^{2301;5} = 0. \quad (404)$$

1.3.48 Pair 23:02

$$A^{2302;0} = 0. \quad (405)$$

$$A^{2302;1} = 0. \quad (406)$$

$$A^{2302;2} = 0. \quad (407)$$

$$\begin{aligned} A_{-3}^{2302;3} &= R_{232} \\ A_{-2}^{2302;3} &= R_{318} \end{aligned} \quad (408)$$

$$A^{2302;4} = 0. \quad (409)$$

$$\begin{aligned} A_{-3}^{2302;5} &= R_{319} \\ A_{-2}^{2302;5} &= R_{320} \end{aligned} \quad (410)$$

$$B^{2302;0} = 0. \quad (411)$$

$$B^{2302;1} = 0. \quad (412)$$

$$B^{2302;2} = 0. \quad (413)$$

$$\begin{aligned} B_{-2}^{2302;3} &= R_{284} \\ B_{-1}^{2302;3} &= R_{321} \end{aligned} \quad (414)$$

$$B^{2302;4} = 0. \quad (415)$$

$$\begin{aligned} B_{-2}^{2302;5} &= R_{322} \\ B_{-1}^{2302;5} &= R_{323} \end{aligned} \quad (416)$$

1.3.49 Pair 23:10

$$A^{2310;0} = 0. \quad (417)$$

$$\begin{aligned} A_{-3}^{2310;1} &= R_{307} \\ A_{-2}^{2310;1} &= R_{308} \end{aligned} \quad (418)$$

$$\begin{aligned} A_{-3}^{2310;2} &= R_{309} \\ A_{-2}^{2310;2} &= R_{310} \end{aligned} \quad (419)$$

$$A^{2310;3} = 0. \quad (420)$$

$$\begin{aligned} A_{-3}^{2310;4} &= R_{311} \\ A_{-2}^{2310;4} &= R_{312} \end{aligned} \quad (421)$$

$$A^{2310;5} = 0. \quad (422)$$

$$B^{2310;0} = 0. \quad (423)$$

$$\begin{aligned} B_{-2}^{2310;1} &= R_{219} \\ B_{-1}^{2310;1} &= R_{313} \end{aligned} \quad (424)$$

$$\begin{aligned} B_{-2}^{2310;2} &= R_{314} \\ B_{-1}^{2310;2} &= R_{315} \end{aligned} \quad (425)$$

$$B^{2310;3} = 0. \quad (426)$$

$$\begin{aligned} B_{-2}^{2310;4} &= R_{316} \\ B_{-1}^{2310;4} &= R_{317} \end{aligned} \quad (427)$$

$$B^{2310;5} = 0. \quad (428)$$

1.3.50 Pair 23:11

$$A^{2311;0} = 0. \quad (429)$$

$$A_{-1}^{2311;1} = R_{324} \quad (430)$$

$$A_{-1}^{2311;2} = R_{325} \quad (431)$$

$$A^{2311;3} = 0. \quad (432)$$

$$A_{-1}^{2311;4} = R_{326} \quad (433)$$

$$A^{2311;5} = 0. \quad (434)$$

$$\begin{aligned} B_{-2}^{2311;0} &= R_{179} \\ B_{-1}^{2311;0} &= R_{327} \end{aligned} \quad (435)$$

$$B^{2311;1} = 0. \quad (436)$$

$$B^{2311;2} = 0. \quad (437)$$

$$B^{2311;3} = 0. \quad (438)$$

$$B^{2311;4} = 0. \quad (439)$$

$$B^{2311;5} = 0. \quad (440)$$

1.3.51 Pair 23:12

$$A^{2312;0} = 0. \quad (441)$$

$$A^{2312;1} = 0. \quad (442)$$

$$A^{2312;2} = 0. \quad (443)$$

$$A^{2312;3} = 0. \quad (444)$$

$$A^{2312;4} = 0. \quad (445)$$

$$A^{2312;5} = 0. \quad (446)$$

$$B^{2312;0} = 0. \quad (447)$$

$$B^{2312;1} = 0. \quad (448)$$

$$B^{2312;2} = 0. \quad (449)$$

$$B^{2312;3} = 0. \quad (450)$$

$$B^{2312;4} = 0. \quad (451)$$

$$B^{2312;5} = 0. \quad (452)$$

1.3.52 Pair 23:20

$$A^{2320;0} = 0. \quad (453)$$

$$A^{2320;1} = 0. \quad (454)$$

$$A^{2320;2} = 0. \quad (455)$$

$$A_{-3}^{2320;3} = R_{284}$$

$$A_{-2}^{2320;3} = R_{321} \quad (456)$$

$$A^{2320;4} = 0. \quad (457)$$

$$A_{-3}^{2320;5} = R_{322}$$

$$A_{-2}^{2320;5} = R_{323} \quad (458)$$

$$B^{2320;0} = 0. \quad (459)$$

$$B^{2320;1} = 0. \quad (460)$$

$$B^{2320;2} = 0. \quad (461)$$

$$B_{-2}^{2320;3} = R_{232}$$

$$B_{-1}^{2320;3} = R_{318} \quad (462)$$

$$B^{2320;4} = 0. \quad (463)$$

$$B_{-2}^{2320;5} = R_{319}$$

$$B_{-1}^{2320;5} = R_{320} \quad (464)$$

1.3.53 Pair 23:21

$$A^{2321;0} = 0. \quad (465)$$

$$A^{2321;1} = 0. \quad (466)$$

$$A^{2321;2} = 0. \quad (467)$$

$$A^{2321;3} = 0. \quad (468)$$

$$A^{2321;4} = 0. \quad (469)$$

$$A^{2321;5} = 0. \quad (470)$$

$$B^{2321;0} = 0. \quad (471)$$

$$B^{2321;1} = 0. \quad (472)$$

$$B^{2321;2} = 0. \quad (473)$$

$$B^{2321;3} = 0. \quad (474)$$

$$B^{2321;4} = 0. \quad (475)$$

$$B^{2321;5} = 0. \quad (476)$$

1.3.54 Pair 23:22

$$A^{2322;0} = 0. \quad (477)$$

$$A_{-1}^{2322;1} = R_{328} \quad (478)$$

$$A_{-1}^{2322;2} = R_{329} \quad (479)$$

$$A^{2322;3} = 0. \quad (480)$$

$$A_{-1}^{2322;4} = R_{330} \quad (481)$$

$$A^{2322;5} = 0. \quad (482)$$

$$B_{-2}^{2322;0} = R_{181}$$

$$B_{-1}^{2322;0} = R_{331} \quad (483)$$

$$B^{2322;1} = 0. \quad (484)$$

$$B^{2322;2} = 0. \quad (485)$$

$$B^{2322;3} = 0. \quad (486)$$

$$B^{2322;4} = 0. \quad (487)$$

$$B^{2322;5} = 0. \quad (488)$$

1.4 Twenty-four time-ordered exchange entries

For each set of external fields, the table below specifies all entries, including those with zero coefficients.

Partition	Later pair	Earlier pair	Internal indices
02 13	02	13	0, 1, 2
02 13	13	02	0, 1, 2
03 12	03	12	0, 1, 2
03 12	12	03	0, 1, 2
01 23	01	23	0, 1, 2, 3, 4, 5
01 23	23	01	0, 1, 2, 3, 4, 5

For a line with nonzero momentum, $\nu_I = c_I |\mathbf{p}_i + \mathbf{p}_j|$. The exchange coefficients at two times are obtained by expanding the product

$$\frac{\tau_I}{2\nu_I \mathcal{N}_I} [A_{a,I}(x) - i\nu_I B_{a,I}(x)] [A_{b,I}(y) + i\nu_I B_{b,I}(y)]. \quad (489)$$

Each coefficient of $x^p y^s$ is multiplied by $-D_{ps}(\Omega_a + \nu_I, \Omega_b - \nu_I)$. The corresponding row of the table is used for the reverse time ordering.

For a global coordinate, set $X_a = A_a$, $Y_a = \sigma_I [(x - x_i)A_a + B_a]$. The covariance bilinear form (402) of the main article is used instead of (489). For the scalar, $\sigma_I = -1$, and there is no division by a negative norm; for the tensors, $\sigma_I = 1$, and division by 2, 6, 2, 2, 2 is retained. The physical covariances C_{QQ}, C_{PP}, C_{QP} are fixed before choosing k , whereas the dimensionless covariances transform as $C_{qq} = kC_{QQ}$, $C_{pp} = C_{PP}/k$, $C_{qp} = C_{QP}$. The additional volume factor and the two spatial delta functions are not included in A, B .

1.5 Finite reconstruction of the angular dependence

The radial basis in Subsections 1.2–1.3 is also a complete basis for the angular coefficients. The following finite procedure computes each coefficient without requiring data in machine-readable form. All operations are performed before taking the limit $r \rightarrow 1$.

For the partition $\varsigma = \pm 1$, substitute $z_\varsigma = (1 + r^2 - q^2)\varsigma/(2r)$. After cancellation of common factors in the rational expression, its denominator is a product of a power of q and a factor depending only on r . Multiplying out and collecting terms with equal powers of q therefore gives the finite series

$$A_{a,I,p}(r, z_\varsigma) = \sum_u a_{a,I,p,u}(r) q^u, \quad B_{a,I,p}(r, z_\varsigma) = \sum_u b_{a,I,p,u}(r) q^u. \quad (490)$$

The left-hand sides are given explicitly in terms of the functions R_s . Obtaining these series requires neither differentiating the action nor solving new constraints. The absence of a nonmonomial denominator in q is checked separately in the computational verification protocol.

Let $m = h_0 - h_1$, $n = h_2 - h_3$, $Z = (1 + r^2)/(2r)$, and

$$d_{mn}^j(\theta) = p_{jmn} W_{jmn}(z), \quad p_{jmn} = \sqrt{(j+m)!(j-m)!(j+n)!(j-n)!}. \quad (491)$$

For each admissible b , set $A_b = j + (n - m)/2 - b$, $B_b = (m - n)/2 + b$. The finite set of coefficients w_{2d} of the polynomial $W_{jmn}(z_\varsigma) = \sum_d w_{2d} q^{2d}$ is given by

$$w_{2d} = \sum_b \frac{(-1)^{m-n+b}}{2^j (j+n-b)! b! (m-n+b)! (j-m-b)!} \times \sum_{u+v=d} \binom{A_b}{u} \binom{B_b}{v} (1 + \varsigma Z)^{A_b-u} (1 - \varsigma Z)^{B_b-v} \left(-\frac{\varsigma}{2r}\right)^u \left(\frac{\varsigma}{2r}\right)^v. \quad (492)$$

The outer sum is restricted by the nonnegativity of the factorial arguments, and the inner sum by the conditions $0 \leq u \leq A_b$, $0 \leq v \leq B_b$. All coefficients w with odd indices are zero. Set $\mathcal{N}_I = n_I q^{d_I}$, where $(n_I, d_I) = (1, 0), (2, 0), (2, 2)$, and

$$v_{a,I,p,u}^\pm = a_{a,I,p,u} \mp i c_I b_{a,I,p,u-1}. \quad (493)$$

For each set of external fields, partition, and time ordering, the exact exchange coefficients in the linear polarization basis are given by the finite convolution

$$C_{ps\ell}^{a|b,I} = \frac{\tau_I}{2c_I n_I} \sum_{t+u+v=\ell+d_I} w_t v_{a,I,p,u}^+ v_{b,I,s,v}^- \quad (494)$$

The individual parts $h_p^{(d)}$ of the contact term are first reconstructed using Subsection 1.5.2. The corresponding momentum transfer is chosen for each crossed partition before adding the contributions of the two partitions. If $h_p^{(d)}(r, z_\zeta) = \sum_u h_{p,u}^{(d)} q^u$, then

$$H_{p\ell}^{(d)} = \sum_{t+u=\ell-1} w_t h_{p,u}^{(d)}. \quad (495)$$

The original contact term and the homogeneous part have no soft momentum-transfer denominator. They can be integrated with respect to z using the rule $\int_{-1}^1 z^v dz = 2/(v+1)$ for even v ; for odd v , the integral is zero. For homogeneous exchange, this rule applies to the products $A_a A_b$, $A_a B_b$, $B_a A_b$, $B_a B_b$ with the appropriate normalization. The fixed covariance bilinear form of the preceding section is then reconstructed using X, Y .

The transformation to helicities is performed by a finite sum over the four linear polarization indices with $C_{00} = 1$, $C_{h1} = 1/2$, $C_{h2} = i \operatorname{sgn}(h)/2$. Coefficients not listed are zero. The overall factors p_{jmn}/r , $2\pi/\sqrt{(1+\delta_{h_0 h_1})(1+\delta_{h_2 h_3})}$, the frequency factors in the normalization of the external one-particle states, and $g_R^2 k^3$ are restored according to the article. No additional tensor normalization is applied after this.

For the boundary contributions, the same convolutions use $f_a V_b^-$, $V_a^+ f_b$, and $f_a f_b$. The nonzero coefficients f in the linear polarization basis at $\zeta = \zeta z$ are

$$\begin{aligned} f_{00} &= \frac{6ic_0 r(r-1)(1+\zeta)}{q^2}, & f_{11} &= \frac{6ir(r-1)(1+\zeta)(1+\zeta^2)}{q^2}, \\ f_{22} &= -\frac{12ir(r-1)\zeta(1+\zeta)}{q^2}, & f_{ab} &= 0 \quad (a \neq b). \end{aligned} \quad (496)$$

The coefficients of their finite series in q replace one or both coefficients v in (494) at $I = 0$. The coefficients are multiplied by exactly the time factors in Eqs. (390)–(391) of the main article [1]. The factors x_i^{-1}, x_f^{-1} are not included in the coefficients a second time. The contributions from both time orderings are summed. The five boundary terms are given in Subsection C.6 of the main article [1], Eqs. (393)–(394). That subsection also shows the exact cancellation of the leading contribution proportional to q^{-3} when the boundary terms are combined with the bulk exchange.

The radial basis given here and the finite convolutions determine all bulk and boundary entries of the following table. Equation (492), the convolutions (494)–(495), and the boundary coefficients (496) define an analytical procedure for any integer $j \geq \max(|m|, |n|)$ allowed for the chosen external channels. If identical particles have equal helicities in the incoming or outgoing pair, j must be even, as stated after Eq. (369) of the main article [1]. The four checked values $j = 0, 2, 3, 4$ determine only the contents of the following table, not the domain of applicability of this procedure.

1.5.1 Table of 74 bulk and 74 boundary entries

In the table, a is the incoming channel number and c is the outgoing channel number, with numbering starting from zero. The columns give the helicity values, so the external indices and the arguments of the Wigner polynomial are uniquely determined. Zero entries are retained.

j	c	a	h_0, h_1	h_2, h_3	m	n	Number of exchanges
0	0	0	0, 0	0, 0	0	0	24
0	1	0	0, 0	2, 2	0	0	24
0	2	0	0, 0	-2, -2	0	0	24
0	0	1	2, 2	0, 0	0	0	24

0	1	1	2,2	2,2	0	0	24
0	2	1	2,2	-2,-2	0	0	24
0	0	2	-2,-2	0,0	0	0	24
0	1	2	-2,-2	2,2	0	0	24
0	2	2	-2,-2	-2,-2	0	0	24
2	0	0	0,0	0,0	0	0	24
2	1	0	0,0	0,2	0	-2	24
2	2	0	0,0	0,-2	0	2	24
2	3	0	0,0	2,2	0	0	24
2	4	0	0,0	-2,-2	0	0	24
2	0	1	0,2	0,0	-2	0	24
2	1	1	0,2	0,2	-2	-2	24
2	2	1	0,2	0,-2	-2	2	24
2	3	1	0,2	2,2	-2	0	24
2	4	1	0,2	-2,-2	-2	0	24
2	0	2	0,-2	0,0	2	0	24
2	1	2	0,-2	0,2	2	-2	24
2	2	2	0,-2	0,-2	2	2	24
2	3	2	0,-2	2,2	2	0	24
2	4	2	0,-2	-2,-2	2	0	24
2	0	3	2,2	0,0	0	0	24
2	1	3	2,2	0,2	0	-2	24
2	2	3	2,2	0,-2	0	2	24
2	3	3	2,2	2,2	0	0	24
2	4	3	2,2	-2,-2	0	0	24
2	0	4	-2,-2	0,0	0	0	24
2	1	4	-2,-2	0,2	0	-2	24
2	2	4	-2,-2	0,-2	0	2	24
2	3	4	-2,-2	2,2	0	0	24
2	4	4	-2,-2	-2,-2	0	0	24
3	0	0	0,2	0,2	-2	-2	24
3	1	0	0,2	0,-2	-2	2	24
3	0	1	0,-2	0,2	2	-2	24
3	1	1	0,-2	0,-2	2	2	24
4	0	0	0,0	0,0	0	0	24
4	1	0	0,0	0,2	0	-2	24
4	2	0	0,0	0,-2	0	2	24
4	3	0	0,0	2,2	0	0	24
4	4	0	0,0	2,-2	0	4	24
4	5	0	0,0	-2,-2	0	0	24
4	0	1	0,2	0,0	-2	0	24
4	1	1	0,2	0,2	-2	-2	24
4	2	1	0,2	0,-2	-2	2	24
4	3	1	0,2	2,2	-2	0	24
4	4	1	0,2	2,-2	-2	4	24
4	5	1	0,2	-2,-2	-2	0	24
4	0	2	0,-2	0,0	2	0	24
4	1	2	0,-2	0,2	2	-2	24
4	2	2	0,-2	0,-2	2	2	24
4	3	2	0,-2	2,2	2	0	24
4	4	2	0,-2	2,-2	2	4	24
4	5	2	0,-2	-2,-2	2	0	24
4	0	3	2,2	0,0	0	0	24

4	1	3	2, 2	0, 2	0	-2	24
4	2	3	2, 2	0, -2	0	2	24
4	3	3	2, 2	2, 2	0	0	24
4	4	3	2, 2	2, -2	0	4	24
4	5	3	2, 2	-2, -2	0	0	24
4	0	4	2, -2	0, 0	4	0	24
4	1	4	2, -2	0, 2	4	-2	24
4	2	4	2, -2	0, -2	4	2	24
4	3	4	2, -2	2, 2	4	0	24
4	4	4	2, -2	2, -2	4	4	24
4	5	4	2, -2	-2, -2	4	0	24
4	0	5	-2, -2	0, 0	0	0	24
4	1	5	-2, -2	0, 2	0	-2	24
4	2	5	-2, -2	0, -2	0	2	24
4	3	5	-2, -2	2, 2	0	0	24
4	4	5	-2, -2	2, -2	0	4	24
4	5	5	-2, -2	-2, -2	0	0	24

1.5.2 Reconstruction of the individual contact partitions

The sums u_p and v_p in Subsection 1.2 do not determine the distribution of regular terms among the partitions. This distribution is specified explicitly below. The labels $0, +, -$ refer, respectively, to the partitions $01|23$, $02|13$, $03|12$ and to the momentum transfers

$$q_0 = 0, \quad q_+^2 = 1 + r^2 - 2rz, \quad q_-^2 = 1 + r^2 + 2rz. \quad (497)$$

The coefficients decompose as

$$\begin{aligned} u_p &= u_p^{(0)} + u_p^{(+)} + u_p^{(-)}, & v_p &= v_p^{(0)} + v_p^{(+)} + v_p^{(-)}, \\ h_p &= b_p + u_p^{(0)} + u_p^{(+)} + u_p^{(-)} + v_p^{(0)} + v_p^{(+)} + v_p^{(-)}. \end{aligned} \quad (498)$$

The external indices $ABCD$ are omitted in these equalities. The seven terms in the second line are the individual parts of the contact term denoted by $h_p^{(d)}$ in (495) and in Eq. (378) of the main article [1].

Legendre-transform contact terms. For $a|b = 02|13$ or $03|12$, use the coefficients $B_{a,I,s}$ from Subsection 1.3, calculated with the internal momentum direction of the corresponding pair taken into account. Their finite convolution gives

$$v_p^{(a|b)} = \sum_{I=0}^2 \frac{\tau_I}{\mathcal{N}_I} \sum_{s+t=p} B_{a,I,s} B_{b,I,t}, \quad (\tau_I) = (1, 1, -1), \quad (\mathcal{N}_I) = (1, 2, 2q_{a|b}^2). \quad (499)$$

Here $q_{02|13} = q_+$, $q_{03|12} = q_-$. The homogeneous partition uses six coordinates and positive squared geometric norms:

$$\begin{aligned} v_p^{(0)} &= \sum_{I=0}^5 \frac{\sigma_I}{\mathcal{N}_I^{(0)}} \sum_{s+t=p} B_{01,I,s} B_{23,I,t}, \\ (\sigma_I) &= (-1, 1, 1, 1, 1, 1), \quad (\mathcal{N}_I^{(0)}) = (1, 2, 6, 2, 2, 2). \end{aligned} \quad (500)$$

In both convolutions, $s, t \in \{-2, -1, 0\}$; coefficients not listed are zero. The minus sign at $I = 0$ in (500) corresponds to the inverse kinetic coefficient of the homogeneous scalar coordinate. No additional factor of $1/2$ is introduced: the two cross terms of the quadratic correction arising from the Legendre transform have already been combined for one unordered partition. Nor is a second reversal of the polarizations at the other end of the internal line required, since it is already accounted for by the factor τ_I .

Homogeneous contact term from constraint elimination. Define the dimensionless Laurent polynomials $\mathcal{J}_{01}^{AB}(x)$ and $\mathcal{J}_{23}^{CD}(x)$ by

$$\begin{aligned}\mathcal{J}_{01}^{00} &= -18x^{-2} - \frac{20\sqrt{3}i}{3}x^{-1} + \frac{4}{3}, \\ \mathcal{J}_{23}^{00} &= -18x^{-2} + \frac{20\sqrt{3}ir}{3}x^{-1} + \frac{4r^2}{3}, \\ \mathcal{J}_{01}^{11} &= -\mathcal{J}_{01}^{22} = -2x^{-2} - 4ix^{-1}, \\ \mathcal{J}_{23}^{11} &= -\mathcal{J}_{23}^{22} = -2x^{-2} + 4irx^{-1}.\end{aligned}\tag{501}$$

All polynomials with mixed external indices are zero. Denoting by $[x^p]$ the operation of extracting the coefficient of x^p , we obtain

$$u_p^{(0),ABCD} = 2[x^p](\mathcal{J}_{01}^{AB}(x)\mathcal{J}_{23}^{CD}(x)).\tag{502}$$

This finite rule determines the homogeneous part independently of the behavior of nonzero momentum transfer as $r \rightarrow 1$ and $z \rightarrow \pm 1$.

Contact terms from constraint elimination in the crossed partitions. The 46 nonzero coefficients $u_p^{(+),ABCD}$ for the 13 representatives in Subsection 1.2 are given below. Coefficients of all other powers, including $p = 0$, are zero. The formulas retain the full rational functions, including the regular terms. For each representative, the coefficients of the second crossed partition are determined from the sum in Subsection 1.2:

$$u_p^{(-),ABCD} = u_p^{ABCD} - u_p^{(0),ABCD} - u_p^{(+),ABCD}.\tag{503}$$

Thus, the coefficients below, together with Eqs. (502) and (503), specify both crossed parts without repeating the constraint elimination.

Permutations and the order of reconstruction of the angular dependence. For $w = u$ or $w = v$, permutations of the external indices also act on the partition label. Let $\iota(0) = 0$, $\iota(+)= -$, $\iota(-) = +$. Then

$$\begin{aligned}w_p^{(d),BACD}(r, z) &= w_p^{(d),ABDC}(r, z) = w_p^{\iota(d),ABCD}(r, -z), \\ w_p^{(d),CDAB}(r, z) &= r^{4+p} \overline{w_p^{(d),ABCD}(1/r, z)}.\end{aligned}\tag{504}$$

The table in Subsection 1.2.14 applies to the entire triple of partitions: an odd number of within-pair exchanges requires the simultaneous replacements $z \mapsto -z$ and $+ \leftrightarrow -$. Exchanging the incoming and outgoing pairs does not change the partition label. All 40 components containing an odd number of fields with the second linear tensor polarization are zero separately in each partition.

For the part labeled $+$, substitute $z = (1+r^2-q^2)/(2r)$ into (495); for the part labeled $-$, substitute $z = -(1+r^2-q^2)/(2r)$. The coefficients in q are extracted separately and only then summed. If the polynomial part is also separated out, division with remainder is performed in the variable q at fixed r : the numerator of the reduced fraction $u_p^{(\pm)}(r, z_\varsigma)$ or $v_p^{(\pm)}(r, z_\varsigma)$ is divided by its denominator after the stated substitution with the corresponding sign ς . As in (490), the denominator contains only a nonnegative integer power of q and a nonzero factor depending on r . The polynomial quotient therefore consists of terms with nonnegative powers of q , while the remainder divided by the denominator consists of terms with negative powers. This is equivalent to separating the finite Laurent series already obtained according to the sign of the power. The division is performed separately within each reconstructed function, and the polynomial quotient retains the same partition label. If the denominator is independent of q , the entire function belongs to the quotient. The sum of negative powers alone does not replace the full function: the regular part is neither discarded nor redistributed between $+$ and $-$.

The terms $u_p^{(0)}$ and $v_p^{(0)}$ are retained separately from b_p and from the regular parts at nonzero momentum transfer. Their volume normalization remains that of the isolated zero mode, specified by Eq. (403) of the main article [1]. This normalization is not obtained by substituting $q = 0$ into the expression for a line with nonzero momentum. The absence of a soft momentum-transfer pole in a regular term does not by itself make that term an exactly homogeneous contribution.

Coefficients $u_p^{(+),0000}$.

$$u_{-4}^{(+),0000} = \frac{1}{q_+^4 r} \begin{bmatrix} -72z^3 + 10r + 10r^5 + 20r^3 + 24z \\ -264r^2z^3 - 264r^4z^3 - 104r^3z^2 \\ -72r^6z^3 + 24r^6z + 60rz^2 \\ +60r^5z^2 + 66r^4z + 66r^5z^4 \\ +72r^2z^5 + 72r^4z^5 + 80r^2z \\ +80r^4z + 132r^3z^4 \end{bmatrix} \quad (505)$$

$$u_{-3}^{(+),0000} = \frac{-4\sqrt{3}ir + 4\sqrt{3}i}{3q_+^4} \begin{bmatrix} 1 + r^4 - 9z^4 - 6z + 2r^2 + 18z^2 + 36z^3 \\ -42rz^4 - 42r^3z^4 - 36r^2z^4 \\ -26r^2z^2 - 18rz^2 - 18r^2z^5 \\ -18r^3z^2 - 12r^2z - 9r^4z^4 - 6rz^3 \\ -6r^4z - 6r^3z^3 + 8rz + 8r^3z \\ +18rz^5 + 18r^3z^5 + 18r^4z^2 \\ +36r^4z^3 + 90r^2z^3 \end{bmatrix} \quad (506)$$

$$u_{-2}^{(+),0000} = \frac{2rz}{3q_+^4} \begin{bmatrix} -8 - 36z^3 - 16r^2 - 8r^4 + 12r + 12r^3 \\ +24z^2 + 36z - 72r^2z^4 - 66rz^2 \\ -66r^3z^2 - 55rz - 55r^3z - 42r^2z^3 \\ -36r^4z^3 + 6r^2z + 9rz^3 + 9r^3z^3 \\ +24r^4z^2 + 36r^4z + 72rz^4 \\ +72r^3z^4 + 148r^2z^2 \end{bmatrix} \quad (507)$$

$$u_{-1}^{(+),0000} = -\frac{4\sqrt{3}ir^2z^2(r-1)(z+1)(3z-1)}{3q_+^2} \quad (508)$$

Coefficients $u_p^{(+),0001}$.

$$u_{-4}^{(+),0001} = \frac{2\sqrt{3}z^2 - 2\sqrt{3}}{q_+^4 r} \begin{bmatrix} -r - 6z - 6r^3 - 5r^5 - 26r^2z - 18r^6z \\ -12r^2z^3 - 9r^5z^2 - 2r^4z + 11rz^2 \\ +26r^3z^2 + 48r^4z^3 \end{bmatrix} \quad (509)$$

$$u_{-3}^{(+),0001} = \frac{-2iz^2 + 2i}{q_+^4} \begin{bmatrix} -r - 6z - 6z^2 - 4r^3 - 3r^5 + 8r^2 \\ +8r^4 + \sqrt{3}r^5 - \sqrt{3}r - 96r^3z^3 \\ -24r^2z - 12r^2z^3 - 10r^3z \\ -9r^5z^2 - 6\sqrt{3}z + 4r^2z^2 + 11rz^2 \\ +12rz^3 + 18r^4z^2 + 18r^4z^3 + 26rz \\ +30r^3z^2 + 36r^5z - \sqrt{3}rz^2 \\ -8\sqrt{3}r^4z - 4\sqrt{3}r^3z^2 \\ -3\sqrt{3}r^5z^2 + 4\sqrt{3}r^2z \\ +6\sqrt{3}r^4z^3 + 12\sqrt{3}r^2z^3 \end{bmatrix} \quad (510)$$

$$u_{-2}^{(+),0001} = \frac{2rz^2 - 2r}{3q_+^4} \begin{bmatrix} -18z^2 + 3r + 3r^2 + 3r^3 + 3r^4 + 18z \\ -39r^2z^2 - 36r^2z^3 - 18r^2z \\ -18r^4z^2 - 15r^3z - 9r^4z - 3\sqrt{3}r^2 \\ -3\sqrt{3}r^4 - 2\sqrt{3}z + 3\sqrt{3}r + 3rz^2 \\ +3\sqrt{3}r^3 + 6\sqrt{3}z^2 + 12rz + 36rz^3 \\ +36r^3z^2 + 36r^3z^3 - 48\sqrt{3}r^2z^3 \\ -27\sqrt{3}rz - 18\sqrt{3}r^4z^2 - 12\sqrt{3}rz^3 \\ -8\sqrt{3}r^2z + 3\sqrt{3}r^3z^2 + 21\sqrt{3}r^4z \\ +21\sqrt{3}r^2z^2 + 28\sqrt{3}rz^2 \\ +36\sqrt{3}r^3z^3 \end{bmatrix} \quad (511)$$

$$u_{-1}^{(+),0001} = \frac{-2ir^2z^3 + 2ir^2z}{3q_+^2} [-3r - 2\sqrt{3} - 6\sqrt{3}z + 3\sqrt{3}r + 9rz + 3\sqrt{3}rz] \quad (512)$$

Coefficients $u_p^{(+),0011}$.

$$u_{-4}^{(+),0011} = \frac{-2z^2 + 2}{q_+^4} \begin{bmatrix} -9 - 45r^4 - 26r^2 + 25z^2 - 324r^3z^3 \\ -108rz^3 - 60r^3z - 20rz + 144r^2z^4 \\ +189r^4z^2 + 234r^2z^2 \end{bmatrix} \quad (513)$$

$$u_{-3}^{(+),0011} = \frac{-4iz^2 + 4i}{q_+^4} \begin{bmatrix} -\sqrt{3} + 9r + 9r^5 + 32r^3 - rz^2 - 42r^4z \\ -28r^2z - 15\sqrt{3}r^4 - 12r^2z^3 \\ -9r^5z^2 - 2\sqrt{3}r^2 + 9\sqrt{3}z^2 \\ +18r^4z^3 + 24r^3z^2 - 108\sqrt{3}r^3z^3 \\ -42\sqrt{3}rz^3 - 28\sqrt{3}r^3z - 14\sqrt{3}rz \\ +48\sqrt{3}r^2z^4 + 63\sqrt{3}r^4z^2 \\ +90\sqrt{3}r^2z^2 \end{bmatrix} \quad (514)$$

$$u_{-2}^{(+),0011} = \frac{2z^2 - 2}{q_+^4} \begin{bmatrix} -17r^4 - 3r^2 + 12z^2 - 120r^3z^3 \\ -48rz^3 - 32r^3z - 24rz + 4\sqrt{3}r \\ +12\sqrt{3}r^5 + 30\sqrt{3}r^3 + 48r^2z^4 \\ +69r^4z^2 + 115r^2z^2 - 40\sqrt{3}r^4z \\ -24\sqrt{3}r^2z - 12\sqrt{3}r^5z^2 \\ -8\sqrt{3}r^2z^3 + 4\sqrt{3}rz^2 \\ +10\sqrt{3}r^3z^2 + 24\sqrt{3}r^4z^3 \end{bmatrix} \quad (515)$$

$$u_{-1}^{(+),0011} = -\frac{4ir^2(3r - \sqrt{3})(z - 1)^2(z + 1)^2}{q_+^2} \quad (516)$$

Coefficients $u_p^{(+),0022}$.

$$u_{-4}^{(+),0022} = \frac{32(z - 1)(z + 1)(3rz - 1)^2}{q_+^2} \quad (517)$$

$$u_{-3}^{(+),0022} = \frac{32i(z - 1)(z + 1)(3rz - 1)(2\sqrt{3}rz - r - \sqrt{3})}{q_+^2} \quad (518)$$

$$u_{-2}^{(+),0022} = -\frac{8(z - 1)(z + 1)(12r^2z^2 - 4\sqrt{3}r^2z + r^2 - 12rz + 2\sqrt{3}r + 3)}{q_+^2} \quad (519)$$

Coefficients $u_p^{(+),0101}$.

$$u_{-4}^{(+),0101} = \frac{2(z^2 + 1)(-18r^4z + 15r^3z^2 - 5r^3 + 18r^2z^3 - 2r^2z + 15rz^2 - 5r - 18z)}{q_+^2 r} \quad (520)$$

$$u_{-3}^{(+),0101} = \frac{2ir - 2i}{q_+^4} \left[\begin{aligned} &-1 + \sqrt{3} - r^4 - 27z^4 - 18z - 18z^3 - 6r \\ &-6r^3 - 4z^2 - 2r^2 + \sqrt{3}r^4 \\ &-36r^2z^3 - 27r^4z^4 - 20rz^3 \\ &-20r^3z^3 - 18r^4z - 18r^4z^3 \\ &-6rz^4 - 6\sqrt{3}z - 6\sqrt{3}z^3 - 6r^3z^4 \\ &-4r^4z^2 + 2\sqrt{3}r^2 + 2r^2z^4 \\ &+3\sqrt{3}z^4 + 4rz + 4r^3z + 4\sqrt{3}z^2 \\ &+18r^2z + 18r^2z^6 + 24rz^5 \\ &+24r^3z^5 + 30r^2z^2 + 36rz^2 \\ &+36r^3z^2 + 42r^2z^5 - 28\sqrt{3}r^2z^3 \\ &-20\sqrt{3}rz^3 - 20\sqrt{3}r^3z^3 \\ &-16\sqrt{3}r^2z^5 - 12\sqrt{3}rz^5 - 12\sqrt{3}r^2z \\ &-12\sqrt{3}r^3z^5 - 8\sqrt{3}rz - 8\sqrt{3}r^3z \\ &-6\sqrt{3}r^4z - 6\sqrt{3}r^4z^3 \\ &+3\sqrt{3}r^4z^4 + 4\sqrt{3}r^4z^2 \\ &+12\sqrt{3}r^2z^6 + 20\sqrt{3}rz^2 + 20\sqrt{3}rz^4 \\ &+20\sqrt{3}r^2z^2 + 20\sqrt{3}r^3z^2 \\ &+20\sqrt{3}r^3z^4 + 30\sqrt{3}r^2z^4 \end{aligned} \right] \quad (521)$$

$$u_{-2}^{(+),0101} = \frac{2r}{q_+^4} \left[\begin{aligned} &1 + r^4 - \sqrt{3} - z^3 - 6z^4 - 5z^2 - 3z^5 \\ &+2z + 2r^2 + \sqrt{3}z^3 - \sqrt{3}r^4 - r^4z^3 \\ &-32r^2z^4 - 18r^2z^2 - 12r^2z^6 \\ &-10r^2z^5 - 6r^2z^3 - 6r^4z^4 \\ &-5r^4z^2 - 3\sqrt{3}z^2 - 3\sqrt{3}z^5 \\ &-3r^4z^5 - 2rz^2 - 2\sqrt{3}r^2 \\ &-2r^3z^2 + 2rz + 2\sqrt{3}r + 2r^3z \\ &+2r^4z + 2\sqrt{3}r^3 + 4rz^4 + 4\sqrt{3}z \\ &+4r^2z + 4r^3z^4 + 6rz^6 + 6\sqrt{3}z^4 \\ &+6r^3z^6 + 18rz^5 + 18r^3z^5 \\ &+20rz^3 + 20r^3z^3 + \sqrt{3}rz^5 \\ &+\sqrt{3}r^3z^5 + \sqrt{3}r^4z^3 - 12\sqrt{3}rz^2 \\ &-12\sqrt{3}r^2z^5 - 12\sqrt{3}r^2z^6 \\ &-12\sqrt{3}r^3z^2 - 10\sqrt{3}r^2z - 7\sqrt{3}rz \\ &-7\sqrt{3}r^3z - 3\sqrt{3}r^4z^2 \\ &-3\sqrt{3}r^4z^5 - 2\sqrt{3}r^2z^2 \\ &+4\sqrt{3}r^4z + 4\sqrt{3}r^2z^4 + 6\sqrt{3}rz^6 \\ &+6\sqrt{3}r^3z^6 + 6\sqrt{3}r^4z^4 \\ &+8\sqrt{3}rz^4 + 8\sqrt{3}r^3z^4 + 10\sqrt{3}rz^3 \\ &+10\sqrt{3}r^2z^3 + 10\sqrt{3}r^3z^3 \end{aligned} \right] \quad (522)$$

$$u_{-1}^{(+),0101} = -\frac{2ir^2z(r-1)(z+1)(z^2+1)(\sqrt{3}z+3z-1+\sqrt{3})}{q_+^2} \quad (523)$$

Coefficients $u_p^{(+),0111}$.

$$u_{-4}^{(+),0111} = \frac{2\sqrt{3}(z-1)(z+1)(z^2+1)(-15r^2+6rz-7)}{q_+^2} \quad (524)$$

$$u_{-3}^{(+),0111} = \frac{-2iz^2 + 2i}{q_+^4} \left[\begin{aligned} &3 + \sqrt{3} + 3z^2 + 15r^4 + 18r^2 - 36r^3z \\ &-36r^3z^3 - 34\sqrt{3}r^3 - 14\sqrt{3}r - 12rz \\ &-12rz^3 - 6\sqrt{3}r^5 + 2\sqrt{3}r^2 + 9\sqrt{3}z^2 \\ &+12r^2z^4 + 15\sqrt{3}r^4 + 15r^4z^2 \\ &+30r^2z^2 - 72\sqrt{3}r^3z^3 - 40\sqrt{3}rz^3 \\ &-28\sqrt{3}r^3z^2 - 18\sqrt{3}r^4z^3 - 8\sqrt{3}rz \\ &-8\sqrt{3}r^3z - 6\sqrt{3}rz^2 + 2\sqrt{3}r^2z^3 \\ &+6\sqrt{3}r^3z^4 + 18\sqrt{3}r^5z^2 \\ &+30\sqrt{3}r^2z^4 + 32\sqrt{3}r^2z^2 \\ &+34\sqrt{3}r^2z + 39\sqrt{3}r^4z^2 + 46\sqrt{3}r^4z \end{aligned} \right] \quad (525)$$

$$u_{-2}^{(+),0111} = \frac{2z^2 - 2}{q_+^4} \left[\begin{aligned} &-33r^3 - 6r - 6r^2 - 6r^5 + 12z^2 \\ &+15r^4 - 72r^3z^3 - 39rz^3 - 18r^4z^3 \\ &-15\sqrt{3}r^3 - 15r^2z^3 - 15r^3z^2 \\ &-12r^3z - 6\sqrt{3}r^5 - 3rz - 2\sqrt{3}r \\ &+6rz^2 + 6r^3z^4 + 7\sqrt{3}r^2 \\ &+18r^5z^2 + 21r^2z + 30r^2z^4 \\ &+36r^2z^2 + 39r^4z^2 + 42r^4z - \sqrt{3}rz \\ &-\sqrt{3}rz^3 - 22\sqrt{3}r^3z^3 - 18\sqrt{3}r^3z \\ &-10\sqrt{3}r^5z - 9\sqrt{3}r^3z^2 \\ &-6\sqrt{3}r^3z^4 - 6\sqrt{3}r^5z^2 \\ &-6\sqrt{3}r^5z^3 + 2\sqrt{3}rz^2 \\ &+2\sqrt{3}r^2z^4 + 6\sqrt{3}r^2z^3 \\ &+10\sqrt{3}r^2z + 11\sqrt{3}r^4z + 12\sqrt{3}r^4z^4 \\ &+13\sqrt{3}r^2z^2 + 15\sqrt{3}r^4z^3 \\ &+24\sqrt{3}r^4z^2 \end{aligned} \right] \quad (526)$$

$$u_{-1}^{(+),0111} = -\frac{2ir^2(z-1)(z+1)^2(z^2+1)(6r-3-\sqrt{3})}{q_+^2} \quad (527)$$

Coefficients $u_p^{(+),0122}$.

$$u_{-3}^{(+),0122} = \frac{16\sqrt{3}i(r+1)(z-1)(z+1)(rz+1)(3rz-1)}{q_+^2} \quad (528)$$

$$u_{-2}^{(+),0122} = -\frac{8(r+1)(z-1)(z+1)(rz+1)(6rz-\sqrt{3}r-3)}{q_+^2} \quad (529)$$

Coefficients $u_p^{(+),0202}$.

$$u_{-4}^{(+),0202} = -\frac{4z(-18r^4z + 15r^3z^2 - 5r^3 + 18r^2z^3 - 2r^2z + 15rz^2 - 5r - 18z)}{q_+^2 r} \quad (530)$$

$$u_{-3}^{(+),0202} = \frac{-4irz + 4iz}{q_+^4} \left[\begin{aligned} &5 + \sqrt{3} - 21z^2 - 18z + 5r^4 + 6r + 6r^3 \\ &+10r^2 + \sqrt{3}r^4 - 21r^4z^2 - 18r^4z \\ &-8rz - 8r^3z - 6\sqrt{3}z - 6r^2z \\ &-4r^2z^2 + 2\sqrt{3}r^2 + 3\sqrt{3}z^2 + 6rz^2 \\ &+6r^3z^2 + 12rz^3 + 12r^3z^3 \\ &+18r^2z^3 + 18r^2z^4 - 16\sqrt{3}r^2z^3 \\ &-12\sqrt{3}rz^3 - 12\sqrt{3}r^2z - 12\sqrt{3}r^3z^3 \\ &-8\sqrt{3}rz - 8\sqrt{3}r^3z - 6\sqrt{3}r^4z \\ &+3\sqrt{3}r^4z^2 + 12\sqrt{3}r^2z^4 \\ &+18\sqrt{3}r^2z^2 + 20\sqrt{3}rz^2 \\ &+20\sqrt{3}r^3z^2 \end{aligned} \right] \quad (531)$$

$$u_{-2}^{(+),0202} = -\frac{4rz}{q_+^4} \left[\begin{aligned} &1 + r^4 - 6z^2 - 3\sqrt{3} - 3z^3 + 2z + 2r^2 \\ &-20r^2z^2 - 12r^2z^4 - 10r^2z^3 \\ &-6\sqrt{3}r^2 - 6r^4z^2 - 3\sqrt{3}r^4 \\ &-3\sqrt{3}z^3 - 3r^4z^3 - 2\sqrt{3}r - 2rz^2 \\ &-2\sqrt{3}r^3 - 2r^3z^2 + 2rz + 2r^3z \\ &+2r^4z + 4\sqrt{3}z + 4r^2z + 4\sqrt{3}z^2 \\ &+6rz^4 + 6r^3z^4 + 18rz^3 + 18r^3z^3 \\ &-12\sqrt{3}r^2z^4 - 4\sqrt{3}r^2z^3 - 3\sqrt{3}rz \\ &-3\sqrt{3}r^3z - 3\sqrt{3}r^4z^3 - 2\sqrt{3}rz^2 \\ &-2\sqrt{3}r^2z - 2\sqrt{3}r^3z^2 + 4\sqrt{3}r^4z \\ &+4\sqrt{3}r^4z^2 + 5\sqrt{3}rz^3 \\ &+5\sqrt{3}r^3z^3 + 6\sqrt{3}rz^4 \\ &+6\sqrt{3}r^3z^4 + 12\sqrt{3}r^2z^2 \end{aligned} \right] \quad (532)$$

$$u_{-1}^{(+),0202} = \frac{4ir^2z^2(r-1)(z+1)(\sqrt{3}z+3z-1+\sqrt{3})}{q_+^2} \quad (533)$$

Coefficients $u_p^{(+),0212}$.

$$u_{-4}^{(+),0212} = -\frac{4\sqrt{3}z(z-1)(z+1)(-15r^2+6rz-7)}{q_+^2} \quad (534)$$

$$u_{-3}^{(+),0212} = \frac{-4iz^2+4i}{q_+^4} \left[\begin{aligned} &-3z - 18r^2z - 15r^4z - 12r^2z^3 \\ &-5\sqrt{3}z + 4\sqrt{3}r + 4\sqrt{3}r^2 + 4\sqrt{3}r^3 \\ &+4\sqrt{3}r^4 + 12rz^2 + 36r^3z^2 \\ &-27\sqrt{3}r^4z - 26\sqrt{3}r^2z - 22\sqrt{3}r^2z^2 \\ &-18\sqrt{3}r^4z^2 - 6\sqrt{3}r^5z \\ &-6\sqrt{3}r^2z^3 + 10\sqrt{3}rz + 10\sqrt{3}r^3z \\ &+18\sqrt{3}r^3z^3 + 20\sqrt{3}rz^2 \\ &+36\sqrt{3}r^3z^2 \end{aligned} \right] \quad (535)$$

$$u_{-2}^{(+),0212} = \frac{-4z^2+4}{q_+^4} \left[\begin{aligned} &-6r - 6r^2 - 6r^3 - 6r^4 + 6z \\ &-36r^3z^2 - 18r^3z^3 - 15rz^2 \\ &-3r^3z - 2\sqrt{3}r^2 - 2\sqrt{3}r^3 - 2\sqrt{3}r^4 \\ &-2\sqrt{3}r^5 + 6r^5z + 6r^2z^3 \\ &+9r^2z^2 + 18r^4z^2 + 24r^2z \\ &+27r^4z - \sqrt{3}rz^2 - 18\sqrt{3}r^3z^2 \\ &-9\sqrt{3}r^3z - 6\sqrt{3}r^5z - 6\sqrt{3}r^3z^3 \\ &-6\sqrt{3}r^5z^2 + 2\sqrt{3}r^2z^3 \\ &+6\sqrt{3}r^4z + 9\sqrt{3}r^2z + 10\sqrt{3}r^2z^2 \\ &+12\sqrt{3}r^4z^3 + 15\sqrt{3}r^4z^2 \end{aligned} \right] \quad (536)$$

$$u_{-1}^{(+),0212} = \frac{4ir^2z(z-1)(z+1)^2(6r-3-\sqrt{3})}{q_+^2} \quad (537)$$

Coefficients $u_p^{(+),1111}$.

$$u_{-4}^{(+),1111} = -30(z^2+1)^2 \quad (538)$$

$$u_{-3}^{(+),1111} = -\frac{12i(r-1)(z^2+1)^2(-r^2+6rz+4r-1)}{q_+^2} \quad (539)$$

$$u_{-2}^{(+),1111} = \frac{-6z-6}{q_+^4} \begin{bmatrix} -4z^3 - 3r^2 - 3r^4 + 4r + 4r^5 + 4z^2 \\ +22r^3 - 50r^2z^3 - 50r^4z^3 \\ -26r^2z^2 - 26r^4z^2 - 19r^2z^5 \\ -19r^4z^5 - 16r^3z^3 - 16r^5z^3 \\ -4r^6z^3 - 3r^2z - 3r^4z + 4r^6z^2 \\ +6r^3z^5 + 8rz + 8rz^2 + 8r^5z \\ +8r^3z^6 + 8r^5z^2 + 12rz^4 \\ +12r^3z^3 + 12r^5z^4 + 13r^2z^4 \\ +13r^4z^4 + 20r^3z^2 + 22r^3z \\ +54r^3z^4 \end{bmatrix} \quad (540)$$

$$u_{-1}^{(+),1111} = -\frac{12ir^2(r-1)(z+1)^2(z^2+1)^2}{q_+^2} \quad (541)$$

Coefficients $u_p^{(+),1122}$.

$$u_{-2}^{(+),1122} = -\frac{24(r+1)^2(z-1)(z+1)(rz+1)^2}{q_+^2} \quad (542)$$

Coefficients $u_p^{(+),1212}$.

$$u_{-4}^{(+),1212} = 60z(z^2+1) \quad (543)$$

$$u_{-3}^{(+),1212} = \frac{24iz(r-1)(z^2+1)(-r^2+6rz+4r-1)}{q_+^2} \quad (544)$$

$$u_{-2}^{(+),1212} = \frac{12z+12}{q_+^4} \begin{bmatrix} -2z^2 + 2r + 2z + 2r^5 + 4r^2 + 4r^3 \\ +4r^4 - 25r^2z^2 - 25r^4z^2 \\ -15r^2z^4 - 15r^4z^4 - 6rz^2 \\ -6r^5z^2 - 5r^2z - 5r^4z - 3r^2z^3 \\ -3r^4z^3 - 2r^6z^2 + 2r^6z \\ +2r^3z^2 + 6rz + 6rz^3 + 6r^5z \\ +6r^5z^3 + 8r^3z^5 + 14r^3z^4 \\ +18r^3z + 26r^3z^3 \end{bmatrix} \quad (545)$$

$$u_{-1}^{(+),1212} = \frac{24ir^2z(r-1)(z+1)^2(z^2+1)}{q_+^2} \quad (546)$$

Coefficients $u_p^{(+),2222}$.

$$u_{-4}^{(+),2222} = -120z^2 \quad (547)$$

$$u_{-3}^{(+),2222} = -\frac{48iz^2(r-1)(-r^2+6rz+4r-1)}{q_+^2} \quad (548)$$

$$u_{-2}^{(+),2222} = \frac{-24z-24}{q_+^4} \begin{bmatrix} 1 + r^6 - z + 2r + 2r^5 + 3r^2 + 3r^4 \\ +4r^3 - r^6z - 15r^2z^3 - 15r^4z^3 \\ -7r^2z^2 - 7r^4z^2 - 4r^3z - 3r^2z \\ -3r^4z - 2rz - 2r^5z + 4rz^2 \\ +4r^5z^2 + 8r^3z^4 + 14r^3z^2 \\ +14r^3z^3 \end{bmatrix} \quad (549)$$

$$u_{-1}^{(+),2222} = -\frac{48ir^2z^2(r-1)(z+1)^2}{q_+^2} \quad (550)$$

1.6 All 27 partial derivatives and five local vertices

This section applies to a general potential $U \in C^6$ on the regular local branch $\chi < 0$, $\Delta_\Phi \neq 0$, not only to the radiation-like background. The function $\mathcal{W}(\Phi, s)$ is obtained by algebraically eliminating L from the local density (162) of the main article [1]. The equation for L is $sL + L^5 U'(L) + 4\Phi = 0$. Here $A_L = s + d(L^5 U')/dL = L^3 \Delta_\Phi$, $f_m = d^m(L^5 U')/dL^m$ for $m = 2, 3, 4, 5$, as in Eq. (417) of the main article [1].

We write $\mathcal{W}_{ab} = \partial_\Phi^a \partial_s^b \mathcal{W}$, where a, b are nonnegative integer indices, $1 \leq a + b \leq 6$. When differentiating with respect to one argument, the other is held fixed. After differentiation, all coefficients are evaluated on the background. In local inertial coordinates, set $\phi = \delta\Phi$, $s = \dot{\Phi} > 0$, $u = \dot{\phi}$, and $G_\phi = \delta^{ij} \partial_i \phi \partial_j \phi$. A dot denotes differentiation with respect to local time. The notation u and G_ϕ applies only to this section. The local densities $\mathcal{L}_n$ with $n = 2, \dots, 6$ contain all terms of the corresponding order in the perturbations and are defined by Eqs. (416) and (423) of the main article [1].

In the check case $U = 0$, we have $f_m = 0$, $A_L = s$, and $\mathcal{W} = -s^4/(768\Phi^3)$. In this local expansion, the fifth- and sixth-order gravitational constraints are not eliminated.

$$\mathcal{W}_{10} = \frac{1}{L^4} \quad (551)$$

$$\mathcal{W}_{01} = \frac{1}{3L^3} \quad (552)$$

$$\mathcal{W}_{20} = \frac{16}{A_L L^5} \quad (553)$$

$$\mathcal{W}_{11} = \frac{4}{A_L L^4} \quad (554)$$

$$\mathcal{W}_{02} = \frac{1}{A_L L^3} \quad (555)$$

$$\mathcal{W}_{30} = \frac{320A_L + 64Lf_2}{A_L^3 L^6} \quad (556)$$

$$\mathcal{W}_{21} = \frac{64A_L + 16Lf_2}{A_L^3 L^5} \quad (557)$$

$$\mathcal{W}_{12} = \frac{12A_L + 4Lf_2}{A_L^3 L^4} \quad (558)$$

$$\mathcal{W}_{03} = \frac{2A_L + Lf_2}{A_L^3 L^3} \quad (559)$$

$$\mathcal{W}_{40} = \frac{7680A_L^2 - 256A_L L^2 f_3 + 3840A_L L f_2 + 768L^2 f_2^2}{A_L^5 L^7} \quad (560)$$

$$\mathcal{W}_{31} = \frac{1280A_L^2 - 64A_L L^2 f_3 + 768A_L L f_2 + 192L^2 f_2^2}{A_L^5 L^6} \quad (561)$$

$$\mathcal{W}_{22} = \frac{192A_L^2 - 16A_L L^2 f_3 + 144A_L L f_2 + 48L^2 f_2^2}{A_L^5 L^5} \quad (562)$$

$$\mathcal{W}_{13} = \frac{24A_L^2 - 4A_L L^2 f_3 + 24A_L L f_2 + 12L^2 f_2^2}{A_L^5 L^4} \quad (563)$$

$$\mathcal{W}_{04} = \frac{2A_L^2 - A_L L^2 f_3 + 3A_L L f_2 + 3L^2 f_2^2}{A_L^5 L^3} \quad (564)$$

$$\mathcal{W}_{50} = \frac{1}{A_L^7 L^8} \left(\begin{aligned} &215040A_L^3 + 15360L^3 f_2^3 - 20480A_L^2 L^2 f_3 \\ &+ 1024A_L^2 L^3 f_4 + 76800A_L L^2 f_2^2 + 184320A_L^2 L f_2 \\ &- 10240A_L L^3 f_2 f_3 \end{aligned} \right) \quad (565)$$

$$\mathcal{W}_{41} = \frac{1}{A_L^7 L^7} \begin{pmatrix} 30720A_L^3 + 3840L^3 f_2^3 - 4096A_L^2 L^2 f_3 \\ +256A_L^2 L^3 f_4 + 15360A_L L^2 f_2^2 + 30720A_L^2 L f_2 \\ -2560A_L L^3 f_2 f_3 \end{pmatrix} \quad (566)$$

$$\mathcal{W}_{32} = \frac{1}{A_L^7 L^6} \begin{pmatrix} 3840A_L^3 + 960L^3 f_2^3 - 768A_L^2 L^2 f_3 \\ +64A_L^2 L^3 f_4 + 2880A_L L^2 f_2^2 + 4608A_L^2 L f_2 \\ -640A_L L^3 f_2 f_3 \end{pmatrix} \quad (567)$$

$$\mathcal{W}_{23} = \frac{1}{A_L^7 L^5} \begin{pmatrix} 384A_L^3 + 240L^3 f_2^3 - 128A_L^2 L^2 f_3 \\ +16A_L^2 L^3 f_4 + 480A_L L^2 f_2^2 + 576A_L^2 L f_2 \\ -160A_L L^3 f_2 f_3 \end{pmatrix} \quad (568)$$

$$\mathcal{W}_{14} = \frac{1}{A_L^7 L^4} \begin{pmatrix} 24A_L^3 + 60L^3 f_2^3 - 16A_L^2 L^2 f_3 \\ +4A_L^2 L^3 f_4 + 48A_L^2 L f_2 + 60A_L L^2 f_2^2 \\ -40A_L L^3 f_2 f_3 \end{pmatrix} \quad (569)$$

$$\mathcal{W}_{05} = \frac{A_L^2 f_4 - 10A_L f_2 f_3 + 15f_2^3}{A_L^7} \quad (570)$$

$$\mathcal{W}_{60} = \frac{1}{A_L^9 L^9} \begin{pmatrix} 6881280A_L^4 + 430080L^4 f_2^4 - 1228800A_L^3 L^2 f_3 \\ -4096A_L^3 L^4 f_5 + 40960A_L^2 L^4 f_3^2 + 102400A_L^3 L^3 f_4 \\ +2150400A_L L^3 f_2^3 + 5529600A_L^2 L^2 f_2^2 + 8601600A_L^3 L f_2 \\ -1228800A_L^2 L^3 f_2 f_3 - 430080A_L L^4 f_2^2 f_3 + 61440A_L^2 L^4 f_2 f_4 \end{pmatrix} \quad (571)$$

$$\mathcal{W}_{51} = \frac{1}{A_L^9 L^8} \begin{pmatrix} 860160A_L^4 + 107520L^4 f_2^4 - 204800A_L^3 L^2 f_3 \\ -1024A_L^3 L^4 f_5 + 10240A_L^2 L^4 f_3^2 + 20480A_L^3 L^3 f_4 \\ +430080A_L L^3 f_2^3 + 921600A_L^2 L^2 f_2^2 + 1228800A_L^3 L f_2 \\ -245760A_L^2 L^3 f_2 f_3 - 107520A_L L^4 f_2^2 f_3 + 15360A_L^2 L^4 f_2 f_4 \end{pmatrix} \quad (572)$$

$$\mathcal{W}_{42} = \frac{1}{A_L^9 L^7} \begin{pmatrix} 92160A_L^4 + 26880L^4 f_2^4 - 30720A_L^3 L^2 f_3 \\ -256A_L^3 L^4 f_5 + 2560A_L^2 L^4 f_3^2 + 3840A_L^3 L^3 f_4 \\ +80640A_L L^3 f_2^3 + 138240A_L^2 L^2 f_2^2 + 153600A_L^3 L f_2 \\ -46080A_L^2 L^3 f_2 f_3 - 26880A_L L^4 f_2^2 f_3 + 3840A_L^2 L^4 f_2 f_4 \end{pmatrix} \quad (573)$$

$$\mathcal{W}_{33} = \frac{1}{A_L^9 L^6} \begin{pmatrix} 7680A_L^4 + 6720L^4 f_2^4 - 3840A_L^3 L^2 f_3 \\ -64A_L^3 L^4 f_5 + 640A_L^3 L^3 f_4 + 640A_L^2 L^4 f_3^2 \\ +13440A_L L^3 f_2^3 + 15360A_L^3 L f_2 + 17280A_L^2 L^2 f_2^2 \\ -7680A_L^2 L^3 f_2 f_3 - 6720A_L L^4 f_2^2 f_3 + 960A_L^2 L^4 f_2 f_4 \end{pmatrix} \quad (574)$$

$$\mathcal{W}_{24} = \frac{1}{A_L^9 L^5} \begin{pmatrix} 384A_L^4 + 1680L^4 f_2^4 - 320A_L^3 L^2 f_3 \\ -16A_L^3 L^4 f_5 + 80A_L^3 L^3 f_4 + 160A_L^2 L^4 f_3^2 \\ +960A_L^3 L f_2 + 1440A_L^2 L^2 f_2^2 + 1680A_L L^3 f_2^3 \\ -1680A_L L^4 f_2^2 f_3 - 960A_L^2 L^3 f_2 f_3 + 240A_L^2 L^4 f_2 f_4 \end{pmatrix} \quad (575)$$

$$\mathcal{W}_{15} = \frac{-4A_L^3 f_5 + 60A_L^2 f_2 f_4 + 40A_L^2 f_3^2 - 420A_L f_2^2 f_3 + 420f_2^4}{A_L^9} \quad (576)$$

$$\mathcal{W}_{06} = \frac{1}{A_L^9} \begin{pmatrix} -105A_L f_2^3 - 5A_L^3 f_4 + 105L f_2^4 \\ -A_L^3 L f_5 + 10A_L^2 L f_3^2 + 60A_L^2 L f_2 f_3 \\ -105A_L L f_2^2 f_3 + 15A_L^2 L f_2 f_4 \end{pmatrix} \quad (577)$$

$$\mathcal{L}_2 = \begin{pmatrix} \frac{\mathcal{W}_{02}u^2}{2} + \frac{\mathcal{W}_{20}\phi^2}{2} + \mathcal{W}_{11}\phi u \\ -\frac{G_\phi \mathcal{W}_{01}}{2s} \end{pmatrix} \quad (578)$$

$$\mathcal{L}_3 = \begin{pmatrix} \frac{\mathcal{W}_{03}u^3}{6} + \frac{\mathcal{W}_{30}\phi^3}{6} + \frac{\mathcal{W}_{12}\phi u^2}{2} \\ + \frac{\mathcal{W}_{21}\phi^2 u}{2} + \frac{G_\phi \mathcal{W}_{01}u}{2s^2} - \frac{G_\phi \mathcal{W}_{02}u}{2s} \\ - \frac{G_\phi \mathcal{W}_{11}\phi}{2s} \end{pmatrix} \quad (579)$$

$$\mathcal{L}_4 = \begin{pmatrix} \frac{\mathcal{W}_{04}u^4}{24} + \frac{\mathcal{W}_{40}\phi^4}{24} - \frac{G_\phi^2 \mathcal{W}_{01}}{8s^3} \\ + \frac{\mathcal{W}_{22}\phi^2 u^2}{4} + \frac{\mathcal{W}_{13}\phi u^3}{6} + \frac{\mathcal{W}_{31}\phi^3 u}{6} \\ + \frac{G_\phi^2 \mathcal{W}_{02}}{8s^2} + \frac{G_\phi \mathcal{W}_{02}u^2}{2s^2} - \frac{G_\phi \mathcal{W}_{01}u^2}{2s^3} \\ - \frac{G_\phi \mathcal{W}_{03}u^2}{4s} - \frac{G_\phi \mathcal{W}_{21}\phi^2}{4s} + \frac{G_\phi \mathcal{W}_{11}\phi u}{2s^2} \\ - \frac{G_\phi \mathcal{W}_{12}\phi u}{2s} \end{pmatrix} \quad (580)$$

$$\mathcal{L}_5 = \begin{pmatrix} \frac{\mathcal{W}_{05}u^5}{120} + \frac{\mathcal{W}_{50}\phi^5}{120} + \frac{\mathcal{W}_{23}\phi^2 u^3}{12} \\ + \frac{\mathcal{W}_{32}\phi^3 u^2}{12} + \frac{\mathcal{W}_{14}\phi u^4}{24} + \frac{\mathcal{W}_{41}\phi^4 u}{24} \\ + \frac{G_\phi \mathcal{W}_{01}u^3}{2s^4} - \frac{3G_\phi^2 \mathcal{W}_{02}u}{8s^3} - \frac{G_\phi \mathcal{W}_{02}u^3}{2s^3} \\ - \frac{G_\phi^2 \mathcal{W}_{11}\phi}{8s^3} - \frac{G_\phi \mathcal{W}_{04}u^3}{12s} - \frac{G_\phi \mathcal{W}_{31}\phi^3}{12s} \\ + \frac{G_\phi \mathcal{W}_{03}u^3}{4s^2} + \frac{G_\phi^2 \mathcal{W}_{03}u}{8s^2} + \frac{G_\phi^2 \mathcal{W}_{12}\phi}{8s^2} \\ + \frac{3G_\phi^2 \mathcal{W}_{01}u}{8s^4} + \frac{G_\phi \mathcal{W}_{12}\phi u^2}{2s^2} - \frac{G_\phi \mathcal{W}_{11}\phi u^2}{2s^3} \\ - \frac{G_\phi \mathcal{W}_{13}\phi u^2}{4s} - \frac{G_\phi \mathcal{W}_{22}\phi^2 u}{4s} + \frac{G_\phi \mathcal{W}_{21}\phi^2 u}{4s^2} \end{pmatrix} \quad (581)$$

$$\mathcal{L}_6 = \left(\begin{aligned} & \frac{\mathcal{W}_{06}u^6}{720} + \frac{\mathcal{W}_{60}\phi^6}{720} - \frac{G_\phi^3\mathcal{W}_{01}}{16s^5} \\ & - \frac{G_\phi^3\mathcal{W}_{03}}{48s^3} + \frac{G_\phi^3\mathcal{W}_{02}}{16s^4} + \frac{\mathcal{W}_{33}\phi^3u^3}{36} \\ & + \frac{\mathcal{W}_{24}\phi^2u^4}{48} + \frac{\mathcal{W}_{42}\phi^4u^2}{48} + \frac{\mathcal{W}_{15}\phi u^5}{120} \\ & + \frac{\mathcal{W}_{51}\phi^5u}{120} + \frac{G_\phi\mathcal{W}_{02}u^4}{2s^4} - \frac{5G_\phi^2\mathcal{W}_{03}u^2}{16s^3} \\ & - \frac{3G_\phi^2\mathcal{W}_{01}u^2}{4s^5} - \frac{G_\phi\mathcal{W}_{01}u^4}{2s^5} - \frac{G_\phi\mathcal{W}_{03}u^4}{4s^3} \\ & - \frac{G_\phi^2\mathcal{W}_{21}\phi^2}{16s^3} - \frac{G_\phi\mathcal{W}_{05}u^4}{48s} - \frac{G_\phi\mathcal{W}_{41}\phi^4}{48s} \\ & + \frac{G_\phi\mathcal{W}_{04}u^4}{12s^2} + \frac{G_\phi^2\mathcal{W}_{04}u^2}{16s^2} + \frac{G_\phi^2\mathcal{W}_{22}\phi^2}{16s^2} \\ & + \frac{3G_\phi^2\mathcal{W}_{02}u^2}{4s^4} + \frac{G_\phi\mathcal{W}_{11}\phi u^3}{2s^4} - \frac{3G_\phi^2\mathcal{W}_{12}\phi u}{8s^3} \\ & - \frac{G_\phi\mathcal{W}_{12}\phi u^3}{2s^3} - \frac{G_\phi\mathcal{W}_{21}\phi^2u^2}{4s^3} - \frac{G_\phi\mathcal{W}_{23}\phi^2u^2}{8s} \\ & - \frac{G_\phi\mathcal{W}_{14}\phi u^3}{12s} - \frac{G_\phi\mathcal{W}_{32}\phi^3u}{12s} + \frac{G_\phi\mathcal{W}_{13}\phi u^3}{4s^2} \\ & + \frac{G_\phi\mathcal{W}_{22}\phi^2u^2}{4s^2} + \frac{G_\phi^2\mathcal{W}_{13}\phi u}{8s^2} + \frac{G_\phi\mathcal{W}_{31}\phi^3u}{12s^2} \\ & + \frac{3G_\phi^2\mathcal{W}_{11}\phi u}{8s^4} \end{aligned} \right) \quad (582)$$

1.7 Explicit algebraic functions

The functions $R_s(r, z)$ given below determine the coefficients on the radiation-like background in Subsections 1.2–1.5. Their arguments satisfy the conditions $r > 0$ and $-1 < z < 1$ of Subsection 1.1. The joint asymptotics in the soft momentum-transfer limit are defined for the combined transition kernel, not for an individual function R_s .

For the chosen crossed partition, set $q = q_+$ at $\varsigma = +1$ and $q = q_-$ at $\varsigma = -1$, where the momentum transfers are defined in (497). The limit is specified by the conditions $q \rightarrow 0^+$, $r = 1 + \mu q$, and $z = \varsigma(1 + r^2 - q^2)/(2r)$. The parameter $\mu \in [-1, 1]$, the scale $k > 0$, the chosen external fields, and the finite time interval $0 < \eta_i < \eta_f$ are held fixed. Hence $x_i = k\eta_i$ and $x_f = k\eta_f$ are also fixed. In this limit, $k' = kr \rightarrow k$ and $z \rightarrow \varsigma$. The values $|\mu| = 1$ are understood as boundary values approached from $-1 < z < 1$. For the momentum transfer q_+ , this is the setup preceding Eq. (195) of the main article [1], where the catalogue partition 02|13 is denoted by 13|24.

Before taking the limit, the contact contribution with coefficients $b_p, u_p^{(\pm)}, v_p^{(\pm)}$ and the scalar and tensor exchange contributions for both time orderings are calculated using Eq. (192) of the main article [1]. They are added to all five boundary corrections in Eq. (199) of the main article [1]. The explicit corrections for each time-ordered partition are given in Eqs. (385)–(387) of the main article [1]. The reverse time ordering is added separately. The resulting expression is $\tilde{\mathcal{F}}_{\neq 0}$ in Eq. (395) of the main article [1]. In the boundary transformation, the upper boundary Q_* of the dimensionful soft band is held fixed with $2Q_* < k, k'$. Its lower boundary ϵ serves as a soft regulator: the contributions are first summed at $\epsilon > 0$, and the regulator is then removed in their sum. The contact terms $u_p^{(0)}, v_p^{(0)}$ and the exactly homogeneous exchange are retained separately with the normalization (403) of the main article and are not determined by the limit $q \rightarrow 0$ of nonzero momentum transfer.

$$R_1(r, z) = \frac{N_1(r, z)}{D_1(r, z)}. \quad (583)$$

$$\begin{aligned}
& -368 - 736r^2 - 368r^4 - 288z^2 + 72z^4 \\
& -288r^4z^2 - 48r^2z^4 + 72r^4z^4 \\
N_1(r, z) = & +1952r^2z^2
\end{aligned} \tag{584}$$

$$D_1(r, z) = (-r^2 + 2rz - 1) (r^2 + 2rz + 1) \tag{585}$$

$$R_2(r, z) = \frac{N_2(r, z)}{D_2(r, z)}. \tag{586}$$

$$\begin{aligned}
& 104\sqrt{3}i - 648\sqrt{3}ir^5 - 440\sqrt{3}ir^3 \\
& -424\sqrt{3}ir^7 - 112\sqrt{3}ir - 104\sqrt{3}ir^9 \\
& -24\sqrt{3}iz^4 + 32\sqrt{3}iz^2 + 112\sqrt{3}ir^8 \\
& +424\sqrt{3}ir^2 + 440\sqrt{3}ir^6 + 648\sqrt{3}ir^4 \\
& -2312\sqrt{3}ir^5z^4 - 1840\sqrt{3}ir^4z^2 \\
& -960\sqrt{3}ir^2z^2 - 704\sqrt{3}ir^6z^2 \\
& -664\sqrt{3}ir^6z^4 - 192\sqrt{3}ir^3z^6 \\
& -160\sqrt{3}ir^4z^6 - 144\sqrt{3}irz^2 \\
& -104\sqrt{3}ir^2z^4 - 96\sqrt{3}ir^7z^6 \\
& -48\sqrt{3}ir^8z^4 - 32\sqrt{3}ir^9z^2 \\
& +24\sqrt{3}ir^9z^4 + 48\sqrt{3}irz^4 \\
& +96\sqrt{3}ir^2z^6 + 104\sqrt{3}ir^7z^4 \\
& +144\sqrt{3}ir^8z^2 + 160\sqrt{3}ir^5z^6 \\
& +192\sqrt{3}ir^6z^6 + 664\sqrt{3}ir^3z^4 \\
& +704\sqrt{3}ir^3z^2 + 960\sqrt{3}ir^7z^2 \\
N_2(r, z) = & +1840\sqrt{3}ir^5z^2 + 2312\sqrt{3}ir^4z^4
\end{aligned} \tag{587}$$

$$D_2(r, z) = (-r^2 + 2rz - 1)^2 (r^2 + 2rz + 1)^2 \tag{588}$$

$$R_3(r, z) = \frac{N_3(r, z)}{D_3(r, z)}. \tag{589}$$

$$\begin{aligned}
& -80 - 836r^4 - 836r^6 - 412r^2 - 412r^8 \\
& -80r^{10} + 256r + 256r^9 + 1048r^3 + 1048r^7 \\
& +1584r^5 - 4624r^5z^2 - 2216r^3z^2 \\
& -2216r^7z^2 - 1280r^4z^4 - 1280r^6z^4 \\
& -768r^5z^6 - 576r^3z^4 - 576r^7z^4 \\
& -144rz^4 - 144r^9z^4 - 48r^2z^4 \\
& -48r^8z^4 + 144rz^2 + 144r^9z^2 \\
& +192r^4z^6 + 192r^6z^6 + 576r^3z^6 \\
& +576r^7z^6 + 628r^2z^2 + 628r^8z^2 \\
N_3(r, z) = & +1836r^4z^2 + 1836r^6z^2 + 5632r^5z^4
\end{aligned} \tag{590}$$

$$D_3(r, z) = 3(-r^2 + 2rz - 1)^2 (r^2 + 2rz + 1)^2 \tag{591}$$

$$R_4(r, z) = \frac{N_4(r, z)}{D_4(r, z)}. \tag{592}$$

$$\begin{aligned}
& -128\sqrt{3}ir^3 - 64\sqrt{3}ir - 64\sqrt{3}ir^5 \\
& +64\sqrt{3}ir^2 + 64\sqrt{3}ir^6 + 128\sqrt{3}ir^4 \\
& -328\sqrt{3}ir^4z^2 - 120\sqrt{3}ir^3z^4 \\
& -72\sqrt{3}ir^2z^4 - 24\sqrt{3}ir^5z^2 \\
& +24\sqrt{3}ir^2z^2 + 72\sqrt{3}ir^5z^4 \\
N_4(r, z) = & +120\sqrt{3}ir^4z^4 + 328\sqrt{3}ir^3z^2
\end{aligned} \tag{593}$$

$$D_4(r, z) = 9(-r^2 + 2rz - 1) (r^2 + 2rz + 1) \tag{594}$$

$$R_5(r, z) = -\frac{4r^2(z^2 - 3)}{9} \quad (595)$$

$$R_6(r, z) = -\frac{24(3r^4z^2 + 10r^2z^2 + 26r^2 + 3z^2)}{r^2} \quad (596)$$

$$R_7(r, z) = -\frac{16\sqrt{3}i(r-1)(9r^2z^2 - 6rz^2 - 38r + 9z^2)}{3r} \quad (597)$$

$$R_8(r, z) = 40r^2z^2 + \frac{104r^2}{3} - 80rz^2 - 176r + 40z^2 + \frac{104}{3} \quad (598)$$

$$R_9(r, z) = \frac{16\sqrt{3}ir(r-1)(3z^2 + 5)}{9} \quad (599)$$

$$R_{10}(r, z) = -\frac{4r^2(z^2 + 1)}{3} \quad (600)$$

$$R_{11}(r, z) = \frac{N_{11}(r, z)}{D_{11}(r, z)}. \quad (601)$$

$$\begin{aligned} & 668 - 444z^4 + 312z^2 + 668r^8 + 2672r^2 \\ & + 2672r^6 + 4008r^4 - 9104r^4z^2 \\ & - 4240r^2z^2 - 4240r^6z^2 - 1680r^2z^4 \\ & - 1680r^6z^4 - 444r^8z^4 + 312r^8z^2 \\ & + 1104r^2z^6 + 1104r^6z^6 + 2208r^4z^6 \\ N_{11}(r, z) = & + 6104r^4z^4 \end{aligned} \quad (602)$$

$$D_{11}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (603)$$

$$R_{12}(r, z) = \frac{N_{12}(r, z)}{D_{12}(r, z)}. \quad (604)$$

$$\begin{aligned} & 728\sqrt{3}i - 4368\sqrt{3}ir^5 - 2912\sqrt{3}ir^3 \\ & - 2912\sqrt{3}ir^7 - 728\sqrt{3}ir - 728\sqrt{3}ir^9 \\ & - 72\sqrt{3}iz^4 + 144\sqrt{3}iz^2 + 728\sqrt{3}ir^8 \\ & + 2912\sqrt{3}ir^2 + 2912\sqrt{3}ir^6 + 4368\sqrt{3}ir^4 \\ & - 10064\sqrt{3}ir^4z^2 - 7136\sqrt{3}ir^5z^4 \\ & - 5056\sqrt{3}ir^2z^2 - 4384\sqrt{3}ir^6z^2 \\ & - 2496\sqrt{3}ir^6z^4 - 2208\sqrt{3}ir^3z^6 \\ & - 1920\sqrt{3}ir^5z^6 - 888\sqrt{3}ir^8z^4 \\ & - 864\sqrt{3}ir^2z^4 - 480\sqrt{3}irz^2 \\ & - 288\sqrt{3}ir^7z^6 - 144\sqrt{3}ir^9z^2 \\ & + 72\sqrt{3}ir^9z^4 + 288\sqrt{3}ir^2z^6 \\ & + 480\sqrt{3}ir^8z^2 + 864\sqrt{3}ir^7z^4 \\ & + 888\sqrt{3}irz^4 + 1920\sqrt{3}ir^4z^6 \\ & + 2208\sqrt{3}ir^6z^6 + 2496\sqrt{3}ir^3z^4 \\ & + 4384\sqrt{3}ir^3z^2 + 5056\sqrt{3}ir^7z^2 \\ N_{12}(r, z) = & + 7136\sqrt{3}ir^4z^4 + 10064\sqrt{3}ir^5z^2 \end{aligned} \quad (605)$$

$$D_{12}(r, z) = 3 \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (606)$$

$$R_{13}(r, z) = \frac{N_{13}(r, z)}{D_{13}(r, z)}. \quad (607)$$

$$\begin{aligned}
& -144 - 1440r^4 - 1440r^6 - 720r^2 - 720r^8 \\
& -144r^{10} + 800r + 800r^9 + 3200r^3 + 3200r^7 \\
& +4800r^5 - 12080r^5z^2 - 5896r^3z^2 \\
& -5896r^7z^2 - 1008r^4z^6 - 1008r^6z^6 \\
& -936r^3z^4 - 936r^7z^4 - 324r^4z^4 \\
& -324r^6z^4 - 144rz^4 - 144r^9z^4 \\
& +144rz^2 + 144r^9z^2 + 420r^2z^4 \\
& +420r^8z^4 + 576r^3z^6 + 576r^7z^6 \\
& +804r^2z^2 + 804r^8z^2 + 1632r^5z^6 \\
N_{13}(r, z) = & +2412r^4z^2 + 2412r^6z^2 + 10160r^5z^4
\end{aligned} \tag{608}$$

$$D_{13}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{609}$$

$$R_{14}(r, z) = \frac{N_{14}(r, z)}{D_{14}(r, z)}. \tag{610}$$

$$\begin{aligned}
& -320\sqrt{3}ir^3 - 160\sqrt{3}ir - 160\sqrt{3}ir^5 \\
& +160\sqrt{3}ir^2 + 160\sqrt{3}ir^6 + 320\sqrt{3}ir^4 \\
& -616\sqrt{3}ir^4z^2 - 72\sqrt{3}ir^2z^4 \\
& -24\sqrt{3}ir^5z^2 - 24\sqrt{3}ir^3z^4 \\
& +24\sqrt{3}ir^2z^2 + 24\sqrt{3}ir^4z^4 \\
N_{14}(r, z) = & +72\sqrt{3}ir^5z^4 + 616\sqrt{3}ir^3z^2
\end{aligned} \tag{611}$$

$$D_{14}(r, z) = 9 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \tag{612}$$

$$R_{15}(r, z) = \frac{32r^2}{9} \tag{613}$$

$$R_{16}(r, z) = \frac{N_{16}(r, z)}{D_{16}(r, z)}. \tag{614}$$

$$\begin{aligned}
& 72z^2 + 324r^2 + 324r^{10} + 1296r^4 + 1296r^8 \\
& +1944r^6 - 5136r^6z^2 - 2640r^4z^4 \\
& -2640r^8z^4 - 2136r^4z^2 - 2136r^8z^2 \\
& -204r^2z^4 - 204r^{10}z^4 + 72r^{12}z^2 \\
& +336r^4z^6 + 336r^8z^6 + 504r^2z^2 \\
N_{16}(r, z) = & +504r^{10}z^2 + 1440r^6z^6 + 6648r^6z^4
\end{aligned} \tag{615}$$

$$D_{16}(r, z) = r^2 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{616}$$

$$R_{17}(r, z) = \frac{N_{17}(r, z)}{D_{17}(r, z)}. \tag{617}$$

$$\begin{aligned}
& -408\sqrt{3}ir^6 - 280\sqrt{3}ir^4 - 264\sqrt{3}ir^8 \\
& -72\sqrt{3}ir^2 - 64\sqrt{3}ir^{10} - 48\sqrt{3}iz^2 \\
& +64\sqrt{3}ir + 72\sqrt{3}ir^9 + 264\sqrt{3}ir^3 \\
& +280\sqrt{3}ir^7 + 408\sqrt{3}ir^5 \\
& -1512\sqrt{3}ir^6z^4 - 1240\sqrt{3}ir^7z^4 \\
& -1056\sqrt{3}ir^5z^2 - 528\sqrt{3}ir^3z^2 \\
& -480\sqrt{3}ir^6z^6 - 456\sqrt{3}ir^3z^4 \\
& -256\sqrt{3}ir^2z^2 - 256\sqrt{3}ir^7z^2 \\
& -224\sqrt{3}ir^4z^6 - 136\sqrt{3}ir^9z^4 \\
& -64\sqrt{3}ir^{10}z^2 + 48\sqrt{3}ir^{11}z^2 \\
& +64\sqrt{3}irz^2 + 136\sqrt{3}ir^2z^4 \\
& +224\sqrt{3}ir^7z^6 + 256\sqrt{3}ir^4z^2 \\
& +256\sqrt{3}ir^9z^2 + 456\sqrt{3}ir^8z^4 \\
& +480\sqrt{3}ir^5z^6 + 528\sqrt{3}ir^8z^2 \\
& +1056\sqrt{3}ir^6z^2 + 1240\sqrt{3}ir^4z^4 \\
N_{17}(r, z) = & +1512\sqrt{3}ir^5z^4
\end{aligned} \tag{618}$$

$$D_{17}(r, z) = r \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (619)$$

$$R_{18}(r, z) = \frac{N_{18}(r, z)}{D_{18}(r, z)}. \quad (620)$$

$$\begin{aligned} & -40 - 436r^4 - 436r^6 - 212r^2 - 212r^8 \\ & -120z^2 - 48r^5 - 40r^3 - 40r^7 - 40r^{10} \\ & -16r - 16r^9 - 1560r^3z^4 - 1560r^7z^4 \\ & -720r^4z^6 - 720r^6z^6 - 120r^{10}z^2 \\ & +56r^2z^2 + 56r^8z^2 + 80r^5z^4 \\ & +240rz^2 + 240r^9z^2 + 260r^4z^4 \\ & +260r^6z^4 + 416r^3z^2 + 416r^7z^2 \\ & +448r^5z^2 + 492r^2z^4 + 492r^8z^4 \\ N_{18}(r, z) = & +720r^4z^2 + 720r^6z^2 + 1440r^5z^6 \end{aligned} \quad (621)$$

$$D_{18}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (622)$$

$$R_{19}(r, z) = \frac{N_{19}(r, z)}{D_{19}(r, z)}. \quad (623)$$

$$\begin{aligned} & -32\sqrt{3}ir^4 - 16\sqrt{3}ir^2 - 16\sqrt{3}ir^6 \\ & +16\sqrt{3}ir + 16\sqrt{3}ir^5 + 32\sqrt{3}ir^3 \\ & -96\sqrt{3}ir^4z^4 - 64\sqrt{3}ir^3z^2 \\ & -48\sqrt{3}irz^2 - 48\sqrt{3}ir^5z^2 \\ & +48\sqrt{3}ir^2z^2 + 48\sqrt{3}ir^6z^2 \\ N_{19}(r, z) = & +64\sqrt{3}ir^4z^2 + 96\sqrt{3}ir^3z^4 \end{aligned} \quad (624)$$

$$D_{19}(r, z) = 9 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \quad (625)$$

$$R_{20}(r, z) = \frac{8r^2(z-1)(z+1)}{9} \quad (626)$$

$$R_{21}(r, z) = \frac{N_{21}(r, z)}{D_{21}(r, z)}. \quad (627)$$

$$\begin{aligned} & -48\sqrt{3} - 588\sqrt{3}r^4 - 492\sqrt{3}r^6 - 292\sqrt{3}r^2 \\ & -148\sqrt{3}r^8 + 24\sqrt{3}z^2 + 24\sqrt{3}z^4 \\ & -1420\sqrt{3}r^4z^4 - 956\sqrt{3}r^6z^4 \\ & -244\sqrt{3}r^2z^4 - 60\sqrt{3}r^8z^4 \\ & -48\sqrt{3}r^2z^6 + 192\sqrt{3}r^6z^6 \\ & +208\sqrt{3}r^8z^2 + 400\sqrt{3}r^4z^6 \\ & +584\sqrt{3}r^2z^2 + 1256\sqrt{3}r^6z^2 \\ N_{21}(r, z) = & +1608\sqrt{3}r^4z^2 \end{aligned} \quad (628)$$

$$D_{21}(r, z) = \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (629)$$

$$R_{22}(r, z) = \frac{N_{22}(r, z)}{D_{22}(r, z)}. \quad (630)$$

$$\begin{aligned}
& -16i - 664ir^4 - 636ir^6 - 252ir^2 \\
& -208ir^8 - 8iz^2 + 24ir + 24iz^4 + 60ir^9 \\
& +136ir^3 + 208ir^7 + 260ir^5 \\
& -2072ir^4z^4 - 1580ir^6z^4 - 704ir^5z^2 \\
& -520ir^7z^2 - 296ir^3z^2 - 172ir^2z^4 \\
& -144ir^5z^6 - 144ir^7z^6 - 120ir^8z^4 \\
& -96ir^9z^2 - 48ir^2z^6 - 24irz^4 \\
& +4\sqrt{3}ir^9 + 36ir^9z^4 + 48\sqrt{3}ir \\
& +48ir^3z^6 + 60\sqrt{3}ir^7 + 112ir^3z^4 \\
& +148\sqrt{3}ir^3 + 156\sqrt{3}ir^5 + 328ir^8z^2 \\
& +384ir^6z^6 + 456ir^7z^4 + 472ir^2z^2 \\
& +588ir^5z^4 + 736ir^4z^6 + 1832ir^6z^2 \\
& +2000ir^4z^2 - 408\sqrt{3}ir^5z^2 \\
& -392\sqrt{3}ir^3z^2 - 104\sqrt{3}ir^7z^2 \\
& -96\sqrt{3}ir^3z^6 - 72\sqrt{3}irz^2 \\
& -48\sqrt{3}ir^7z^6 - 16\sqrt{3}ir^9z^2 \\
& -16\sqrt{3}ir^5z^6 + 12\sqrt{3}ir^9z^4 \\
& +24\sqrt{3}irz^4 + 92\sqrt{3}ir^7z^4 \\
& +268\sqrt{3}ir^5z^4 + 340\sqrt{3}ir^3z^4
\end{aligned} \tag{631}$$

$$D_{22}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \tag{632}$$

$$R_{23}(r, z) = \frac{N_{23}(r, z)}{D_{23}(r, z)}. \tag{633}$$

$$\begin{aligned}
& -156r^5 - 144r^3 - 72r^7 - 48r - 12r^9 - 8\sqrt{3} \\
& +72r^2 + 84r^8 + 228r^4 + 240r^6 \\
& -708r^4z^2 - 600r^6z^2 - 552r^3z^4 \\
& -456r^7z^4 - 288r^4z^6 - 276\sqrt{3}r^5 \\
& -260\sqrt{3}r^7 - 240r^5z^4 - 144r^2z^2 \\
& -108\sqrt{3}r^3 - 96r^5z^6 - 84\sqrt{3}r^9 \\
& -84r^8z^2 - 72rz^4 - 72r^9z^4 - 8\sqrt{3}r \\
& +8\sqrt{3}z^2 + 28\sqrt{3}r^2 + 52\sqrt{3}r^8 \\
& +72r^2z^4 + 84r^9z^2 + 120rz^2 \\
& +132\sqrt{3}r^4 + 148\sqrt{3}r^6 + 240r^7z^2 \\
& +288r^3z^6 + 288r^7z^6 + 360r^6z^4 \\
& +408r^3z^2 + 492r^5z^2 + 768r^4z^4 \\
& -872\sqrt{3}r^5z^4 - 804\sqrt{3}r^7z^4 \\
& -460\sqrt{3}r^6z^2 - 408\sqrt{3}r^4z^2 \\
& -176\sqrt{3}r^4z^6 - 96\sqrt{3}r^6z^6 \\
& -88\sqrt{3}r^8z^2 - 72\sqrt{3}r^9z^4 \\
& -68\sqrt{3}r^3z^4 - 48\sqrt{3}r^3z^6 \\
& -28\sqrt{3}r^2z^2 - 16\sqrt{3}rz^2 + 24\sqrt{3}rz^4 \\
& +36\sqrt{3}r^8z^4 + 156\sqrt{3}r^9z^2 \\
& +224\sqrt{3}r^3z^2 + 288\sqrt{3}r^5z^6 \\
& +288\sqrt{3}r^7z^6 + 408\sqrt{3}r^6z^4 \\
& +452\sqrt{3}r^4z^4 + 776\sqrt{3}r^7z^2 \\
& +860\sqrt{3}r^5z^2
\end{aligned} \tag{634}$$

$$D_{23}(r, z) = 3 \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \tag{635}$$

$$R_{24}(r, z) = \frac{N_{24}(r, z)}{D_{24}(r, z)}. \tag{636}$$

$$\begin{aligned}
& -8ir + 8ir^3 + 16ir^5 - 52ir^5z^2 \\
& -24ir^4z^2 - 12ir^3z^2 - 8\sqrt{3}ir^2 \\
& -8\sqrt{3}ir^4 + 4ir^3z^4 + 8irz^2 \\
& +24ir^4z^4 + 36ir^5z^4 - 24\sqrt{3}ir^2z^4 \\
& -20\sqrt{3}ir^3z^4 - 12\sqrt{3}ir^5z^2 \\
& +8\sqrt{3}ir^4z^2 + 12\sqrt{3}ir^5z^4 \\
N_{24}(r, z) = & +20\sqrt{3}ir^3z^2 + 32\sqrt{3}ir^2z^2
\end{aligned} \tag{637}$$

$$D_{24}(r, z) = 3 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \tag{638}$$

$$R_{25}(r, z) = -40\sqrt{3}(z-1)(z+1) \tag{639}$$

$$R_{26}(r, z) = 16\sqrt{3}ir(z-1)(z+1) + 8i(3r-5)(z-1)(z+1) \tag{640}$$

$$R_{27}(r, z) = -16r(z-1)(z+1) - \frac{8\sqrt{3}(3r-1)(z-1)(z+1)}{3} \tag{641}$$

$$R_{28}(r, z) = -\frac{8ir(z-1)(z+1)}{3} \tag{642}$$

$$R_{29}(r, z) = \frac{N_{29}(r, z)}{D_{29}(r, z)}. \tag{643}$$

$$\begin{aligned}
& 4\sqrt{3} - 52\sqrt{3}z^4 + 20\sqrt{3}r^8 + 32\sqrt{3}r^2 \\
& +48\sqrt{3}z^2 + 64\sqrt{3}r^6 + 72\sqrt{3}r^4 \\
& -1320\sqrt{3}r^4z^4 - 992\sqrt{3}r^6z^4 \\
& -324\sqrt{3}r^8z^4 - 320\sqrt{3}r^2z^4 \\
& -16\sqrt{3}r^2z^6 + 256\sqrt{3}r^4z^2 \\
& +304\sqrt{3}r^2z^2 + 304\sqrt{3}r^6z^2 \\
& +304\sqrt{3}r^8z^2 + 624\sqrt{3}r^6z^6 \\
N_{29}(r, z) = & +992\sqrt{3}r^4z^6
\end{aligned} \tag{644}$$

$$D_{29}(r, z) = \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{645}$$

$$R_{30}(r, z) = \frac{N_{30}(r, z)}{D_{30}(r, z)}. \tag{646}$$

$$\begin{aligned}
& -48ir^5 - 40ir^7 - 24iz^2 - 24ir^3 \\
& -12ir^9 - 4ir + 24iz^4 + 32ir^2 + 32ir^8 \\
& +96ir^4 + 96ir^6 - 1952ir^6z^4 \\
& -1744ir^4z^4 - 648ir^8z^4 - 576ir^5z^6 \\
& -288ir^2z^4 - 264ir^3z^2 - 152ir^5z^2 \\
& -144ir^7z^6 - 96ir^2z^6 - 48irz^2 \\
& -24ir^9z^2 - 8\sqrt{3}ir^3 - 4\sqrt{3}ir \\
& +4\sqrt{3}ir^9 + 8\sqrt{3}ir^7 + 16ir^3z^6 \\
& +36ir^9z^4 + 40ir^7z^2 + 52irz^4 \\
& +144ir^7z^4 + 272ir^3z^4 + 352ir^2z^2 \\
& +368ir^4z^2 + 608ir^6z^2 + 616ir^8z^2 \\
& +776ir^5z^4 + 1248ir^6z^6 + 1280ir^4z^6 \\
& -224\sqrt{3}ir^5z^6 - 176\sqrt{3}ir^3z^6 \\
& -160\sqrt{3}ir^7z^2 - 112\sqrt{3}ir^5z^2 \\
& -96\sqrt{3}irz^2 - 64\sqrt{3}ir^3z^2 \\
& -48\sqrt{3}ir^7z^6 - 16\sqrt{3}ir^9z^2 \\
& +12\sqrt{3}ir^9z^4 + 100\sqrt{3}irz^4 \\
& +200\sqrt{3}ir^7z^4 + 248\sqrt{3}ir^3z^4 \\
N_{30}(r, z) = & +336\sqrt{3}ir^5z^4
\end{aligned} \tag{647}$$

$$D_{30}(r, z) = \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{648}$$

$$R_{31}(r, z) = \frac{N_{31}(r, z)}{D_{31}(r, z)}. \quad (649)$$

$$\begin{aligned} & -36r^4 - 36r^5 - 36r^6 - 36r^7 - 12r^2 \\ & -12r^3 - 12r^8 - 12r^9 - 984r^5z^4 \\ & -780r^7z^4 - 528r^4z^6 - 396r^3z^4 \\ & -288r^2z^2 - 180r^4z^2 - 84r^6z^4 \\ & -72rz^4 - 72r^9z^4 - 36\sqrt{3}r^4 - 36\sqrt{3}r^6 \\ & -12\sqrt{3}r^2 - 12\sqrt{3}r^8 + 12\sqrt{3}r^3 \\ & +12\sqrt{3}r^9 + 12r^8z^2 + 36\sqrt{3}r^5 \\ & +36\sqrt{3}r^7 + 72rz^2 + 84r^9z^2 \\ & +120r^3z^2 + 120r^6z^2 + 288r^3z^6 \\ & +288r^7z^6 + 300r^2z^4 + 492r^5z^2 \\ & +528r^7z^2 + 528r^5z^6 + 744r^4z^4 \\ & -1080\sqrt{3}r^5z^4 - 720\sqrt{3}r^6z^6 \\ & -396\sqrt{3}r^7z^4 - 356\sqrt{3}r^6z^2 \\ & -336\sqrt{3}r^8z^2 - 320\sqrt{3}r^4z^6 \\ & -204\sqrt{3}r^3z^4 - 96\sqrt{3}r^3z^6 \\ & -88\sqrt{3}r^4z^2 - 72\sqrt{3}r^9z^4 \\ & -68\sqrt{3}r^2z^2 - 24\sqrt{3}rz^2 + 24\sqrt{3}rz^4 \\ & +60\sqrt{3}r^9z^2 + 72\sqrt{3}r^7z^2 \\ & +80\sqrt{3}r^2z^4 + 288\sqrt{3}r^3z^2 \\ & +288\sqrt{3}r^7z^6 + 324\sqrt{3}r^5z^2 \\ & +348\sqrt{3}r^8z^4 + 444\sqrt{3}r^4z^4 \\ N_{31}(r, z) = & +720\sqrt{3}r^5z^6 + 1112\sqrt{3}r^6z^4 \end{aligned} \quad (650)$$

$$D_{31}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (651)$$

$$R_{32}(r, z) = \frac{N_{32}(r, z)}{D_{32}(r, z)}. \quad (652)$$

$$\begin{aligned} & -36ir^3z^2 - 36ir^5z^2 - 24ir^4z^4 \\ & +24ir^4z^2 + 36ir^3z^4 + 36ir^5z^4 \\ & -24\sqrt{3}ir^2z^4 - 12\sqrt{3}ir^5z^2 \\ & -4\sqrt{3}ir^3z^4 + 4\sqrt{3}ir^3z^2 \\ N_{32}(r, z) = & +12\sqrt{3}ir^5z^4 + 24\sqrt{3}ir^2z^2 \end{aligned} \quad (653)$$

$$D_{32}(r, z) = 3 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \quad (654)$$

$$R_{33}(r, z) = \frac{N_{33}(r, z)}{D_{33}(r, z)}. \quad (655)$$

$$\begin{aligned} & -92\sqrt{3} - 900\sqrt{3}r^4 - 716\sqrt{3}r^6 - 484\sqrt{3}r^2 \\ & -208\sqrt{3}r^8 + 16\sqrt{3}z^2 + 76\sqrt{3}z^4 \\ & -1380\sqrt{3}r^4z^4 - 432\sqrt{3}r^6z^6 \\ & -284\sqrt{3}r^6z^4 - 244\sqrt{3}r^2z^4 \\ & -56\sqrt{3}r^8z^2 - 32\sqrt{3}r^2z^6 \\ & +48\sqrt{3}r^4z^6 + 264\sqrt{3}r^8z^4 \\ & +760\sqrt{3}r^2z^2 + 1432\sqrt{3}r^6z^2 \\ N_{33}(r, z) = & +2232\sqrt{3}r^4z^2 \end{aligned} \quad (656)$$

$$D_{33}(r, z) = \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (657)$$

$$R_{34}(r, z) = \frac{N_{34}(r, z)}{D_{34}(r, z)}. \quad (658)$$

$$\begin{aligned}
& -56i - 1000ir^4 - 892ir^6 - 444ir^2 \\
& -280ir^8 + 52ir + 56iz^2 + 96ir^9 \\
& +256ir^3 + 344ir^7 + 452ir^5 \\
& -1608ir^4z^4 - 1080ir^5z^2 - 864ir^6z^6 \\
& -848ir^7z^2 - 320ir^3z^2 - 248ir^8z^2 \\
& -204ir^2z^4 - 96ir^9z^2 - 76irz^4 \\
& +16\sqrt{3}ir^9 + 24irz^2 + 32ir^3z^4 \\
& +32ir^3z^6 + 48ir^2z^6 + 48ir^5z^6 \\
& +52ir^6z^4 + 68\sqrt{3}ir + 96ir^4z^6 \\
& +116\sqrt{3}ir^7 + 220\sqrt{3}ir^3 + 252\sqrt{3}ir^5 \\
& +504ir^7z^4 + 528ir^8z^4 + 580ir^5z^4 \\
& +600ir^2z^2 + 1704ir^6z^2 + 2512ir^4z^2 \\
& -648\sqrt{3}ir^5z^2 - 520\sqrt{3}ir^3z^2 \\
& -136\sqrt{3}ir^7z^2 - 76\sqrt{3}irz^4 \\
& -48\sqrt{3}ir^5z^6 - 16\sqrt{3}ir^9z^2 \\
& +8\sqrt{3}irz^2 + 20\sqrt{3}ir^7z^4 \\
& +80\sqrt{3}ir^3z^6 + 220\sqrt{3}ir^3z^4 \\
N_{34}(r, z) = & +444\sqrt{3}ir^5z^4
\end{aligned} \tag{659}$$

$$D_{34}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \tag{660}$$

$$R_{35}(r, z) = \frac{N_{35}(r, z)}{D_{35}(r, z)}. \tag{661}$$

$$\begin{aligned}
& -408r^5 - 324r^3 - 228r^7 - 96r - 48r^9 \\
& +84r^2 + 96r^8 + 264r^4 + 276r^6 \\
& -792r^5z^4 - 720r^6z^2 - 540r^3z^4 \\
& -528r^4z^2 - 456\sqrt{3}r^5 - 392\sqrt{3}r^7 \\
& -228r^2z^4 - 216\sqrt{3}r^3 - 120\sqrt{3}r^9 \\
& -96r^8z^2 - 60r^7z^4 - 32\sqrt{3}r \\
& +24r^4z^4 + 48r^9z^2 + 72\sqrt{3}r^2 \\
& +72\sqrt{3}r^8 + 96rz^2 + 144r^2z^2 \\
& +144r^5z^6 + 216\sqrt{3}r^4 + 216\sqrt{3}r^6 \\
& +240r^4z^6 + 288r^7z^2 + 444r^6z^4 \\
& +864r^3z^2 + 1056r^5z^2 - 640\sqrt{3}r^6z^4 \\
& -600\sqrt{3}r^7z^4 - 560\sqrt{3}r^5z^4 \\
& -496\sqrt{3}r^4z^2 - 312\sqrt{3}r^8z^4 \\
& -200\sqrt{3}r^6z^2 - 56\sqrt{3}r^2z^2 \\
& -56\sqrt{3}r^3z^4 - 48\sqrt{3}r^5z^6 \\
& -16\sqrt{3}r^2z^4 + 16\sqrt{3}r^4z^6 + 32\sqrt{3}rz^2 \\
& +48\sqrt{3}r^3z^6 + 120\sqrt{3}r^9z^2 \\
& +224\sqrt{3}r^3z^2 + 240\sqrt{3}r^8z^2 \\
& +264\sqrt{3}r^4z^4 + 624\sqrt{3}r^6z^6 \\
N_{35}(r, z) = & +992\sqrt{3}r^7z^2 + 1064\sqrt{3}r^5z^2
\end{aligned} \tag{662}$$

$$D_{35}(r, z) = 3 \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \tag{663}$$

$$R_{36}(r, z) = \frac{N_{36}(r, z)}{D_{36}(r, z)}. \tag{664}$$

$$\begin{aligned}
& 24ir^3 + 24ir^5 - 48ir^4z^2 - 24ir^3z^2 \\
& -24ir^5z^2 - 8\sqrt{3}ir^2 - 8\sqrt{3}ir^4 \\
& +48ir^4z^4 - 16\sqrt{3}ir^3z^4 \\
& +8\sqrt{3}ir^2z^2 + 8\sqrt{3}ir^4z^2 \\
N_{36}(r, z) = & +16\sqrt{3}ir^3z^2
\end{aligned} \tag{665}$$

$$D_{36}(r, z) = 3 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \quad (666)$$

$$R_{37}(r, z) = \frac{N_{37}(r, z)}{D_{37}(r, z)}. \quad (667)$$

$$\begin{aligned} & 16 - 48z^2 + 72z^4 + 76r^8 + 136r^2 + 256r^6 \\ & + 300r^4 - 1176r^4z^6 - 1072r^4z^2 \\ & - 648r^6z^4 - 524r^6z^2 - 428r^2z^2 \\ & - 252r^8z^4 - 156r^2z^6 + 144r^4z^8 \\ & + 216r^8z^2 + 288r^2z^4 + 756r^6z^6 \\ N_{37}(r, z) = & + 2044r^4z^4 \end{aligned} \quad (668)$$

$$D_{37}(r, z) = \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (669)$$

$$R_{38}(r, z) = \frac{N_{38}(r, z)}{D_{38}(r, z)}. \quad (670)$$

$$\begin{aligned} & -240ir^3 - 216ir^5 - 96ir - 96ir^7 \\ & -24ir^9 - 8\sqrt{3}i - 2592ir^5z^6 \\ & -960ir^7z^2 - 864ir^7z^6 - 432ir^9z^2 \\ & -336ir^5z^2 - 144irz^4 - 96ir^3z^2 \\ & -48\sqrt{3}iz^2 + 72\sqrt{3}iz^4 + 88\sqrt{3}ir^8 \\ & + 112\sqrt{3}ir^2 + 216ir^9z^4 + 288ir^3z^6 \\ & + 304\sqrt{3}ir^6 + 336\sqrt{3}ir^4 + 1008ir^3z^4 \\ & + 1704ir^5z^4 + 2880ir^7z^4 \\ & - 2832\sqrt{3}ir^4z^6 - 1552\sqrt{3}ir^4z^2 \\ & - 1152\sqrt{3}ir^6z^4 - 728\sqrt{3}ir^6z^2 \\ & - 632\sqrt{3}ir^2z^2 - 504\sqrt{3}ir^8z^4 \\ & - 120\sqrt{3}ir^2z^6 + 288\sqrt{3}ir^4z^8 \\ & + 432\sqrt{3}ir^8z^2 + 576\sqrt{3}ir^2z^4 \\ N_{38}(r, z) = & + 1512\sqrt{3}ir^6z^6 + 3856\sqrt{3}ir^4z^4 \end{aligned} \quad (671)$$

$$D_{38}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (672)$$

$$R_{39}(r, z) = \frac{N_{39}(r, z)}{D_{39}(r, z)}. \quad (673)$$

$$\begin{aligned} & 48 - 48z^2 + 84r^8 + 192r^2 + 264r^6 + 324r^4 \\ & - 1044r^6z^6 - 912r^4z^2 - 900r^6z^2 \\ & - 432\sqrt{3}r^5 - 408r^8z^2 - 376\sqrt{3}r^7 \\ & - 324r^2z^2 - 184\sqrt{3}r^3 - 144r^4z^8 \\ & - 112\sqrt{3}r^9 - 36r^2z^6 - 16\sqrt{3}r \\ & + 132r^4z^4 + 168r^2z^4 + 324r^8z^4 \\ & + 600r^4z^6 + 1680r^6z^4 - 2128\sqrt{3}r^5z^4 \\ & - 1752\sqrt{3}r^7z^4 - 408\sqrt{3}r^3z^4 \\ & - 144\sqrt{3}r^9z^4 - 96\sqrt{3}r^3z^6 \\ & + 48\sqrt{3}rz^4 + 288\sqrt{3}r^9z^2 \\ & + 560\sqrt{3}r^3z^2 + 576\sqrt{3}r^7z^6 \\ & + 1152\sqrt{3}r^5z^6 + 1424\sqrt{3}r^7z^2 \\ N_{39}(r, z) = & + 1600\sqrt{3}r^5z^2 \end{aligned} \quad (674)$$

$$D_{39}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (675)$$

$$R_{40}(r, z) = \frac{N_{40}(r, z)}{D_{40}(r, z)}. \quad (676)$$

$$\begin{aligned}
& 24ir^3 + 24ir^5 - 48ir^3z^2 - 48ir^5z^2 \\
& -8\sqrt{3}ir^2 - 8\sqrt{3}ir^4 + 24ir^3z^4 \\
& +24ir^5z^4 - 8\sqrt{3}ir^2z^4 \\
& -8\sqrt{3}ir^4z^4 + 16\sqrt{3}ir^2z^2 \\
N_{40}(r, z) = & +16\sqrt{3}ir^4z^2
\end{aligned} \tag{677}$$

$$D_{40}(r, z) = (-r^2 + 2rz - 1)(r^2 + 2rz + 1) \tag{678}$$

$$R_{41}(r, z) = -144r^2z^2 + 144z^4 - 144z^2 - 16 \tag{679}$$

$$R_{42}(r, z) = 32ir - 96\sqrt{3}iz^2(r^2 - z^2 + 1) \tag{680}$$

$$R_{43}(r, z) = 16(3z^2 + 1)(r^2 - z^2 + 1) \tag{681}$$

$$R_{44}(r, z) = \frac{N_{44}(r, z)}{D_{44}(r, z)}. \tag{682}$$

$$\begin{aligned}
& 36 - 176r^6 - 108r^8 - 100z^4 + 8r^4 + 112r^2 \\
& +136z^2 - 2352r^6z^4 - 2304r^4z^8 \\
& -1896r^4z^4 - 1424r^2z^4 - 756r^8z^4 \\
& +272r^2z^2 + 304r^4z^2 + 656r^6z^2 \\
& +752r^2z^6 + 936r^8z^2 + 1584r^6z^6 \\
N_{44}(r, z) = & +4320r^4z^6
\end{aligned} \tag{683}$$

$$D_{44}(r, z) = (-r^2 + 2rz - 1)^2(r^2 + 2rz + 1)^2 \tag{684}$$

$$R_{45}(r, z) = \frac{N_{45}(r, z)}{D_{45}(r, z)}. \tag{685}$$

$$\begin{aligned}
& -624ir^5 - 528ir^3 - 528ir^7 - 216ir \\
& -216ir^9 + 56\sqrt{3}i - 2880ir^5z^6 \\
& -1296ir^5z^4 - 864ir^7z^6 - 768ir^7z^2 \\
& -448\sqrt{3}ir^6 - 432ir^9z^2 - 280\sqrt{3}ir^8 \\
& -240irz^2 - 216\sqrt{3}iz^4 + 24irz^4 \\
& +48ir^3z^4 + 216ir^9z^4 + 224\sqrt{3}ir^2 \\
& +240\sqrt{3}iz^2 + 960ir^3z^2 + 1248ir^3z^6 \\
& +2208ir^5z^2 + 3888ir^7z^4 \\
& -4608\sqrt{3}ir^4z^8 - 4224\sqrt{3}ir^6z^4 \\
& -4000\sqrt{3}ir^4z^4 - 3168\sqrt{3}ir^2z^4 \\
& -1512\sqrt{3}ir^8z^4 + 608\sqrt{3}ir^2z^2 \\
& +1024\sqrt{3}ir^4z^2 + 1184\sqrt{3}ir^6z^2 \\
& +1872\sqrt{3}ir^8z^2 + 2016\sqrt{3}ir^2z^6 \\
N_{45}(r, z) = & +3168\sqrt{3}ir^6z^6 + 8064\sqrt{3}ir^4z^6
\end{aligned} \tag{686}$$

$$D_{45}(r, z) = 3(-r^2 + 2rz - 1)^2(r^2 + 2rz + 1)^2 \tag{687}$$

$$R_{46}(r, z) = \frac{N_{46}(r, z)}{D_{46}(r, z)}. \tag{688}$$

$$\begin{aligned}
& -16 - 144z^2 - 28r^2 + 144z^4 + 180r^4 \\
& + 188r^8 + 380r^6 - 3696r^4z^6 - 1872r^6z^6 \\
& - 1152r^2z^6 - 1032r^8z^2 - 1000r^6z^2 \\
& - 920r^4z^2 - 424r^2z^2 - 8\sqrt{3}r^7 \\
& + 16\sqrt{3}r^9 + 48\sqrt{3}r^5 + 112\sqrt{3}r + 184\sqrt{3}r^3 \\
& + 828r^8z^4 + 1668r^2z^4 + 2036r^4z^4 \\
& + 2304r^4z^8 + 2556r^6z^4 - 2088\sqrt{3}r^7z^4 \\
& - 552\sqrt{3}r^3z^4 - 192\sqrt{3}r^3z^6 \\
& - 176\sqrt{3}r^5z^4 - 160\sqrt{3}r^5z^2 \\
& - 144\sqrt{3}r^9z^4 - 80\sqrt{3}r^3z^2 \\
& + 48\sqrt{3}rz^4 + 288\sqrt{3}r^9z^2 \\
& + 576\sqrt{3}r^7z^6 + 880\sqrt{3}r^7z^2 \\
N_{46}(r, z) = & +1248\sqrt{3}r^5z^6
\end{aligned} \tag{689}$$

$$D_{46}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{690}$$

$$R_{47}(r, z) = \frac{N_{47}(r, z)}{D_{47}(r, z)}. \tag{691}$$

$$\begin{aligned}
& -32ir + 8ir^3 + 40ir^5 - 144ir^5z^2 \\
& - 24\sqrt{3}ir^2 - 24\sqrt{3}ir^4 - 16ir^3z^2 \\
& + 72ir^3z^4 + 72ir^5z^4 - 24\sqrt{3}ir^2z^4 \\
& - 24\sqrt{3}ir^4z^4 + 48\sqrt{3}ir^2z^2 \\
N_{47}(r, z) = & +48\sqrt{3}ir^4z^2
\end{aligned} \tag{692}$$

$$D_{47}(r, z) = 3 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \tag{693}$$

$$R_{48}(r, z) = \frac{N_{48}(r, z)}{D_{48}(r, z)}. \tag{694}$$

$$\begin{aligned}
& -4 - 40z^2 + 28z^4 + 88r^2 + 200r^8 + 388r^4 \\
& + 496r^6 - 2328r^6z^4 - 888r^4z^6 \\
& - 792r^8z^4 - 192r^4z^2 - 124r^4z^4 \\
& - 108r^2z^2 - 16r^2z^4 + 132r^6z^2 \\
& + 144r^{10}z^2 + 144r^4z^8 + 244r^2z^6 \\
N_{48}(r, z) = & +2628r^6z^6
\end{aligned} \tag{695}$$

$$D_{48}(r, z) = \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{696}$$

$$R_{49}(r, z) = \frac{N_{49}(r, z)}{D_{49}(r, z)}. \tag{697}$$

$$\begin{aligned}
& -168ir^5 - 96ir^3 - 64\sqrt{3}i + 24ir + 48ir^7 \\
& + 96ir^9 - 1008ir^5z^2 - 1008ir^7z^4 \\
& - 960ir^3z^6 - 288ir^3z^2 - 168irz^4 \\
& - 112\sqrt{3}ir^2 + 240irz^2 + 288ir^5z^6 \\
& + 336\sqrt{3}ir^4 + 368\sqrt{3}ir^8 + 576ir^7z^2 \\
& + 752\sqrt{3}ir^6 + 960ir^3z^4 + 1464ir^5z^4 \\
& - 4992\sqrt{3}ir^6z^4 - 1680\sqrt{3}ir^4z^6 \\
& - 1584\sqrt{3}ir^8z^4 - 784\sqrt{3}ir^4z^4 \\
& + 168\sqrt{3}ir^2z^6 + 200\sqrt{3}ir^2z^2 \\
& + 288\sqrt{3}ir^{10}z^2 + 288\sqrt{3}ir^2z^4 \\
& + 288\sqrt{3}ir^4z^8 + 304\sqrt{3}ir^4z^2 \\
N_{49}(r, z) = & +968\sqrt{3}ir^6z^2 + 5256\sqrt{3}ir^6z^6
\end{aligned} \tag{698}$$

$$D_{49}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{699}$$

$$R_{50}(r, z) = \frac{N_{50}(r, z)}{D_{50}(r, z)}. \quad (700)$$

$$\begin{aligned} & 16 - 596r^6 - 344r^8 - 336r^4 - 48r^{10} - 20r^2 \\ & - 2628r^6z^6 - 480\sqrt{3}r^5 - 368\sqrt{3}r^3 \\ & - 368\sqrt{3}r^7 - 156r^2z^4 - 144r^{10}z^2 \\ & - 144r^4z^8 - 128\sqrt{3}r - 128\sqrt{3}r^9 \\ & - 44r^2z^2 - 36r^2z^6 + 4r^6z^2 \\ & + 8r^4z^2 + 336r^8z^2 + 456r^4z^6 \\ & + 792r^8z^4 + 880r^4z^4 + 2004r^6z^4 \\ & - 1952\sqrt{3}r^5z^4 - 96\sqrt{3}r^5z^6 \\ & + 96\sqrt{3}r^3z^6 + 144\sqrt{3}r^3z^4 \\ & + 336\sqrt{3}r^7z^4 + 544\sqrt{3}r^7z^2 \\ N_{50}(r, z) = & + 640\sqrt{3}r^3z^2 + 1760\sqrt{3}r^5z^2 \end{aligned} \quad (701)$$

$$D_{50}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \quad (702)$$

$$R_{51}(r, z) = -\frac{32ir}{3} \quad (703)$$

$$R_{52}(r, z) = \frac{N_{52}(r, z)}{D_{52}(r, z)}. \quad (704)$$

$$\begin{aligned} & -56 - 160r^2 - 104r^4 + 96z^2 - 432r^4z^4 \\ N_{52}(r, z) = & -48r^2z^4 + 128r^2z^2 + 576r^4z^2 \end{aligned} \quad (705)$$

$$D_{52}(r, z) = \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \quad (706)$$

$$R_{53}(r, z) = \frac{N_{53}(r, z)}{D_{53}(r, z)}. \quad (707)$$

$$\begin{aligned} & -240ir^5 - 192ir^3 - 80\sqrt{3}i + 48ir \\ & - 352\sqrt{3}ir^2 - 288irz^2 - 272\sqrt{3}ir^4 \\ & + 96\sqrt{3}iz^2 + 96ir^3z^2 + 576ir^3z^4 \\ & - 864\sqrt{3}ir^4z^4 + 96\sqrt{3}ir^2z^4 \\ N_{53}(r, z) = & + 224\sqrt{3}ir^2z^2 + 1152\sqrt{3}ir^4z^2 \end{aligned} \quad (708)$$

$$D_{53}(r, z) = 3 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \quad (709)$$

$$R_{54}(r, z) = \frac{N_{54}(r, z)}{D_{54}(r, z)}. \quad (710)$$

$$\begin{aligned} & 48 - 48z^2 + 192r^4 + 240r^2 - 624r^4z^2 \\ & - 288r^2z^2 - 64\sqrt{3}r - 32\sqrt{3}r^3 + 32\sqrt{3}r^5 \\ & + 48r^2z^4 + 432r^4z^4 - 192\sqrt{3}r^3z^4 \\ N_{54}(r, z) = & + 96\sqrt{3}rz^2 + 160\sqrt{3}r^3z^2 \end{aligned} \quad (711)$$

$$D_{54}(r, z) = 3 \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \quad (712)$$

$$R_{55}(r, z) = 144r^2z^2 - 144z^2 + 160 \quad (713)$$

$$R_{56}(r, z) = -32ir + 96\sqrt{3}i \left(r^2z^2 - z^2 + 1 \right) \quad (714)$$

$$R_{57}(r, z) = -48r^2z^2 - 16r^2 + 64z^2 - 64 \quad (715)$$

$$R_{58}(r, z) = \frac{N_{58}(r, z)}{D_{58}(r, z)}. \quad (716)$$

$$\begin{aligned} & 136 - 64z^2 + 72r^4 + 208r^2 - 576r^4z^4 \\ N_{58}(r, z) = & -544r^2z^2 + 192r^2z^4 + 576r^4z^2 \end{aligned} \quad (717)$$

$$D_{58}(r, z) = (-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (718)$$

$$R_{59}(r, z) = \frac{N_{59}(r, z)}{D_{59}(r, z)}. \quad (719)$$

$$\begin{aligned} & -672ir^3 - 432ir^5 - 240ir + 272\sqrt{3}i \\ & -192irz^2 - 192\sqrt{3}iz^2 + 80\sqrt{3}ir^4 \\ & + 352\sqrt{3}ir^2 + 384ir^3z^2 + 1152ir^3z^4 \\ & - 1280\sqrt{3}ir^2z^2 - 1152\sqrt{3}ir^4z^4 \\ N_{59}(r, z) = & +768\sqrt{3}ir^2z^4 + 1152\sqrt{3}ir^4z^2 \end{aligned} \quad (720)$$

$$D_{59}(r, z) = 3(-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (721)$$

$$R_{60}(r, z) = \frac{N_{60}(r, z)}{D_{60}(r, z)}. \quad (722)$$

$$\begin{aligned} & -160 - 224r^2 - 64r^4 + 144z^2 - 576r^2z^4 \\ & -528r^4z^2 + 64\sqrt{3}r + 160\sqrt{3}r^5 \\ & + 224\sqrt{3}r^3 + 576r^4z^4 + 832r^2z^2 \\ & - 384\sqrt{3}r^3z^4 - 160\sqrt{3}r^3z^2 \\ N_{60}(r, z) = & +96\sqrt{3}rz^2 \end{aligned} \quad (723)$$

$$D_{60}(r, z) = 3(-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (724)$$

$$R_{61}(r, z) = \frac{N_{61}(r, z)}{D_{61}(r, z)}. \quad (725)$$

$$\begin{aligned} & -32 - 48r^2 - 16r^4 + 16z^2 - 432r^4z^4 \\ & -112r^2z^2 + 144r^4z^2 + 144r^6z^2 \\ N_{61}(r, z) = & +336r^2z^4 \end{aligned} \quad (726)$$

$$D_{61}(r, z) = (-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (727)$$

$$R_{62}(r, z) = \frac{N_{62}(r, z)}{D_{62}(r, z)}. \quad (728)$$

$$\begin{aligned} & -64\sqrt{3}i + 96ir^5 + 192ir + 288ir^3 \\ & -576ir^3z^4 - 128\sqrt{3}ir^2 - 96irz^2 \\ & -64\sqrt{3}ir^4 + 96ir^3z^2 - 864\sqrt{3}ir^4z^4 \\ & + 64\sqrt{3}ir^2z^2 + 288\sqrt{3}ir^4z^2 \\ N_{62}(r, z) = & +288\sqrt{3}ir^6z^2 + 480\sqrt{3}ir^2z^4 \end{aligned} \quad (729)$$

$$D_{62}(r, z) = 3(-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (730)$$

$$R_{63}(r, z) = \frac{N_{63}(r, z)}{D_{63}(r, z)}. \quad (731)$$

$$\begin{aligned} & 16 - 48r^6 - 32r^4 + 32r^2 - 256\sqrt{3}r^3 \\ & -144r^6z^2 - 144r^2z^4 - 128\sqrt{3}r \\ & -128\sqrt{3}r^5 - 112r^2z^2 + 432r^4z^4 \\ N_{63}(r, z) = & +192\sqrt{3}r^3z^4 + 320\sqrt{3}r^3z^2 \end{aligned} \quad (732)$$

$$D_{63}(r, z) = 3(-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (733)$$

$$R_{64}(r, z) = \frac{32ir}{3} \quad (734)$$

$$R_{65}(r, z) = \frac{N_{65}(r, z)}{D_{65}(r, z)}. \quad (735)$$

$$\begin{aligned}
& -68 - 198r^2 - 198r^4 - 68r^6 - 36z^4 - 32z^2 \\
& -652r^3z^3 - 396r^3z^7 - 364rz^3 \\
& -364r^5z^3 - 36r^6z^4 - 32r^6z^2 + 38rz \\
& +38r^5z + 38r^2z^4 + 38r^4z^4 \\
& +42r^2z^6 + 42r^4z^6 + 54rz^5 \\
& +54r^5z^5 + 64r^3z + 254r^2z^2 \\
N_{65}(r, z) = & +254r^4z^2 + 1528r^3z^5
\end{aligned} \tag{736}$$

$$D_{65}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right)^2 \tag{737}$$

$$R_{66}(r, z) = \frac{N_{66}(r, z)}{D_{66}(r, z)}. \tag{738}$$

$$\begin{aligned}
& -4i - 206ir^3 - 204ir^5 - 68ir - 62ir^7 \\
& -16iz^2 + 4\sqrt{3}i + 4ir^9 + 12iz^4 + 62ir^2 \\
& + 68ir^8 + 204ir^4 + 206ir^6 - 810ir^6z^5 \\
& - 706ir^6z^4 - 584ir^5z^6 - 456ir^5z^3 \\
& - 368ir^4z^2 - 240ir^3z^7 - 228ir^3z^6 \\
& - 168irz^3 - 156ir^4z - 152ir^2z^2 \\
& - 144ir^4z^8 - 132ir^3z^3 - 128irz^2 \\
& - 128ir^6z^2 - 114ir^2z - 108ir^5z^4 \\
& - 72ir^8z^5 - 72ir^5z^7 - 60ir^8z^4 \\
& - 60ir^5z^5 - 48irz - 34ir^2z^4 \\
& - 28\sqrt{3}ir^5 - 24\sqrt{3}ir^3 - 18ir^2z^5 \\
& - 16\sqrt{3}ir^7 - 12ir^2z^3 - 12ir^9z^4 \\
& - 12ir^7z^6 - 8\sqrt{3}ir - 8\sqrt{3}iz^2 \\
& - 6ir^3z - 4\sqrt{3}ir^9 + 6ir^6z + 8\sqrt{3}ir^8 \\
& + 12\sqrt{3}iz^4 + 12ir^7z^3 + 12ir^2z^6 \\
& + 16\sqrt{3}ir^2 + 16ir^9z^2 + 18ir^7z^5 \\
& + 24\sqrt{3}ir^6 + 28\sqrt{3}ir^4 + 34ir^7z^4 \\
& + 48ir^8z + 60irz^4 + 60ir^4z^5 \\
& + 72irz^5 + 72ir^4z^7 + 108ir^4z^4 \\
& + 114ir^7z + 128ir^3z^2 + 128ir^8z^2 \\
& + 132ir^6z^3 + 144ir^5z^8 + 152ir^7z^2 \\
& + 156ir^5z + 168ir^8z^3 + 228ir^6z^6 \\
& + 240ir^6z^7 + 368ir^5z^2 + 456ir^4z^3 \\
& + 584ir^4z^6 + 706ir^3z^4 + 810ir^3z^5 \\
& - 788\sqrt{3}ir^5z^6 - 560\sqrt{3}ir^4z^4 \\
& - 560\sqrt{3}ir^5z^5 - 540\sqrt{3}ir^4z^3 \\
& - 530\sqrt{3}ir^3z^5 - 496\sqrt{3}ir^6z^3 \\
& - 264\sqrt{3}ir^4z^8 - 216\sqrt{3}ir^6z^7 \\
& - 164\sqrt{3}ir^5z - 160\sqrt{3}ir^4z^7 \\
& - 158\sqrt{3}ir^3z - 136\sqrt{3}ir^2z^4 \\
& - 136\sqrt{3}ir^6z^4 - 112\sqrt{3}ir^2z^3 \\
& - 108\sqrt{3}ir^8z^3 - 80\sqrt{3}ir^5z^2 \\
& - 62\sqrt{3}ir^7z - 60\sqrt{3}irz^5 - 56\sqrt{3}irz \\
& - 50\sqrt{3}ir^7z^2 - 42\sqrt{3}ir^7z^5 \\
& - 34\sqrt{3}ir^3z^2 - 30\sqrt{3}ir^7z^6 \\
& - 22\sqrt{3}ir^3z^6 - 12\sqrt{3}irz^4 \\
& - 12\sqrt{3}ir^9z^4 - 4\sqrt{3}ir^8z^2 \\
& + 4\sqrt{3}irz^2 + 8\sqrt{3}ir^9z^2 \\
& + 12\sqrt{3}ir^8z^4 + 22\sqrt{3}ir^6z^6 \\
& + 30\sqrt{3}ir^2z^6 + 34\sqrt{3}ir^6z^2 \\
& + 42\sqrt{3}ir^2z^5 + 50\sqrt{3}ir^2z^2 \\
& + 56\sqrt{3}ir^8z + 60\sqrt{3}ir^8z^5 \\
& + 62\sqrt{3}ir^2z + 80\sqrt{3}ir^4z^2 \\
& + 108\sqrt{3}irz^3 + 112\sqrt{3}ir^7z^3 \\
& + 136\sqrt{3}ir^3z^4 + 136\sqrt{3}ir^7z^4 \\
& + 158\sqrt{3}ir^6z + 160\sqrt{3}ir^5z^7 \\
& + 164\sqrt{3}ir^4z + 216\sqrt{3}ir^3z^7 \\
& + 264\sqrt{3}ir^5z^8 + 496\sqrt{3}ir^3z^3 \\
& + 530\sqrt{3}ir^6z^5 + 540\sqrt{3}ir^5z^3 \\
& + 560\sqrt{3}ir^5z^4 + 560\sqrt{3}ir^4z^5 \\
& N_{66}(r, z) = +788\sqrt{3}ir^4z^6
\end{aligned} \tag{739}$$

$$D_{66}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (740)$$

$$R_{67}(r, z) = \frac{N_{67}(r, z)}{D_{67}(r, z)}. \quad (741)$$

$$\begin{aligned}
& -48r^5 - 30r^3 - 30r^7 - 6r - 6r^9 \\
& -828r^5z^4 - 492r^4z^5 - 492r^6z^5 \\
& -292r^5z^5 - 264r^5z^8 - 162r^3z^4 \\
& -162r^7z^4 - 84r^4z - 84r^6z - 60r^2z^5 \\
& -60r^8z^5 - 54\sqrt{3}r^4 - 54\sqrt{3}r^6 - 48r^5z \\
& -48r^4z^4 - 48r^6z^4 - 40r^3z^5 \\
& -40r^7z^5 - 36r^2z - 36r^8z - 36r^4z^6 \\
& -36r^6z^6 - 32r^3z - 32r^7z - 18\sqrt{3}r^2 \\
& -18\sqrt{3}r^8 - 12r^2z^6 - 12r^8z^6 - 8rz \\
& -8r^9z - 6rz^2 - 6rz^5 - 6r^9z^2 \\
& -6r^9z^5 + 2\sqrt{3}r + 2rz^3 + 2\sqrt{3}r^9 \\
& + 2r^9z^3 + 8\sqrt{3}r^3 + 8\sqrt{3}r^7 + 12\sqrt{3}r^5 \\
& + 12r^2z^2 + 12r^8z^2 + 24r^3z^7 \\
& + 24r^7z^7 + 36r^4z^2 + 36r^6z^2 \\
& + 48r^4z^8 + 48r^6z^8 + 78r^3z^6 \\
& + 78r^7z^6 + 80r^5z^7 + 96r^2z^3 \\
& + 96r^3z^3 + 96r^7z^3 + 96r^8z^3 \\
& + 162r^3z^2 + 162r^7z^2 + 188r^5z^3 \\
& + 216r^4z^7 + 216r^6z^7 + 312r^5z^2 \\
& + 360r^4z^3 + 360r^6z^3 + 756r^5z^6 \\
& - 220\sqrt{3}r^4z^3 - 220\sqrt{3}r^6z^3 \\
& - 124\sqrt{3}r^4z^6 - 124\sqrt{3}r^6z^6 \\
& - 104\sqrt{3}r^3z^4 - 104\sqrt{3}r^7z^4 \\
& - 104\sqrt{3}r^4z^7 - 104\sqrt{3}r^6z^7 \\
& - 96\sqrt{3}r^5z - 96\sqrt{3}r^5z^8 - 84\sqrt{3}r^5z^2 \\
& - 60\sqrt{3}r^2z^3 - 60\sqrt{3}r^8z^3 \\
& - 56\sqrt{3}r^3z^5 - 56\sqrt{3}r^7z^5 - 48\sqrt{3}r^3z \\
& - 48\sqrt{3}r^7z - 32\sqrt{3}r^3z^2 - 32\sqrt{3}r^7z^2 \\
& - 20\sqrt{3}r^2z^2 - 20\sqrt{3}r^8z^2 \\
& - 12\sqrt{3}r^2z^6 - 12\sqrt{3}r^8z^6 - 8\sqrt{3}rz^4 \\
& - 8\sqrt{3}r^9z^4 - 6\sqrt{3}rz^5 - 6\sqrt{3}r^9z^5 \\
& - 4\sqrt{3}r^4z^2 - 4\sqrt{3}r^6z^2 \\
& - 4\sqrt{3}r^5z^5 + 6\sqrt{3}r^2z + 6\sqrt{3}r^8z \\
& + 10\sqrt{3}rz^2 + 10\sqrt{3}rz^3 + 10\sqrt{3}r^9z^2 \\
& + 10\sqrt{3}r^9z^3 + 16\sqrt{3}r^3z^3 \\
& + 16\sqrt{3}r^7z^3 + 18\sqrt{3}r^4z + 18\sqrt{3}r^6z \\
& + 24\sqrt{3}r^3z^7 + 24\sqrt{3}r^7z^7 \\
& + 26\sqrt{3}r^2z^4 + 26\sqrt{3}r^8z^4 \\
& + 30\sqrt{3}r^2z^5 + 30\sqrt{3}r^8z^5 \\
& + 32\sqrt{3}r^5z^4 + 48\sqrt{3}r^5z^7 \\
& + 48\sqrt{3}r^4z^8 + 48\sqrt{3}r^6z^8 \\
& + 64\sqrt{3}r^3z^6 + 64\sqrt{3}r^7z^6 \\
& + 158\sqrt{3}r^4z^4 + 158\sqrt{3}r^6z^4 \\
& + 172\sqrt{3}r^5z^3 + 256\sqrt{3}r^5z^6 \\
& + 330\sqrt{3}r^4z^5 + 330\sqrt{3}r^6z^5
\end{aligned} \quad (742)$$

$$D_{67}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (743)$$

$$R_{68}(r, z) = \frac{N_{68}(r, z)}{D_{68}(r, z)}. \quad (744)$$

$$\begin{aligned} & -2ir^2 - 2ir^4 + 2ir^3 + 2ir^5 \\ & -28ir^3z^3 - 18ir^4z - 14ir^2z^4 \\ & -12ir^3z^6 - 8ir^5z^2 - 6ir^4z^4 \\ & -6ir^2z^5 - 4ir^3z^2 - 4ir^2z^3 \\ & -2ir^5z - 2\sqrt{3}ir^2 - 2\sqrt{3}ir^4 \\ & -2ir^4z^5 + 2ir^2z + 2\sqrt{3}ir^3 \\ & +2\sqrt{3}ir^5 + 2ir^3z^5 + 4ir^4z^2 \\ & +4ir^5z^3 + 6ir^3z^4 + 6ir^5z^5 \\ & +8ir^2z^2 + 12ir^4z^6 + 14ir^5z^4 \\ & +18ir^3z + 28ir^4z^3 - 12\sqrt{3}ir^3z^3 \\ & -6\sqrt{3}ir^4z - 6\sqrt{3}ir^2z^4 \\ & -4\sqrt{3}ir^3z^2 - 4\sqrt{3}ir^2z^3 \\ & -4\sqrt{3}ir^3z^6 - 2\sqrt{3}ir^2z \\ & -2\sqrt{3}ir^3z^4 - 2\sqrt{3}ir^2z^5 \\ & -2\sqrt{3}ir^3z^5 + 2\sqrt{3}ir^5z \\ & +2\sqrt{3}ir^4z^4 + 2\sqrt{3}ir^4z^5 \\ & +2\sqrt{3}ir^5z^5 + 4\sqrt{3}ir^4z^2 \\ & +4\sqrt{3}ir^5z^3 + 4\sqrt{3}ir^4z^6 \\ & +6\sqrt{3}ir^3z + 6\sqrt{3}ir^5z^4 \\ N_{68}(r, z) = & +12\sqrt{3}ir^4z^3 \end{aligned} \quad (745)$$

$$D_{68}(r, z) = (-r^2 + 2rz - 1)(r^2 + 2rz + 1) \quad (746)$$

$$R_{69}(r, z) = \frac{4(3r^2z^3 + 3r^2z - 27rz^4 - 8rz^2 - 17r + 3z^3 + 3z)}{r} \quad (747)$$

$$R_{70}(r, z) = 4i(r-1)(3z+5)(z^2+1) + 4\sqrt{3}i(r-1)(9z^4+z^3+z^2+z+4) \quad (748)$$

$$R_{71}(r, z) = -8\sqrt{3}r(z+1)(z^2+1) - 4r(9z^4-2z^3+2z^2+2z+5) \quad (749)$$

$$R_{72}(r, z) = \frac{N_{72}(r, z)}{D_{72}(r, z)}. \quad (750)$$

$$\begin{aligned} & 36z + 36z^3 + 52r + 52r^7 + 212r^3 + 212r^5 \\ & -880r^4z^5 - 644r^3z^4 - 644r^5z^4 \\ & -344r^4z^3 - 240r^2z^5 - 240r^6z^5 \\ & +36r^8z + 36r^8z^3 + 60rz^4 + 60r^7z^4 \\ & +64r^3z^2 + 64r^5z^2 + 120r^3z^6 \\ & +120r^5z^6 + 136rz^2 + 136r^7z^2 \\ & +208r^2z + 208r^6z + 232r^4z + 240r^2z^3 \\ N_{72}(r, z) = & +240r^6z^3 + 432r^4z^7 \end{aligned} \quad (751)$$

$$D_{72}(r, z) = r(-r^2 + 2rz - 1)(r^2 + 2rz + 1)^2 \quad (752)$$

$$R_{73}(r, z) = \frac{N_{73}(r, z)}{D_{73}(r, z)}. \quad (753)$$

$$\begin{aligned}
N_{73}(r, z) = & -4i - 132ir^6 - 112ir^4 - 52ir^8 - 36ir^2 \\
& - 8\sqrt{3}i + 4ir^9 + 20iz^2 + 36iz + 36iz^3 \\
& + 36ir^7 + 48iz^4 + 52ir + 112ir^5 \\
& + 132ir^3 - 876ir^4z^5 - 536ir^3z^3 \\
& - 444ir^6z^5 - 392ir^5z^3 - 384ir^3z^7 \\
& - 384ir^5z^7 - 376ir^7z^3 - 324ir^2z^5 \\
& - 300ir^3z^6 - 252ir^3z^2 - 240ir^5z^2 \\
& - 192ir^8z^4 - 188ir^4z - 180ir^8z^5 \\
& - 168ir^4z^4 - 152ir^4z^6 - 144ir^4z^8 \\
& - 128\sqrt{3}ir^4 - 100ir^2z - 84\sqrt{3}ir^6 \\
& - 68irz^2 - 68\sqrt{3}ir^2 - 48ir^9z^4 \\
& - 44ir^6z - 44\sqrt{3}iz^2 - 44ir^3z^4 \\
& - 44ir^7z^4 - 36ir^9z - 36ir^2z^2 \\
& - 36ir^9z^3 - 20ir^9z^2 - 16\sqrt{3}ir^8 \\
& - 12ir^7z^6 - 8irz - 4irz^3 + 4ir^8z^3 \\
& + 8ir^8z + 8\sqrt{3}ir^9 + 12\sqrt{3}iz \\
& + 12\sqrt{3}iz^3 + 12ir^2z^6 + 16\sqrt{3}ir \\
& + 36\sqrt{3}iz^4 + 36ir^7z^2 + 44ir^3z \\
& + 44ir^2z^4 + 44ir^6z^4 + 68\sqrt{3}ir^7 \\
& + 68ir^8z^2 + 84\sqrt{3}ir^3 + 100ir^7z \\
& + 128\sqrt{3}ir^5 + 144ir^5z^8 + 152ir^5z^6 \\
& + 168ir^5z^4 + 180irz^5 + 188ir^5z \\
& + 192irz^4 + 240ir^4z^2 + 252ir^6z^2 \\
& + 300ir^6z^6 + 324ir^7z^5 + 376ir^2z^3 \\
& + 384ir^4z^7 + 384ir^6z^7 + 392ir^4z^3 \\
& + 444ir^3z^5 + 536ir^6z^3 + 876ir^5z^5 \\
& - 592\sqrt{3}ir^4z^6 - 340\sqrt{3}ir^3z^4 \\
& - 288\sqrt{3}ir^5z^8 - 276\sqrt{3}ir^7z^4 \\
& - 224\sqrt{3}ir^3z^5 - 216\sqrt{3}ir^2z^6 \\
& - 200\sqrt{3}ir^6z^6 - 192\sqrt{3}ir^5z^4 \\
& - 164\sqrt{3}ir^5z^5 - 152\sqrt{3}ir^6z^3 \\
& - 144\sqrt{3}ir^6z^7 - 136\sqrt{3}ir^4z^3 \\
& - 120\sqrt{3}ir^2z^5 - 96\sqrt{3}ir^5z^2 \\
& - 88\sqrt{3}ir^7z^2 - 84\sqrt{3}ir^8z^3 \\
& - 64\sqrt{3}ir^5z^7 - 56\sqrt{3}irz \\
& - 56\sqrt{3}ir^3z^2 - 48\sqrt{3}ir^7z \\
& - 36\sqrt{3}irz^5 - 36\sqrt{3}ir^9z^4 \\
& - 24\sqrt{3}ir^3z - 20\sqrt{3}ir^8z^2 \\
& - 12\sqrt{3}ir^9z - 12\sqrt{3}ir^9z^3 \\
& - 8\sqrt{3}ir^2z^3 - 4\sqrt{3}ir^5z - 4\sqrt{3}irz^4
\end{aligned}$$

$$\begin{aligned}
& + 4\sqrt{3}ir^4z + 4\sqrt{3}ir^8z^4 \\
& + 8\sqrt{3}ir^7z^3 + 20\sqrt{3}irz^2 \\
& + 24\sqrt{3}ir^6z + 36\sqrt{3}ir^8z^5 \\
& + 44\sqrt{3}ir^9z^2 + 48\sqrt{3}ir^2z \\
& + 56\sqrt{3}ir^8z + 56\sqrt{3}ir^6z^2 \\
& + 64\sqrt{3}ir^4z^7 + 84\sqrt{3}irz^3 \\
& + 88\sqrt{3}ir^2z^2 + 96\sqrt{3}ir^4z^2 \\
& + 120\sqrt{3}ir^7z^5 + 136\sqrt{3}ir^5z^3 \\
& + 144\sqrt{3}ir^3z^7 + 152\sqrt{3}ir^3z^3 \\
& + 164\sqrt{3}ir^4z^5 + 192\sqrt{3}ir^4z^4 \\
& + 200\sqrt{3}ir^3z^6 + 216\sqrt{3}ir^7z^6 \\
& + 224\sqrt{3}ir^6z^5 + 276\sqrt{3}ir^2z^4 \\
& + 288\sqrt{3}ir^4z^8 + 340\sqrt{3}ir^6z^4 \\
& + 592\sqrt{3}ir^5z^6
\end{aligned} \tag{754}$$

$$D_{73}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \tag{755}$$

$$R_{74}(r, z) = \frac{N_{74}(r, z)}{D_{74}(r, z)}. \tag{756}$$

$$\begin{aligned}
& -172r^5 - 96r^3 - 96r^7 - 10r - 10r^9 \\
& -464r^5z^6 - 204r^4z^5 - 204r^6z^5 \\
& -168r^3z^6 - 168r^7z^6 - 60\sqrt{3}r^5 \\
& -48r^4z^4 - 48r^6z^4 - 44r^2z - 44r^8z \\
& -40\sqrt{3}r^3 - 40\sqrt{3}r^7 - 36r^2z^5 \\
& -36r^5z^5 - 36r^8z^5 - 36r^4z^6 \\
& -36r^6z^6 - 34r^2z^2 - 34r^9z^2 \\
& -32r^4z^3 - 32r^6z^3 - 28r^5z^3 \\
& -16r^3z^3 - 16r^7z^3 - 12r^2z^6 \\
& -12r^8z^6 - 10\sqrt{3}r - 10\sqrt{3}r^9 - 8r^3z^5 \\
& -8r^7z^5 - 6rz^5 - 6r^9z^5 - 2rz^3 \\
& -2r^9z^3 + 4rz + 4r^9z + 12r^2z^2 \\
& +12r^8z^2 + 16r^3z + 16r^7z + 16r^5z^7 \\
& +18\sqrt{3}r^2 + 18\sqrt{3}r^8 + 24r^5z + 24r^4z^4 \\
& +24r^9z^4 + 24r^3z^7 + 24r^7z^7 \\
& +36r^4z^2 + 36r^6z^2 + 48r^4z^8 \\
& +48r^6z^8 + 54\sqrt{3}r^4 + 54\sqrt{3}r^6 \\
& +80r^2z^3 + 80r^8z^3 + 92r^4z + 92r^6z \\
& +144r^4z^7 + 144r^6z^7 + 152r^3z^4 \\
& +152r^7z^4 + 192r^3z^2 + 192r^7z^2 \\
& +228r^5z^2 + 288r^5z^8 - 412\sqrt{3}r^5z^5 \\
& -196\sqrt{3}r^4z^3 - 196\sqrt{3}r^6z^3 \\
& -180\sqrt{3}r^4z^2 - 180\sqrt{3}r^6z^2 \\
& -152\sqrt{3}r^5z - 152\sqrt{3}r^4z^7 \\
& -152\sqrt{3}r^6z^7 - 116\sqrt{3}r^3z^5 \\
& -116\sqrt{3}r^7z^5 - 96\sqrt{3}r^5z^8 \\
& -76\sqrt{3}r^3z - 76\sqrt{3}r^7z - 36\sqrt{3}r^2z^2 \\
& -36\sqrt{3}r^8z^2 - 36\sqrt{3}r^2z^3 \\
& -36\sqrt{3}r^8z^3 - 28\sqrt{3}r^4z^6 \\
& -28\sqrt{3}r^6z^6 - 18\sqrt{3}r^4z - 18\sqrt{3}r^6z \\
& -12\sqrt{3}r^2z^6 - 12\sqrt{3}r^8z^6 \\
& -8\sqrt{3}r^3z^4 - 8\sqrt{3}r^7z^4 - 6\sqrt{3}r^2z \\
& -6\sqrt{3}r^8z - 6\sqrt{3}rz^5 - 6\sqrt{3}r^9z^5 \\
& +4\sqrt{3}rz^4 + 4\sqrt{3}r^9z^4 + 8\sqrt{3}r^5z^4 \\
& +10\sqrt{3}rz^2 + 10\sqrt{3}rz^3 + 10\sqrt{3}r^9z^2 \\
& +10\sqrt{3}r^9z^3 + 12\sqrt{3}r^5z^2 \\
& +16\sqrt{3}r^3z^2 + 16\sqrt{3}r^7z^2 \\
& +24\sqrt{3}r^3z^7 + 24\sqrt{3}r^7z^7 \\
& +48\sqrt{3}r^4z^8 + 48\sqrt{3}r^6z^8 \\
& +54\sqrt{3}r^2z^4 + 54\sqrt{3}r^8z^4 \\
& +64\sqrt{3}r^3z^6 + 64\sqrt{3}r^5z^6 \\
& +64\sqrt{3}r^7z^6 + 66\sqrt{3}r^2z^5 \\
& +66\sqrt{3}r^8z^5 + 82\sqrt{3}r^4z^4 \\
& +82\sqrt{3}r^6z^4 + 200\sqrt{3}r^3z^3 \\
& +200\sqrt{3}r^7z^3 + 236\sqrt{3}r^5z^3 \\
& +256\sqrt{3}r^5z^7 + 342\sqrt{3}r^4z^5 \\
& +342\sqrt{3}r^6z^5
\end{aligned}$$

$$N_{74}(r, z) = +342\sqrt{3}r^6z^5 \quad (757)$$

$$D_{74}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (758)$$

$$R_{75}(r, z) = \frac{N_{75}(r, z)}{D_{75}(r, z)}. \quad (759)$$

$$\begin{aligned}
& -6ir^2 - 6ir^4 + 6ir^3 + 6ir^5 \\
& -28ir^3z^3 - 12ir^3z^6 - 10ir^4z \\
& -10ir^2z^4 - 10ir^4z^5 - 8ir^5z^2 \\
& -6ir^2z^5 - 4ir^3z^2 - 4ir^2z^3 \\
& -2ir^5z - 2\sqrt{3}ir^2 - 2\sqrt{3}ir^4 \\
& -2ir^4z^4 + 2ir^2z + 2\sqrt{3}ir^3 \\
& +2\sqrt{3}ir^5 + 2ir^3z^4 + 4ir^4z^2 \\
& +4ir^5z^3 + 6ir^5z^5 + 8ir^2z^2 \\
& +10ir^3z + 10ir^5z^4 + 10ir^3z^5 \\
& +12ir^4z^6 + 28ir^4z^3 - 12\sqrt{3}ir^3z^3 \\
& -6\sqrt{3}ir^4z - 6\sqrt{3}ir^2z^4 \\
& -4\sqrt{3}ir^3z^2 - 4\sqrt{3}ir^2z^3 \\
& -4\sqrt{3}ir^3z^6 - 2\sqrt{3}ir^2z \\
& -2\sqrt{3}ir^3z^4 - 2\sqrt{3}ir^2z^5 \\
& -2\sqrt{3}ir^3z^5 + 2\sqrt{3}ir^5z \\
& +2\sqrt{3}ir^4z^4 + 2\sqrt{3}ir^4z^5 \\
& +2\sqrt{3}ir^5z^5 + 4\sqrt{3}ir^4z^2 \\
& +4\sqrt{3}ir^5z^3 + 4\sqrt{3}ir^4z^6 \\
& +6\sqrt{3}ir^3z + 6\sqrt{3}ir^5z^4 \\
N_{75}(r, z) = & +12\sqrt{3}ir^4z^3
\end{aligned} \tag{760}$$

$$D_{75}(r, z) = (-r^2 + 2rz - 1) (r^2 + 2rz + 1) \tag{761}$$

$$R_{76}(r, z) = \frac{N_{76}(r, z)}{D_{76}(r, z)}. \tag{762}$$

$$\begin{aligned}
& -614r^3 - 614r^5 - 188r - 188r^7 - 24z - 24z^3 \\
& -668r^2z^3 - 668r^6z^3 - 368r^4z \\
& -258r^2z - 258r^6z - 180r^4z^4 - 180r^7z^4 \\
& -176r^2z^2 - 176r^7z^2 - 24r^8z - 24r^8z^3 \\
& +30r^2z^5 + 30r^6z^5 + 36r^4z^7 \\
& +84r^4z^3 + 258r^3z^6 + 258r^5z^6 \\
& +438r^3z^2 + 438r^5z^2 + 462r^3z^4 \\
N_{76}(r, z) = & +462r^5z^4 + 2136r^4z^5
\end{aligned} \tag{763}$$

$$D_{76}(r, z) = r (-r^2 + 2rz - 1) (r^2 + 2rz + 1)^2 \tag{764}$$

$$R_{77}(r, z) = \frac{N_{77}(r, z)}{D_{77}(r, z)}. \tag{765}$$

$$\begin{aligned}
& 20i - 436ir^5 - 418ir^3 - 178ir^7 - 140ir \\
& - 36iz^4 - 24iz - 24iz^3 - 20ir^9 - 16iz^2 \\
& + 28\sqrt{3}i + 140ir^8 + 178ir^2 + 418ir^6 \\
& + 436ir^4 - 1056ir^5z^6 - 936ir^5z^5 \\
& - 910ir^6z^4 - 808ir^4z^2 - 462ir^6z^5 \\
& - 460ir^6z^2 - 452ir^6z^3 - 436ir^2z^3 \\
& - 276ir^5z^4 - 252\sqrt{3}ir^5 - 238ir^2z^4 \\
& - 210ir^7z^5 - 196ir^2z^2 - 176irz^3 \\
& - 172\sqrt{3}ir^3 - 148\sqrt{3}ir^7 - 144ir^6z^7 \\
& - 132irz^4 - 120ir^4z^7 - 108irz^5 \\
& - 104ir^5z - 98ir^3z - 80irz^2 \\
& - 72ir^6z^6 - 56ir^4z^3 - 52irz \\
& - 40\sqrt{3}ir - 34ir^7z - 28\sqrt{3}ir^9 - 8\sqrt{3}iz \\
& - 8\sqrt{3}iz^3 + 12\sqrt{3}iz^4 + 16ir^9z^2 \\
& + 24ir^9z + 24ir^9z^3 + 34ir^2z \\
& + 36ir^9z^4 + 40\sqrt{3}iz^2 + 40\sqrt{3}ir^8 \\
& + 52ir^8z + 56ir^5z^3 + 72ir^3z^6 \\
& + 80ir^8z^2 + 98ir^6z + 104ir^4z \\
& + 108ir^8z^5 + 120ir^5z^7 + 132ir^8z^4 \\
& + 144ir^3z^7 + 148\sqrt{3}ir^2 + 172\sqrt{3}ir^6 \\
& + 176ir^8z^3 + 196ir^7z^2 + 210ir^2z^5 \\
& + 238ir^7z^4 + 252\sqrt{3}ir^4 + 276ir^4z^4 \\
& + 436ir^7z^3 + 452ir^3z^3 + 460ir^3z^2 \\
& + 462ir^3z^5 + 808ir^5z^2 + 910ir^3z^4 \\
& + 936ir^4z^5 + 1056ir^4z^6 \\
& - 868\sqrt{3}ir^5z^6 - 444\sqrt{3}ir^4z^3 \\
& - 396\sqrt{3}ir^5z^5 - 364\sqrt{3}ir^6z^4 \\
& - 360\sqrt{3}ir^6z^3 - 344\sqrt{3}ir^4z^4 \\
& - 300\sqrt{3}ir^2z^4 - 274\sqrt{3}ir^3z^5 \\
& - 248\sqrt{3}ir^4z^2 - 184\sqrt{3}ir^5z \\
& - 160\sqrt{3}ir^4z^7 - 150\sqrt{3}ir^3z \\
& - 150\sqrt{3}ir^2z^2 - 134\sqrt{3}ir^6z^2 \\
& - 130\sqrt{3}ir^7z^5 - 120\sqrt{3}ir^2z^3 \\
& - 72\sqrt{3}ir^6z^7 - 66\sqrt{3}ir^6z^6 \\
& - 44\sqrt{3}irz^4 - 42\sqrt{3}ir^2z^6 \\
& - 40\sqrt{3}ir^9z^2 - 30\sqrt{3}ir^7z \\
& - 24\sqrt{3}irz^5 - 24\sqrt{3}ir^5z^8 \\
& - 20\sqrt{3}irz^2 - 20\sqrt{3}ir^8z^3 \\
& - 12\sqrt{3}ir^9z^4 - 4\sqrt{3}irz + 4\sqrt{3}ir^8z \\
& + 8\sqrt{3}ir^9z + 8\sqrt{3}ir^9z^3 + 20\sqrt{3}irz^3 \\
& + 20\sqrt{3}ir^8z^2 + 24\sqrt{3}ir^8z^5 \\
& + 24\sqrt{3}ir^4z^8 + 30\sqrt{3}ir^2z \\
& + 42\sqrt{3}ir^7z^6 + 44\sqrt{3}ir^8z^4 \\
& + 66\sqrt{3}ir^3z^6 + 72\sqrt{3}ir^3z^7 \\
& + 120\sqrt{3}ir^7z^3 + 130\sqrt{3}ir^2z^5 \\
& + 134\sqrt{3}ir^3z^2 + 150\sqrt{3}ir^6z \\
& + 150\sqrt{3}ir^7z^2 + 160\sqrt{3}ir^5z^7 \\
& + 184\sqrt{3}ir^4z + 248\sqrt{3}ir^5z^2 \\
& + 274\sqrt{3}ir^6z^5 + 300\sqrt{3}ir^7z^4 \\
& + 344\sqrt{3}ir^5z^4 + 360\sqrt{3}ir^3z^3 \\
& + 364\sqrt{3}ir^3z^4 + 396\sqrt{3}ir^4z^5 \\
& N_{77}(r, z) = +444\sqrt{3}ir^5z^3 + 868\sqrt{3}ir^4z^6
\end{aligned} \tag{766}$$

$$D_{77}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (767)$$

$$R_{78}(r, z) = \frac{N_{78}(r, z)}{D_{78}(r, z)}. \quad (768)$$

$$\begin{aligned} & 24r + 24r^9 + 146r^3 + 146r^7 + 244r^5 \\ & -420r^5z^4 - 288r^4z^5 - 288r^6z^5 \\ & -234r^3z^4 - 234r^7z^4 - 188r^5z^2 \\ & -176r^4z - 176r^6z - 158r^3z^2 \\ & -158r^7z^2 - 108\sqrt{3}r^4 - 108\sqrt{3}r^6 \\ & -64r^5z^7 - 42r^3z^6 - 42r^7z^6 \\ & -36\sqrt{3}r^2 - 36\sqrt{3}r^8 - 24r^5z - 24r^2z^5 \\ & -24r^8z^5 - 16r^3z - 16r^7z - 4rz \\ & -4r^9z - 4rz^3 - 4r^9z^3 + 8r^2z \\ & +8r^8z + 12rz^4 + 12r^9z^4 + 16r^2z^3 \\ & +16r^3z^3 + 16r^7z^3 + 16r^8z^3 \\ & +20\sqrt{3}r + 20\sqrt{3}r^9 + 24r^5z^8 + 32r^3z^5 \\ & +32r^7z^5 + 36rz^2 + 36r^9z^2 \\ & +40r^5z^3 + 72r^4z^7 + 72r^6z^7 \\ & +80\sqrt{3}r^3 + 80\sqrt{3}r^7 + 120\sqrt{3}r^5 \\ & +392r^4z^3 + 392r^6z^3 + 772r^5z^6 \\ & -216\sqrt{3}r^3z^3 - 216\sqrt{3}r^7z^3 \\ & -176\sqrt{3}r^5z^2 - 160\sqrt{3}r^3z^4 \\ & -160\sqrt{3}r^7z^4 - 144\sqrt{3}r^5z^3 \\ & -96\sqrt{3}r^4z^6 - 96\sqrt{3}r^6z^6 \\ & -80\sqrt{3}r^3z^2 - 80\sqrt{3}r^7z^2 \\ & -80\sqrt{3}r^5z^7 - 36\sqrt{3}r^2z^5 \\ & -36\sqrt{3}r^8z^5 - 28\sqrt{3}r^2z^4 \\ & -28\sqrt{3}r^8z^4 - 24\sqrt{3}r^2z^3 \\ & -24\sqrt{3}r^4z^3 - 24\sqrt{3}r^6z^3 \\ & -24\sqrt{3}r^8z^3 - 12\sqrt{3}r^4z^4 - 12\sqrt{3}r^9z^4 \\ & -12\sqrt{3}r^4z^5 - 12\sqrt{3}r^6z^5 \\ & -4\sqrt{3}r^3z^5 - 4\sqrt{3}r^7z^5 + 8\sqrt{3}rz \\ & +8\sqrt{3}r^9z + 8\sqrt{3}rz^2 + 8\sqrt{3}rz^3 \\ & +8\sqrt{3}r^9z^2 + 8\sqrt{3}r^9z^3 + 12\sqrt{3}r^2z \\ & +12\sqrt{3}r^8z + 16\sqrt{3}r^2z^2 + 16\sqrt{3}r^8z^2 \\ & +24\sqrt{3}r^5z^4 + 36\sqrt{3}r^4z + 36\sqrt{3}r^6z \\ & +48\sqrt{3}r^4z^7 + 48\sqrt{3}r^6z^7 + 60\sqrt{3}r^3z \\ & +60\sqrt{3}r^7z + 76\sqrt{3}r^4z^4 + 76\sqrt{3}r^6z^4 \\ & +104\sqrt{3}r^5z + 176\sqrt{3}r^4z^2 \\ & +176\sqrt{3}r^6z^2 + 320\sqrt{3}r^5z^6 \end{aligned}$$

$$N_{78}(r, z) = +408\sqrt{3}r^5z^5 \quad (769)$$

$$D_{78}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (770)$$

$$R_{79}(r, z) = -\frac{4ir^2(r-1)(z-1)(z+1)(z^2+1)}{r^2-2rz+1} \quad (771)$$

$$R_{80}(r, z) = \frac{N_{80}(r, z)}{D_{80}(r, z)}. \quad (772)$$

$$\begin{aligned} & -4\sqrt{3} + 4\sqrt{3}z^4 + 8\sqrt{3}r^2 + 12\sqrt{3}r^4 \\ & -48\sqrt{3}r^2z^6 - 12\sqrt{3}r^4z^4 \\ N_{80}(r, z) = & -8\sqrt{3}r^2z^4 + 48\sqrt{3}r^2z^2 \end{aligned} \quad (773)$$

$$D_{80}(r, z) = (-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (774)$$

$$R_{81}(r, z) = \frac{N_{81}(r, z)}{D_{81}(r, z)}. \quad (775)$$

$$\begin{aligned} & 12i - 24ir^6 - 12iz^4 - 12ir^8 + 4\sqrt{3}i \\ & + 24ir^2 - 192ir^4z^8 - 112\sqrt{3}ir^5 \\ & - 96ir^2z^2 - 96ir^4z^2 - 88\sqrt{3}ir^7 \\ & - 56\sqrt{3}ir^3 - 40\sqrt{3}ir^4 - 40\sqrt{3}ir^6 \\ & - 24\sqrt{3}ir^9 - 24ir^2z^4 - 12\sqrt{3}ir^8 \\ & - 8\sqrt{3}ir - 8\sqrt{3}ir^2 - 4\sqrt{3}iz^4 \\ & + 12ir^8z^4 + 24ir^6z^4 + 96ir^2z^6 \\ & + 96ir^4z^6 + 192ir^4z^4 \\ & - 752\sqrt{3}ir^5z^4 - 440\sqrt{3}ir^4z^4 \\ & - 408\sqrt{3}ir^6z^4 - 360\sqrt{3}ir^7z^4 \\ & - 120\sqrt{3}ir^3z^6 - 80\sqrt{3}ir^5z^6 \\ & - 56\sqrt{3}ir^2z^4 - 52\sqrt{3}ir^8z^4 \\ & - 40\sqrt{3}ir^9z^4 - 8\sqrt{3}ir^3z^4 \\ & + 8\sqrt{3}irz^4 + 8\sqrt{3}ir^7z^6 \\ & + 24\sqrt{3}ir^2z^6 + 32\sqrt{3}ir^4z^8 \\ & + 40\sqrt{3}ir^2z^2 + 64\sqrt{3}ir^8z^2 \\ & + 64\sqrt{3}ir^9z^2 + 152\sqrt{3}ir^6z^6 \\ & + 184\sqrt{3}ir^3z^2 + 208\sqrt{3}ir^4z^6 \\ & + 240\sqrt{3}ir^4z^2 + 296\sqrt{3}ir^6z^2 \\ & + 416\sqrt{3}ir^5z^8 + 440\sqrt{3}ir^7z^2 \\ N_{81}(r, z) = & + 528\sqrt{3}ir^5z^2 \end{aligned} \quad (776)$$

$$D_{81}(r, z) = (-r^2 + 2rz - 1)^2 (r^2 + 2rz + 1)^2 \quad (777)$$

$$R_{82}(r, z) = \frac{N_{82}(r, z)}{D_{82}(r, z)}. \quad (778)$$

$$\begin{aligned}
& -48z^2 + 36r^8 + 48z^4 + 72r + 72r^2 + 72r^9 \\
& +144r^6 + 180r^4 + 324r^3 + 324r^7 + 504r^5 \\
& -1728r^5z^2 - 1272r^7z^2 - 1248r^5z^8 \\
& -840r^3z^2 - 840r^6z^2 - 720r^4z^2 \\
& -456r^6z^6 - 384r^4z^6 - 264r^2z^2 \\
& -192r^8z^2 - 192r^9z^2 - 168r^2z^6 \\
& -96r^4z^8 - 48rz^2 - 24rz^4 - 24r^7z^6 \\
& -16\sqrt{3}z^2 + 16\sqrt{3}z^4 + 24\sqrt{3}r + 36\sqrt{3}r^2 \\
& +48\sqrt{3}r^{10} + 72\sqrt{3}r^9 + 108r^3z^4 \\
& +120r^9z^4 + 156\sqrt{3}r^3 + 156r^8z^4 \\
& +168\sqrt{3}r^4 + 192\sqrt{3}r^8 + 252\sqrt{3}r^7 \\
& +276\sqrt{3}r^6 + 312\sqrt{3}r^5 + 360r^2z^4 \\
& +408r^3z^6 + 624r^5z^6 + 972r^7z^4 \\
& +1020r^4z^4 + 1152r^6z^4 + 1848r^5z^4 \\
& -1064\sqrt{3}r^6z^6 - 896\sqrt{3}r^4z^6 \\
& -712\sqrt{3}r^6z^2 - 528\sqrt{3}r^4z^2 \\
& -480\sqrt{3}r^5z^2 - 400\sqrt{3}r^8z^2 \\
& -312\sqrt{3}r^7z^2 - 300\sqrt{3}r^7z^4 \\
& -288\sqrt{3}r^5z^8 - 264\sqrt{3}r^3z^2 \\
& -264\sqrt{3}r^5z^4 - 152\sqrt{3}r^2z^6 \\
& -144\sqrt{3}r^8z^6 - 136\sqrt{3}r^2z^2 \\
& -72\sqrt{3}r^9z^4 - 48\sqrt{3}rz^2 \\
& -48\sqrt{3}r^{10}z^2 + 24\sqrt{3}rz^4 \\
& +24\sqrt{3}r^3z^6 + 84\sqrt{3}r^3z^4 \\
& +252\sqrt{3}r^2z^4 + 352\sqrt{3}r^8z^4 \\
& +352\sqrt{3}r^4z^8 + 360\sqrt{3}r^7z^6 \\
& +576\sqrt{3}r^6z^8 + 720\sqrt{3}r^5z^6 \\
& +904\sqrt{3}r^4z^4 + 924\sqrt{3}r^6z^4
\end{aligned} \tag{779}$$

$$D_{82}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right)^2 \tag{780}$$

$$R_{83}(r, z) = \frac{N_{83}(r, z)}{D_{83}(r, z)}. \tag{781}$$

$$\begin{aligned}
& -24ir^3 - 24ir^5 + 12ir^2 + 12ir^4 \\
& -48ir^4z^2 - 24ir^3z^6 - 12ir^2z^4 \\
& -12ir^4z^4 + 4\sqrt{3}ir^2 + 4\sqrt{3}ir^4 \\
& +24ir^3z^2 + 24ir^3z^4 + 24ir^5z^4 \\
& +48ir^4z^6 - 8\sqrt{3}ir^3z^6 \\
& -4\sqrt{3}ir^2z^4 - 4\sqrt{3}ir^4z^4 \\
& +8\sqrt{3}ir^3z^2
\end{aligned} \tag{782}$$

$$D_{83}(r, z) = \left(-r^2 + 2rz - 1 \right) \left(r^2 + 2rz + 1 \right) \tag{783}$$

$$R_{84}(r, z) = 32\sqrt{3}iz^2 (r+1)(z-1)(z+1) \tag{784}$$

$$R_{85}(r, z) = -32z^2 (r+1)(z-1)(z+1) - \frac{16\sqrt{3}(3r^2 - z^2)(z-1)(z+1)}{3} \tag{785}$$

$$R_{86}(r, z) = \frac{N_{86}(r, z)}{D_{86}(r, z)}. \tag{786}$$

$$\begin{aligned}
& -28\sqrt{3} - 88\sqrt{3}r^2 - 60\sqrt{3}r^4 + 28\sqrt{3}z^4 \\
& -48\sqrt{3}r^2z^6 + 48\sqrt{3}r^2z^2 \\
& +60\sqrt{3}r^4z^4 + 88\sqrt{3}r^2z^4
\end{aligned} \tag{787}$$

$$D_{86}(r, z) = (-r^2 + 2rz - 1) (r^2 + 2rz + 1) \quad (788)$$

$$R_{87}(r, z) = \frac{N_{87}(r, z)}{D_{87}(r, z)}. \quad (789)$$

$$\begin{aligned} & 12i - 12iz^4 + 4\sqrt{3}i + 60ir^8 + 96ir^2 \\ & + 192ir^6 + 216ir^4 - 384ir^4z^2 \\ & - 352\sqrt{3}ir^5 - 288ir^6z^2 - 248\sqrt{3}ir^3 \\ & - 192ir^6z^4 - 192ir^4z^8 - 184\sqrt{3}ir^7 \\ & - 96ir^2z^2 - 96ir^2z^4 - 60ir^8z^4 \\ & - 56\sqrt{3}ir - 36\sqrt{3}iz^4 - 24\sqrt{3}ir^9 \\ & - 24ir^4z^4 + 16\sqrt{3}ir^2 + 32\sqrt{3}iz^2 \\ & + 60\sqrt{3}ir^8 + 80\sqrt{3}ir^4 + 96ir^2z^6 \\ & + 128\sqrt{3}ir^6 + 288ir^6z^6 + 384ir^4z^6 \\ & - 1264\sqrt{3}ir^4z^4 - 1216\sqrt{3}ir^5z^4 \\ & - 960\sqrt{3}ir^6z^4 - 648\sqrt{3}ir^7z^4 \\ & - 480\sqrt{3}ir^4z^8 - 464\sqrt{3}ir^2z^4 \\ & - 200\sqrt{3}ir^3z^4 - 156\sqrt{3}ir^8z^4 \\ & - 96\sqrt{3}ir^5z^8 - 72\sqrt{3}ir^9z^4 \\ & - 24\sqrt{3}ir^7z^6 + 24\sqrt{3}irz^4 \\ & + 32\sqrt{3}irz^2 + 40\sqrt{3}ir^3z^6 \\ & + 72\sqrt{3}ir^2z^2 + 96\sqrt{3}ir^8z^2 \\ & + 96\sqrt{3}ir^9z^2 + 144\sqrt{3}ir^4z^2 \\ & + 376\sqrt{3}ir^2z^6 + 408\sqrt{3}ir^3z^2 \\ & + 408\sqrt{3}ir^6z^6 + 424\sqrt{3}ir^6z^2 \\ & + 752\sqrt{3}ir^5z^6 + 856\sqrt{3}ir^7z^2 \\ N_{87}(r, z) = & + 912\sqrt{3}ir^5z^2 + 1520\sqrt{3}ir^4z^6 \end{aligned} \quad (790)$$

$$D_{87}(r, z) = (-r^2 + 2rz - 1)^2 (r^2 + 2rz + 1)^2 \quad (791)$$

$$R_{88}(r, z) = \frac{N_{88}(r, z)}{D_{88}(r, z)}. \quad (792)$$

$$\begin{aligned}
& -96r^6 - 60r^8 - 48z^2 - 12r^4 + 24r + 24r^2 \\
& + 24r^9 + 48z^4 + 180r^3 + 180r^7 + 312r^5 \\
& - 1440r^4z^6 - 840r^7z^2 - 816r^5z^6 \\
& - 768r^5z^2 - 408r^3z^2 - 408r^6z^2 \\
& - 408r^6z^6 - 312r^2z^6 - 216r^2z^2 \\
& - 144r^4z^2 - 120r^3z^6 - 96r^8z^2 \\
& - 96r^9z^2 - 56\sqrt{3}r^4 - 48rz^2 - 28\sqrt{3}r^2 \\
& - 28\sqrt{3}r^6 + 8\sqrt{3}r + 24rz^4 + 24\sqrt{3}r^9 \\
& + 24r^7z^6 + 72r^9z^4 + 76\sqrt{3}r^3 \\
& + 96r^5z^8 + 108\sqrt{3}r^7 + 152\sqrt{3}r^5 \\
& + 156r^8z^4 + 348r^3z^4 + 480r^4z^8 \\
& + 504r^2z^4 + 636r^7z^4 + 912r^6z^4 \\
& + 1116r^4z^4 + 1176r^5z^4 - 244\sqrt{3}r^6z^4 \\
& - 184\sqrt{3}r^3z^2 - 160\sqrt{3}r^5z^2 \\
& - 152\sqrt{3}r^6z^6 - 136\sqrt{3}r^5z^4 \\
& - 128\sqrt{3}r^4z^6 - 124\sqrt{3}r^7z^4 \\
& - 104\sqrt{3}r^7z^2 - 96\sqrt{3}r^5z^8 \\
& - 48\sqrt{3}r^8z^6 - 24\sqrt{3}r^9z^4 - 16\sqrt{3}rz^2 \\
& - 16\sqrt{3}r^8z^4 - 8\sqrt{3}r^2z^2 \\
& - 8\sqrt{3}r^2z^6 + 4\sqrt{3}r^3z^4 + 8\sqrt{3}rz^4 \\
& + 32\sqrt{3}r^4z^8 + 44\sqrt{3}r^2z^4 \\
& + 48\sqrt{3}r^4z^2 + 64\sqrt{3}r^8z^2 \\
& + 104\sqrt{3}r^4z^4 + 104\sqrt{3}r^3z^6 \\
& + 120\sqrt{3}r^7z^6 + 192\sqrt{3}r^6z^8 \\
N_{88}(r, z) = & + 232\sqrt{3}r^6z^2 + 240\sqrt{3}r^5z^6
\end{aligned} \tag{793}$$

$$D_{88}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \tag{794}$$

$$R_{89}(r, z) = \frac{N_{89}(r, z)}{D_{89}(r, z)}. \tag{795}$$

$$\begin{aligned}
& 24\sqrt{3} - 24\sqrt{3}z^4 + 72\sqrt{3}r^4 + 96\sqrt{3}r^2 \\
N_{89}(r, z) = & -96\sqrt{3}r^2z^4 - 72\sqrt{3}r^4z^4
\end{aligned} \tag{796}$$

$$D_{89}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \tag{797}$$

$$R_{90}(r, z) = \frac{N_{90}(r, z)}{D_{90}(r, z)}. \tag{798}$$

$$\begin{aligned}
& -216ir^4 - 216ir^6 - 72ir^2 - 72ir^8 \\
& - 288ir^4z^6 - 288ir^6z^6 - 168\sqrt{3}ir^6 \\
& - 120\sqrt{3}ir^4 - 72\sqrt{3}ir^8 - 24\sqrt{3}ir^2 \\
& + 48\sqrt{3}ir + 72ir^2z^4 + 72ir^8z^4 \\
& + 96\sqrt{3}ir^7 + 192\sqrt{3}ir^3 + 216ir^4z^4 \\
& + 216ir^6z^4 + 240\sqrt{3}ir^5 + 288ir^4z^2 \\
& + 288ir^6z^2 - 288\sqrt{3}ir^7z^2 \\
& - 288\sqrt{3}ir^4z^6 - 240\sqrt{3}ir^5z^4 \\
& - 192\sqrt{3}ir^5z^2 - 192\sqrt{3}ir^3z^4 \\
& - 96\sqrt{3}ir^3z^2 - 96\sqrt{3}ir^7z^4 \\
& - 96\sqrt{3}ir^2z^6 - 48\sqrt{3}irz^4 \\
& + 24\sqrt{3}ir^2z^4 + 72\sqrt{3}ir^8z^4 \\
& + 96\sqrt{3}ir^2z^2 + 96\sqrt{3}ir^3z^6 \\
& + 120\sqrt{3}ir^4z^4 + 168\sqrt{3}ir^6z^4 \\
& + 192\sqrt{3}ir^5z^6 + 288\sqrt{3}ir^4z^2 \\
N_{90}(r, z) = & + 288\sqrt{3}ir^7z^6
\end{aligned} \tag{799}$$

$$D_{90}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (800)$$

$$R_{91}(r, z) = \frac{N_{91}(r, z)}{D_{91}(r, z)}. \quad (801)$$

$$\begin{aligned} & -144r^5 - 72r^3 - 72r^7 + 72r^4 + 72r^8 \\ & +144r^6 - 288r^4z^2 - 288r^7z^6 \\ & -144r^6z^4 - 72r^4z^4 - 72r^8z^4 \\ & -48\sqrt{3}r^5 - 24\sqrt{3}r^3 - 24\sqrt{3}r^7 \\ & +24\sqrt{3}r^2 + 24\sqrt{3}r^6 + 48\sqrt{3}r^4 \\ & +72r^3z^4 + 72r^7z^4 + 144r^5z^4 \\ & +288r^7z^2 + 288r^4z^6 - 96\sqrt{3}r^6z^2 \\ & -96\sqrt{3}r^3z^6 - 48\sqrt{3}r^4z^4 \\ & -24\sqrt{3}r^2z^4 - 24\sqrt{3}r^6z^4 \\ & +24\sqrt{3}r^3z^4 + 24\sqrt{3}r^7z^4 \\ & +48\sqrt{3}r^5z^4 + 96\sqrt{3}r^3z^2 \\ N_{91}(r, z) = & +96\sqrt{3}r^6z^6 \end{aligned} \quad (802)$$

$$D_{91}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (803)$$

$$R_{92}(r, z) = \frac{N_{92}(r, z)}{D_{92}(r, z)}. \quad (804)$$

$$\begin{aligned} & 32\sqrt{3}ir^2 + 32\sqrt{3}ir^3 + 32\sqrt{3}ir^4 \\ & +32\sqrt{3}ir^5 - 96\sqrt{3}ir^2z^4 \\ & -96\sqrt{3}ir^3z^4 - 96\sqrt{3}ir^4z^4 \\ & -96\sqrt{3}ir^5z^4 + 64\sqrt{3}ir^2z^2 \\ & +64\sqrt{3}ir^3z^2 + 64\sqrt{3}ir^4z^2 \\ N_{92}(r, z) = & +64\sqrt{3}ir^5z^2 \end{aligned} \quad (805)$$

$$D_{92}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (806)$$

$$R_{93}(r, z) = \frac{N_{93}(r, z)}{D_{93}(r, z)}. \quad (807)$$

$$\begin{aligned} & 48 - 96r^4 - 96r^5 - 48z^2 - 48r^2 - 48r^3 \\ & +16\sqrt{3} + 48r - 192r^4z^2 - 192r^5z^2 \\ & -144r^2z^2 - 144r^3z^2 - 48rz^2 \\ & -16\sqrt{3}z^2 + 48\sqrt{3}r + 48\sqrt{3}r^3 + 48\sqrt{3}r^6 \\ & +128\sqrt{3}r^2 + 160\sqrt{3}r^4 + 192r^2z^4 \\ & +192r^3z^4 + 288r^4z^4 + 288r^5z^4 \\ & -256\sqrt{3}r^4z^2 - 192\sqrt{3}r^2z^2 \\ & -96\sqrt{3}r^3z^4 - 48\sqrt{3}rz^2 - 48\sqrt{3}r^6z^2 \\ & +48\sqrt{3}r^3z^2 + 64\sqrt{3}r^2z^4 \\ N_{93}(r, z) = & +96\sqrt{3}r^4z^4 \end{aligned} \quad (808)$$

$$D_{93}(r, z) = 3 \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (809)$$

$$R_{94}(r, z) = 32\sqrt{3}i (r + 1) (z - 1) (z + 1) \quad (810)$$

$$R_{95}(r, z) = -32 (r + 1) (z - 1) (z + 1) + \frac{16\sqrt{3} (3r^2 + 1) (z - 1) (z + 1)}{3} \quad (811)$$

$$R_{96}(r, z) = \frac{N_{96}(r, z)}{D_{96}(r, z)}. \quad (812)$$

$$\begin{aligned}
& -32\sqrt{3}i - 32\sqrt{3}ir - 32\sqrt{3}ir^2 - 32\sqrt{3}ir^3 \\
& + 32\sqrt{3}iz^2 - 224\sqrt{3}ir^2z^4 \\
& - 224\sqrt{3}ir^3z^4 - 96\sqrt{3}ir^4z^4 \\
& - 96\sqrt{3}ir^5z^4 + 32\sqrt{3}irz^2 \\
& + 96\sqrt{3}ir^4z^2 + 96\sqrt{3}ir^5z^2 \\
N_{96}(r, z) = & + 256\sqrt{3}ir^2z^2 + 256\sqrt{3}ir^3z^2
\end{aligned} \tag{813}$$

$$D_{96}(r, z) = (-r^2 + 2rz - 1)(r^2 + 2rz + 1) \tag{814}$$

$$R_{97}(r, z) = \frac{N_{97}(r, z)}{D_{97}(r, z)}. \tag{815}$$

$$\begin{aligned}
& 48 - 48z^2 + 48r + 48r^2 + 48r^3 - 240r^2z^2 \\
& - 240r^3z^2 - 96r^4z^2 - 96r^5z^2 \\
& - 48rz^2 + 16\sqrt{3}r + 16\sqrt{3}r^2 + 16\sqrt{3}r^3 \\
& + 16\sqrt{3}r^4 + 96r^4z^4 + 96r^5z^4 \\
& + 192r^2z^4 + 192r^3z^4 - 32\sqrt{3}r^3z^4 \\
& - 32\sqrt{3}r^4z^4 - 16\sqrt{3}rz^2 - 16\sqrt{3}r^2z^2 \\
N_{97}(r, z) = & + 16\sqrt{3}r^3z^2 + 16\sqrt{3}r^4z^2
\end{aligned} \tag{816}$$

$$D_{97}(r, z) = (-r^2 + 2rz - 1)(r^2 + 2rz + 1) \tag{817}$$

$$R_{98}(r, z) = \frac{N_{98}(r, z)}{D_{98}(r, z)}. \tag{818}$$

$$\begin{aligned}
& -24r - 24r^3 + 36z^3 + 100z - 520r^2z^3 \\
& + 24rz^4 + 24r^3z^4 + 36r^4z^3 + 100r^4z \\
N_{98}(r, z) = & + 248r^2z
\end{aligned} \tag{819}$$

$$D_{98}(r, z) = (-r^2 + 2rz - 1)(r^2 + 2rz + 1) \tag{820}$$

$$R_{99}(r, z) = \frac{N_{99}(r, z)}{D_{99}(r, z)}. \tag{821}$$

$$\begin{aligned}
& -48ir^3 - 24ir - 24ir^5 - 12iz^3 + 20iz \\
& + 24ir^2 + 24ir^6 + 48ir^4 - 728ir^3z^4 \\
& - 580ir^3z^3 - 376ir^4z^2 - 276ir^4z \\
& - 272ir^2z^2 - 200ir^6z^2 - 196ir^6z \\
& - 144ir^4z^6 - 120ir^5z^4 - 116ir^2z^3 \\
& - 60ir^2z - 60irz^3 - 48irz^4 \\
& - 32\sqrt{3}ir^4 - 24ir^4z^5 - 24ir^5z^5 \\
& - 20ir^7z - 16\sqrt{3}ir^2 - 16\sqrt{3}ir^6 \\
& - 12\sqrt{3}iz^3 + 4\sqrt{3}iz + 12ir^7z^3 \\
& + 16\sqrt{3}ir + 16\sqrt{3}ir^5 + 24ir^2z^5 \\
& + 24ir^3z^5 + 32\sqrt{3}ir^3 + 48ir^6z^4 \\
& + 60ir^5z + 60ir^6z^3 + 116ir^5z^3 \\
& + 120ir^2z^4 + 144ir^3z^6 + 196irz \\
& + 200irz^2 + 272ir^5z^2 + 276ir^3z \\
& + 376ir^3z^2 + 580ir^4z^3 + 728ir^4z^4 \\
& - 108\sqrt{3}ir^5z^3 - 96\sqrt{3}ir^3z^4 \\
& - 76\sqrt{3}ir^3z^3 - 68\sqrt{3}ir^4z \\
& - 60\sqrt{3}ir^2z - 48\sqrt{3}ir^6z^4 \\
& - 48\sqrt{3}ir^2z^5 - 40\sqrt{3}irz^2 \\
& - 16\sqrt{3}ir^4z^2 - 16\sqrt{3}ir^2z^4 \\
& - 16\sqrt{3}ir^3z^5 - 12\sqrt{3}ir^6z^3 \\
& - 8\sqrt{3}ir^5z^2 - 4\sqrt{3}ir^6z - 4\sqrt{3}ir^7z \\
& + 4\sqrt{3}irz + 8\sqrt{3}ir^2z^2 + 12\sqrt{3}irz^3 \\
& + 12\sqrt{3}ir^7z^3 + 16\sqrt{3}ir^3z^2 \\
& + 16\sqrt{3}ir^5z^4 + 16\sqrt{3}ir^4z^5 \\
& + 40\sqrt{3}ir^6z^2 + 48\sqrt{3}irz^4 \\
& + 48\sqrt{3}ir^5z^5 + 60\sqrt{3}ir^5z \\
& + 68\sqrt{3}ir^3z + 76\sqrt{3}ir^4z^3 \\
& + 96\sqrt{3}ir^4z^4 + 108\sqrt{3}ir^2z^3
\end{aligned} \tag{822}$$

$$D_{99}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{823}$$

$$R_{100}(r, z) = \frac{N_{100}(r, z)}{D_{100}(r, z)}. \tag{824}$$

$$\begin{aligned}
& 8r + 8r^7 + 24r^2 + 24r^3 + 24r^5 + 24r^6 \\
& + 48r^4 - 96r^2z^2 - 96r^4z^2 - 96r^6z^2 \\
& - 96r^3z^5 - 96r^5z^5 - 60r^3z - 60r^5z \\
& - 56r^3z^2 - 56r^5z^2 - 48r^3z^6 \\
& - 48r^5z^6 - 32r^4z - 16r^2z - 16r^6z \\
& - 8r^2z^2 - 8r^7z^2 - 8r^2z^3 - 8r^6z^3 \\
& + 8\sqrt{3}r + 8\sqrt{3}r^7 + 12rz + 12r^7z + 12rz^4 \\
& + 12r^7z^4 + 24\sqrt{3}r^3 + 24\sqrt{3}r^5 \\
& + 32r^4z^5 + 48r^4z^3 + 48r^2z^4 \\
& + 48r^6z^4 + 68r^3z^4 + 68r^5z^4 \\
& + 96r^4z^4 + 144r^3z^3 + 144r^5z^3 \\
& - 280\sqrt{3}r^4z^4 - 160\sqrt{3}r^4z^3 \\
& - 96\sqrt{3}r^3z^2 - 96\sqrt{3}r^5z^2 \\
& - 48\sqrt{3}r^3z^6 - 48\sqrt{3}r^5z^6 \\
& - 40\sqrt{3}r^3z^5 - 40\sqrt{3}r^5z^5 - 36\sqrt{3}r^3z \\
& - 36\sqrt{3}r^5z - 28\sqrt{3}r^2z^4 - 28\sqrt{3}r^6z^4 \\
& - 24\sqrt{3}rz^2 - 24\sqrt{3}r^7z^2 - 12\sqrt{3}rz \\
& - 12\sqrt{3}r^7z - 8\sqrt{3}r^2z^3 - 8\sqrt{3}r^6z^3 \\
& + 8\sqrt{3}rz^3 + 8\sqrt{3}r^7z^3 + 12\sqrt{3}rz^4 \\
& + 12\sqrt{3}r^7z^4 + 16\sqrt{3}r^4z^5 + 40\sqrt{3}r^2z \\
& + 40\sqrt{3}r^6z + 60\sqrt{3}r^2z^2 + 60\sqrt{3}r^6z^2 \\
& + 80\sqrt{3}r^4z + 80\sqrt{3}r^3z^3 + 80\sqrt{3}r^5z^3 \\
& + 96\sqrt{3}r^4z^6 + 120\sqrt{3}r^4z^2 \\
N_{100}(r, z) = & + 124\sqrt{3}r^3z^4 + 124\sqrt{3}r^5z^4
\end{aligned} \tag{825}$$

$$D_{100}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{826}$$

$$R_{101}(r, z) = \frac{N_{101}(r, z)}{D_{101}(r, z)}. \tag{827}$$

$$\begin{aligned}
& -16ir^3z^3 - 12ir^3z^4 - 8ir^2z \\
& -4ir^2z^2 + 4ir^3z^2 + 8ir^3z \\
& + 12ir^2z^4 + 16ir^2z^3 - 8\sqrt{3}ir^3z^3 \\
& - 4\sqrt{3}ir^3z^2 - 4\sqrt{3}ir^3z^4 \\
& + 4\sqrt{3}ir^2z^2 + 4\sqrt{3}ir^2z^4 \\
N_{101}(r, z) = & + 8\sqrt{3}ir^2z^3
\end{aligned} \tag{828}$$

$$D_{101}(r, z) = -r^2 + 2rz - 1 \tag{829}$$

$$R_{102}(r, z) = -\frac{8z(3r^2z - 18rz^2 - 8r + 3z)}{r} \tag{830}$$

$$R_{103}(r, z) = -8iz(r-1)(3z+5) - 8\sqrt{3}iz(r-1)(6z^2+z+1) \tag{831}$$

$$R_{104}(r, z) = 16\sqrt{3}rz(z+1) + 8r(3z+1)(2z^2-z+1) \tag{832}$$

$$R_{105}(r, z) = \frac{N_{105}(r, z)}{D_{105}(r, z)}. \tag{833}$$

$$\begin{aligned}
& -72z^2 - 32r^2 - 32r^4 - 432r^3z^5 \\
& - 184r^2z^2 - 184r^4z^2 - 168r^3z - 116rz \\
& - 116r^5z - 72r^6z^2 + 12rz^3 + 12r^5z^3 \\
N_{105}(r, z) = & + 288r^2z^4 + 288r^4z^4 + 808r^3z^3
\end{aligned} \tag{834}$$

$$D_{105}(r, z) = r \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \tag{835}$$

$$R_{106}(r, z) = \frac{N_{106}(r, z)}{D_{106}(r, z)}. \tag{836}$$

$$\begin{aligned}
& -84iz^3 - 72iz^2 - 32ir^3 - 16ir - 16ir^5 \\
& +16ir^2 + 16ir^6 + 20iz + 32ir^4 \\
& -432ir^5z^4 - 324ir^4z^3 - 264ir^2z^2 \\
& -224ir^4z^2 - 216ir^3z^5 - 184ir^3z^4 \\
& -168ir^5z^5 - 144ir^4z^6 - 120irz^4 \\
& -96ir^6z^2 - 84irz^3 - 60ir^2z^3 \\
& -52ir^5z - 52ir^6z - 36\sqrt{3}iz^3 \\
& -32\sqrt{3}ir^4 - 24\sqrt{3}iz^2 - 20ir^4z \\
& -20ir^7z - 16\sqrt{3}ir^2 - 16\sqrt{3}ir^6 \\
& +16\sqrt{3}ir + 16\sqrt{3}ir^5 + 20ir^3z \\
& +32\sqrt{3}ir^3 + 52irz + 52\sqrt{3}iz + 52ir^2z \\
& +60ir^5z^3 + 72ir^7z^2 + 84ir^6z^3 \\
& +84ir^7z^3 + 96irz^2 + 120ir^6z^4 \\
& +144ir^3z^6 + 168ir^2z^5 + 184ir^4z^4 \\
& +216ir^4z^5 + 224ir^3z^2 + 264ir^5z^2 \\
& +324ir^3z^3 + 432ir^2z^4 \\
& -400\sqrt{3}ir^4z^4 - 348\sqrt{3}ir^4z^3 \\
& -336\sqrt{3}ir^3z^5 - 288\sqrt{3}ir^3z^6 \\
& -268\sqrt{3}ir^2z^3 - 144\sqrt{3}ir^5z^5 \\
& -112\sqrt{3}irz^2 - 88\sqrt{3}ir^5z^4 \\
& -76\sqrt{3}ir^5z - 72\sqrt{3}ir^3z^2 \\
& -72\sqrt{3}ir^6z^4 - 60\sqrt{3}ir^3z \\
& -52\sqrt{3}ir^7z - 52\sqrt{3}ir^6z^3 - 36\sqrt{3}irz \\
& +24\sqrt{3}ir^7z^2 + 36\sqrt{3}ir^6z \\
& +36\sqrt{3}ir^7z^3 + 52\sqrt{3}irz^3 \\
& +60\sqrt{3}ir^4z + 72\sqrt{3}irz^4 \\
& +72\sqrt{3}ir^4z^2 + 76\sqrt{3}ir^2z \\
& +88\sqrt{3}ir^2z^4 + 112\sqrt{3}ir^6z^2 \\
& +144\sqrt{3}ir^2z^5 + 268\sqrt{3}ir^5z^3 \\
& +288\sqrt{3}ir^4z^6 + 336\sqrt{3}ir^4z^5 \\
& N_{106}(r, z) = +348\sqrt{3}ir^3z^3 + 400\sqrt{3}ir^3z^4
\end{aligned} \tag{837}$$

$$D_{106}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{838}$$

$$R_{107}(r, z) = \frac{N_{107}(r, z)}{D_{107}(r, z)}. \tag{839}$$

$$\begin{aligned}
& 32r^2 + 32r^6 + 64r^4 - 288r^4z^6 \\
& -160r^4z^2 - 144r^2z^2 - 144r^6z^2 \\
& -120r^3z^3 - 120r^5z^3 - 48r^3z^6 \\
& -48r^5z^6 - 24rz^3 - 24r^3z^2 \\
& -24r^5z^2 - 24r^7z^3 - 16r^4z^3 \\
& -8rz^2 - 8r^7z^2 - 8r^2z^3 - 8r^6z^3 \\
& +4r^3z + 4r^5z + 8\sqrt{3}r + 8\sqrt{3}r^7 \\
& +12r^4 + 12r^7z^4 + 24\sqrt{3}r^3 + 24\sqrt{3}r^5 \\
& +32r^4z^5 + 44rz + 44r^7z + 68r^3z^4 \\
& +68r^5z^4 + 72r^2z^4 + 72r^6z^4 \\
& +96r^3z^5 + 96r^5z^5 + 464r^4z^4 \\
& -144\sqrt{3}r^4z^5 - 136\sqrt{3}r^4z^4 \\
& -112\sqrt{3}r^3z^2 - 112\sqrt{3}r^5z^2 \\
& -52\sqrt{3}r^2z^4 - 52\sqrt{3}r^6z^4 \\
& -48\sqrt{3}r^3z^6 - 48\sqrt{3}r^5z^6 \\
& -40\sqrt{3}r^3z^3 - 40\sqrt{3}r^5z^3 - 24\sqrt{3}rz^2 \\
& -24\sqrt{3}r^7z^2 - 24\sqrt{3}r^2z^3 \\
& -24\sqrt{3}r^6z^3 - 16\sqrt{3}rz^3 - 16\sqrt{3}r^7z^3 \\
& +8\sqrt{3}r^2z + 8\sqrt{3}r^6z + 8\sqrt{3}r^3z^5 \\
& +8\sqrt{3}r^5z^5 + 12\sqrt{3}rz + 12\sqrt{3}r^7z \\
& +12\sqrt{3}rz^4 + 12\sqrt{3}r^7z^4 + 16\sqrt{3}r^4z \\
& +36\sqrt{3}r^3z + 36\sqrt{3}r^5z + 36\sqrt{3}r^2z^2 \\
& +36\sqrt{3}r^6z^2 + 72\sqrt{3}r^4z^2 \\
& +96\sqrt{3}r^4z^6 + 140\sqrt{3}r^3z^4 \\
N_{107}(r, z) = & +140\sqrt{3}r^5z^4 + 160\sqrt{3}r^4z^3
\end{aligned} \tag{840}$$

$$D_{107}(r, z) = (-r^2 + 2rz - 1)^2 (r^2 + 2rz + 1) \tag{841}$$

$$R_{108}(r, z) = \frac{N_{108}(r, z)}{D_{108}(r, z)}. \tag{842}$$

$$\begin{aligned}
& -12ir^3z^4 - 8ir^3z^3 - 4ir^2z^2 \\
& +4ir^3z^2 + 8ir^2z^3 + 12ir^2z^4 \\
& -8\sqrt{3}ir^3z^3 - 4\sqrt{3}ir^3z^2 \\
& -4\sqrt{3}ir^3z^4 + 4\sqrt{3}ir^2z^2 \\
N_{108}(r, z) = & +4\sqrt{3}ir^2z^4 + 8\sqrt{3}ir^2z^3
\end{aligned} \tag{843}$$

$$D_{108}(r, z) = -r^2 + 2rz - 1 \tag{844}$$

$$R_{109}(r, z) = \frac{N_{109}(r, z)}{D_{109}(r, z)}. \tag{845}$$

$$\begin{aligned}
& 8r^2 + 8r^4 + 48z^2 - 1296r^3z^3 \\
& -168r^2z^4 - 168r^4z^4 - 144r^3z^5 \\
& +48r^6z^2 + 112r^2z^2 + 112r^4z^2 \\
& +168rz^3 + 168r^5z^3 + 280rz + 280r^5z \\
N_{109}(r, z) = & +544r^3z
\end{aligned} \tag{846}$$

$$D_{109}(r, z) = r(-r^2 + 2rz - 1)(r^2 + 2rz + 1) \tag{847}$$

$$R_{110}(r, z) = \frac{N_{110}(r, z)}{D_{110}(r, z)}. \tag{848}$$

$$\begin{aligned}
& -40iz - 16ir^3 - 8ir - 8ir^5 + 8ir^2 \\
& + 8ir^6 + 16ir^4 + 48iz^2 + 72iz^3 \\
& - 968ir^3z^3 - 960ir^3z^4 - 384ir^4z^2 \\
& - 376ir^4z - 232ir^2z - 216ir^2z^4 \\
& - 208ir^6z^2 - 184ir^6z - 160ir^2z^2 \\
& - 144ir^2z^5 - 72ir^6z^3 - 72ir^7z^3 \\
& - 72ir^6z^4 - 56\sqrt{3}iz - 56ir^5z^3 \\
& - 48ir^7z^2 - 48ir^4z^5 - 24\sqrt{3}iz^3 \\
& + 16\sqrt{3}iz^2 + 40ir^7z + 48ir^3z^5 \\
& + 56ir^2z^3 + 72irz^3 + 72irz^4 \\
& + 144ir^5z^5 + 160ir^5z^2 + 184irz \\
& + 208irz^2 + 216ir^5z^4 + 232ir^5z \\
& + 376ir^3z + 384ir^3z^2 + 960ir^4z^4 \\
& + 968ir^4z^3 - 400\sqrt{3}ir^3z^4 \\
& - 280\sqrt{3}ir^3z^3 - 248\sqrt{3}ir^5z^3 \\
& - 168\sqrt{3}ir^2z^4 - 160\sqrt{3}ir^2z \\
& - 152\sqrt{3}ir^4z - 144\sqrt{3}ir^4z^2 \\
& - 96\sqrt{3}ir^6z^2 - 96\sqrt{3}ir^3z^6 \\
& - 72\sqrt{3}ir^6z^4 - 64\sqrt{3}ir^4z^5 \\
& - 48\sqrt{3}ir^6z - 32\sqrt{3}ir^2z^2 \\
& - 24\sqrt{3}ir^6z^3 - 16\sqrt{3}ir^7z^2 \\
& + 24\sqrt{3}irz^3 + 24\sqrt{3}ir^7z^3 \\
& + 32\sqrt{3}ir^5z^2 + 48\sqrt{3}irz + 56\sqrt{3}ir^7z \\
& + 64\sqrt{3}ir^3z^5 + 72\sqrt{3}irz^4 \\
& + 96\sqrt{3}irz^2 + 96\sqrt{3}ir^4z^6 \\
& + 144\sqrt{3}ir^3z^2 + 152\sqrt{3}ir^3z \\
& + 160\sqrt{3}ir^5z + 168\sqrt{3}ir^5z^4 \\
& + 248\sqrt{3}ir^2z^3 + 280\sqrt{3}ir^4z^3 \\
& N_{110}(r, z) = +400\sqrt{3}ir^4z^4
\end{aligned} \tag{849}$$

$$D_{110}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{850}$$

$$R_{111}(r, z) = \frac{N_{111}(r, z)}{D_{111}(r, z)}. \tag{851}$$

$$\begin{aligned}
& -16r^4 - 8r^2 - 8r^6 - 304r^4z^4 \\
& -112r^3z - 112r^5z - 96r^4z^6 - 48rz \\
& -48r^7z - 32r^4z^3 - 32r^3z^4 \\
& -32r^5z^4 - 24rz^3 - 24r^7z^3 \\
& -16r^2z^3 - 16r^6z^3 + 8rz^2 + 8r^7z^2 \\
& +24r^3z^2 + 24r^5z^2 + 64r^4z^5 \\
& +72r^2z^4 + 72r^6z^4 + 80r^2z^2 \\
& +80r^6z^2 + 128r^4z^2 + 184r^3z^3 \\
& +184r^5z^3 - 272\sqrt{3}r^4z^4 \\
& -256\sqrt{3}r^4z^3 - 120\sqrt{3}r^3z - 120\sqrt{3}r^5z \\
& -48\sqrt{3}r^3z^5 - 48\sqrt{3}r^5z^5 - 40\sqrt{3}rz \\
& -40\sqrt{3}r^7z - 32\sqrt{3}r^3z^2 - 32\sqrt{3}r^5z^2 \\
& -16\sqrt{3}rz^2 - 16\sqrt{3}r^7z^2 + 24\sqrt{3}rz^3 \\
& +24\sqrt{3}r^7z^3 + 24\sqrt{3}r^2z^4 \\
& +24\sqrt{3}r^6z^4 + 32\sqrt{3}r^2z + 32\sqrt{3}r^6z \\
& +32\sqrt{3}r^4z^5 + 48\sqrt{3}r^2z^3 \\
& +48\sqrt{3}r^6z^3 + 48\sqrt{3}r^3z^4 \\
& +48\sqrt{3}r^5z^4 + 56\sqrt{3}r^2z^2 \\
& +56\sqrt{3}r^6z^2 + 64\sqrt{3}r^4z \\
& +112\sqrt{3}r^4z^2 + 184\sqrt{3}r^3z^3 \\
N_{111}(r, z) = & +184\sqrt{3}r^5z^3
\end{aligned} \tag{852}$$

$$D_{111}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{853}$$

$$R_{112}(r, z) = \frac{8ir^2z(r-1)(z-1)(z+1)}{r^2 - 2rz + 1} \tag{854}$$

$$R_{113}(r, z) = -\frac{4\sqrt{3}z(z-1)(z+1)(3r^2 + 6rz - 1)}{r^2 - 2rz + 1} \tag{855}$$

$$R_{114}(r, z) = \frac{N_{114}(r, z)}{D_{114}(r, z)}. \tag{856}$$

$$\begin{aligned}
& -12iz + 12iz^3 - 144ir^3z^4 - 60ir^4z^3 \\
& -48ir^2z^5 - 24irz^4 - 24ir^5z^4 \\
& -12ir^2z - 12ir^6z^3 - 4\sqrt{3}iz \\
& +4\sqrt{3}iz^3 + 12ir^4z + 12ir^6z + 24irz^2 \\
& +24ir^5z^2 + 32\sqrt{3}ir + 32\sqrt{3}ir^2 \\
& +32\sqrt{3}ir^5 + 32\sqrt{3}ir^6 + 48ir^3z^2 \\
& +48ir^4z^5 + 60ir^2z^3 + 64\sqrt{3}ir^3 \\
& +64\sqrt{3}ir^4 + 96ir^3z^6 - 480\sqrt{3}ir^4z^2 \\
& -384\sqrt{3}ir^3z^2 - 384\sqrt{3}ir^6z^2 \\
& -376\sqrt{3}ir^4z^5 - 336\sqrt{3}ir^5z^2 \\
& -328\sqrt{3}ir^5z^5 - 208\sqrt{3}ir^4z^6 \\
& -112\sqrt{3}ir^2z^2 - 88\sqrt{3}ir^2z^5 \\
& -64\sqrt{3}irz^2 - 56\sqrt{3}ir^7z^3 \\
& -44\sqrt{3}ir^6z^3 - 40\sqrt{3}ir^3z^5 \\
& -20\sqrt{3}ir^2z - 16\sqrt{3}ir^3z^6 \\
& -8\sqrt{3}irz^3 + 8\sqrt{3}irz + 16\sqrt{3}ir^3z \\
& +24\sqrt{3}ir^3z^3 + 28\sqrt{3}ir^4z \\
& +32\sqrt{3}ir^4z + 44\sqrt{3}ir^6z + 56\sqrt{3}ir^7z \\
& +64\sqrt{3}ir^5z + 80\sqrt{3}ir^2z^4 \\
& +108\sqrt{3}ir^2z^3 + 264\sqrt{3}ir^5z^3 \\
& +304\sqrt{3}ir^5z^4 + 336\sqrt{3}ir^3z^4 \\
& +348\sqrt{3}ir^4z^3 + 352\sqrt{3}ir^6z^4 \\
& N_{114}(r, z) = +624\sqrt{3}ir^4z^4
\end{aligned} \tag{857}$$

$$D_{114}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{858}$$

$$R_{115}(r, z) = \frac{N_{115}(r, z)}{D_{115}(r, z)}. \tag{859}$$

$$\begin{aligned}
& -288r^3 - 288r^4 - 144r - 144r^2 - 144r^5 \\
& -144r^6 - 48z^3 + 48z - 1764r^4z^4 \\
& -1164r^4z^3 - 1056r^6z^4 - 912r^5z^4 \\
& -900r^3z^4 - 876r^5z^3 - 408r^2z^3 \\
& -240\sqrt{3}r^5 - 192\sqrt{3}r^3 - 168r^7z \\
& -156r^2z^4 - 132r^6z - 108r^5z - 96\sqrt{3}r^4 \\
& -96\sqrt{3}r^7 - 96r^3z^5 - 48\sqrt{3}r - 48\sqrt{3}r^2 \\
& -48\sqrt{3}r^6 - 24rz - 16\sqrt{3}z^3 - 12rz^4 \\
& +16\sqrt{3}z + 24rz^3 + 36r^3z + 36r^4z \\
& +48r^3z^6 + 60r^3z^3 + 132r^6z^3 \\
& +156r^2z^2 + 168r^7z^3 + 192r^2z^5 \\
& +216r^2z + 300r^2z^2 + 624r^4z^6 \\
& +984r^5z^5 + 1056r^5z^2 + 1128r^4z^5 \\
& +1140r^3z^2 + 1200r^6z^2 + 1428r^4z^2 \\
& -916\sqrt{3}r^4z^3 - 288\sqrt{3}r^5z^5 \\
& -288\sqrt{3}r^5z^6 - 268\sqrt{3}r^2z^3 \\
& -256\sqrt{3}r^6z^3 - 176\sqrt{3}r^3z^6 \\
& -120\sqrt{3}r^3z^5 - 108\sqrt{3}r^4z^4 \\
& -72\sqrt{3}r^7z - 36\sqrt{3}r^5z - 36\sqrt{3}r^6z^4 \\
& -24\sqrt{3}rz^3 + 4\sqrt{3}rz^2 + 24\sqrt{3}rz \\
& +24\sqrt{3}r^7z^2 + 44\sqrt{3}rz^4 + 48\sqrt{3}r^2z^4 \\
& +60\sqrt{3}r^3z + 60\sqrt{3}r^4z^2 + 60\sqrt{3}r^3z^3 \\
& +64\sqrt{3}r^2z^5 + 72\sqrt{3}r^7z^3 \\
& +72\sqrt{3}r^7z^4 + 84\sqrt{3}r^6z^2 \\
& +136\sqrt{3}r^5z^2 + 140\sqrt{3}r^3z^2 \\
& +144\sqrt{3}r^4z^6 + 204\sqrt{3}r^2z \\
& +228\sqrt{3}r^3z^4 + 256\sqrt{3}r^6z \\
& +324\sqrt{3}r^5z^3 + 392\sqrt{3}r^5z^4 \\
& +444\sqrt{3}r^4z + 472\sqrt{3}r^4z^5
\end{aligned} \tag{860}$$

$$D_{115}(r, z) = 3 \left(-r^2 + 2rz - 1 \right)^2 \left(r^2 + 2rz + 1 \right) \tag{861}$$

$$R_{116}(r, z) = \frac{N_{116}(r, z)}{D_{116}(r, z)}. \tag{862}$$

$$\begin{aligned}
& -24ir^3z^3 - 24ir^3z^4 - 12ir^2z \\
& -12ir^2z^2 + 12ir^2z^3 + 12ir^2z^4 \\
& +24ir^3z + 24ir^3z^2 - 4\sqrt{3}ir^2z \\
& -4\sqrt{3}ir^2z^2 + 4\sqrt{3}ir^2z^3 \\
& N_{116}(r, z) = +4\sqrt{3}ir^2z^4
\end{aligned} \tag{863}$$

$$D_{116}(r, z) = -r^2 + 2rz - 1 \tag{864}$$

$$R_{117}(r, z) = -32\sqrt{3}iz (r+1)(z-1)(z+1) \tag{865}$$

$$R_{118}(r, z) = 32z (r+1)(z-1)(z+1) + \frac{16\sqrt{3}(3r-z)(z-1)(z+1)}{3} \tag{866}$$

$$R_{119}(r, z) = \frac{4\sqrt{3}z(z-1)(z+1)(15r^2 - 6rz + 7)}{r^2 - 2rz + 1} \tag{867}$$

$$R_{120}(r, z) = \frac{N_{120}(r, z)}{D_{120}(r, z)}. \tag{868}$$

$$\begin{aligned}
& -12iz + 12iz^3 - 240ir^4z^5 - 144ir^3z^4 \\
& -132ir^4z - 84ir^2z - 60ir^6z \\
& -48ir^2z^5 - 36\sqrt{3}iz - 24irz^4 \\
& -24ir^5z^4 + 24irz^2 + 24ir^5z^2 \\
& +32\sqrt{3}ir + 32\sqrt{3}ir^2 + 32\sqrt{3}ir^5 \\
& +32\sqrt{3}ir^6 + 36\sqrt{3}iz^3 + 48ir^3z^2 \\
& +60ir^6z^3 + 64\sqrt{3}ir^3 + 64\sqrt{3}ir^4 \\
& +96ir^3z^6 + 132ir^2z^3 + 372ir^4z^3 \\
& -648\sqrt{3}ir^4z^5 - 392\sqrt{3}ir^6z^2 \\
& -344\sqrt{3}ir^5z^2 - 312\sqrt{3}ir^5z^5 \\
& -304\sqrt{3}ir^4z^2 - 264\sqrt{3}ir^2z^5 \\
& -208\sqrt{3}ir^3z^2 - 164\sqrt{3}ir^4z \\
& -140\sqrt{3}ir^2z - 96\sqrt{3}ir^3z^4 \\
& -60\sqrt{3}ir^6z - 56\sqrt{3}irz^4 \\
& -40\sqrt{3}ir^3z^3 - 24\sqrt{3}irz^3 \\
& -24\sqrt{3}ir^2z^2 - 24\sqrt{3}ir^7z^3 \\
& -24\sqrt{3}ir^3z^5 - 8\sqrt{3}ir^2z^4 \\
& +24\sqrt{3}irz + 24\sqrt{3}ir^7z + 24\sqrt{3}irz^2 \\
& +48\sqrt{3}ir^4z^6 + 60\sqrt{3}ir^6z^3 \\
& +64\sqrt{3}ir^3z + 64\sqrt{3}ir^5z \\
& +192\sqrt{3}ir^4z^4 + 240\sqrt{3}ir^3z^6 \\
& +248\sqrt{3}ir^5z^3 + 312\sqrt{3}ir^5z^4 \\
& +360\sqrt{3}ir^6z^4 + 404\sqrt{3}ir^2z^3 \\
N_{120}(r, z) = & +812\sqrt{3}ir^4z^3 \tag{869}
\end{aligned}$$

$$D_{120}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{870}$$

$$R_{121}(r, z) = \frac{N_{121}(r, z)}{D_{121}(r, z)}. \tag{871}$$

$$\begin{aligned}
& -96r^3 - 96r^4 - 48r - 48z^3 - 48r^2 - 48r^5 \\
& -48r^6 + 48z - 828r^4z^3 - 360r^2z^3 \\
& -360r^6z^4 - 312r^5z^4 - 300r^5z^3 \\
& -240r^3z^6 - 132r^3z^3 - 132r^4z^4 \\
& -60r^6z^3 - 48r^4z^6 - 32\sqrt{3}r^4 \\
& -32\sqrt{3}r^5 - 24r^7z - 24r^3z^3 - 16\sqrt{3}r^2 \\
& -16\sqrt{3}r^3 - 16\sqrt{3}r^6 - 16\sqrt{3}r^7 - 12r^5z \\
& -12r^2z^2 + 12r^2z^4 + 24rz + 24r^7z^3 \\
& +36r^3z + 36r^2z^2 + 60r^6z + 60r^4z^4 \\
& +96r^3z^5 + 156r^3z^4 + 168r^2z \\
& +180r^4z + 180r^3z^2 + 192r^2z^5 \\
& +276r^4z^2 + 312r^5z^5 + 360r^5z^2 \\
& +408r^6z^2 + 648r^4z^5 - 148\sqrt{3}r^4z^3 \\
& -96\sqrt{3}r^5z^5 - 96\sqrt{3}r^5z^6 \\
& -88\sqrt{3}r^3z^5 - 80\sqrt{3}r^5z^2 \\
& -48\sqrt{3}r^6z^3 - 44\sqrt{3}r^2z^3 - 36\sqrt{3}r^5z \\
& -24\sqrt{3}r^7z - 20\sqrt{3}r^3z^2 - 16\sqrt{3}r^3z^6 \\
& -12\sqrt{3}r^4z^4 - 12\sqrt{3}r^6z^4 - 8\sqrt{3}r^3z^3 \\
& -8\sqrt{3}r^7z^2 - 8\sqrt{3}r^2z^4 - 4\sqrt{3}r^3z \\
& -4\sqrt{3}r^2z^2 - 4\sqrt{3}r^4z^2 + 4\sqrt{3}r^4z^4 \\
& +8\sqrt{3}rz + 24\sqrt{3}r^2z^2 + 24\sqrt{3}r^7z^3 \\
& +24\sqrt{3}r^7z^4 + 28\sqrt{3}r^6z^2 + 44\sqrt{3}r^2z \\
& +48\sqrt{3}r^6z + 48\sqrt{3}r^4z^6 + 52\sqrt{3}r^3z^4 \\
& +56\sqrt{3}r^4z^5 + 92\sqrt{3}r^4z + 92\sqrt{3}r^3z^3 \\
& +132\sqrt{3}r^5z^3 + 208\sqrt{3}r^5z^4
\end{aligned}$$

$$N_{121}(r, z) = +132\sqrt{3}r^5z^3 + 208\sqrt{3}r^5z^4 \quad (872)$$

$$D_{121}(r, z) = (-r^2 + 2rz - 1)^2 (r^2 + 2rz + 1) \quad (873)$$

$$R_{122}(r, z) = -\frac{24\sqrt{3}z(3r^2 + 1)(z - 1)(z + 1)}{r^2 - 2rz + 1} \quad (874)$$

$$R_{123}(r, z) = \frac{N_{123}(r, z)}{D_{123}(r, z)}. \quad (875)$$

$$\begin{aligned}
& -144ir^3z^2 - 72ir^2z^3 - 72ir^4z^3 \\
& +72ir^2z + 72ir^4z + 144ir^3z^4 \\
& -96\sqrt{3}ir^3z - 72\sqrt{3}ir^3z^2 \\
& -72\sqrt{3}ir^4z^3 - 72\sqrt{3}ir^2z^4 \\
& -72\sqrt{3}ir^4z^4 - 48\sqrt{3}irz - 24\sqrt{3}irz^2 \\
& -24\sqrt{3}ir^2z^3 + 24\sqrt{3}ir^2z \\
& +24\sqrt{3}irz^4 + 48\sqrt{3}irz^3 + 72\sqrt{3}ir^4z \\
& +72\sqrt{3}ir^2z^2 + 72\sqrt{3}ir^4z^2 \\
& +72\sqrt{3}ir^3z^4 + 96\sqrt{3}ir^3z^3
\end{aligned}$$

$$N_{123}(r, z) = +72\sqrt{3}ir^3z^4 + 96\sqrt{3}ir^3z^3 \quad (876)$$

$$D_{123}(r, z) = (-r^2 + 2rz - 1)^2 \quad (877)$$

$$R_{124}(r, z) = \frac{N_{124}(r, z)}{D_{124}(r, z)}. \quad (878)$$

$$\begin{aligned}
& -72r^4z - 72r^4z^2 - 72r^3z^3 \\
& -72r^3z^4 + 72r^3z + 72r^3z^2 \\
& +72r^4z^3 + 72r^4z^4 - 24\sqrt{3}r^2z \\
& -24\sqrt{3}r^2z^2 - 24\sqrt{3}r^3z^3 \\
& -24\sqrt{3}r^3z^4 + 24\sqrt{3}r^3z + 24\sqrt{3}r^3z^2 \\
& +24\sqrt{3}r^2z^3 + 24\sqrt{3}r^2z^4
\end{aligned}$$

$$N_{124}(r, z) = +24\sqrt{3}r^2z^3 + 24\sqrt{3}r^2z^4 \quad (879)$$

$$D_{124}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \quad (880)$$

$$R_{125}(r, z) = 12z^4 + 24z^2 + 4 \quad (881)$$

$$R_{126}(r, z) = \frac{N_{126}(r, z)}{D_{126}(r, z)}. \quad (882)$$

$$\begin{aligned} & 40i - 40ir^5 - 32ir^3 - 8ir^4 + 8ir \\ & + 24iz^4 + 32ir^2 + 48iz^2 - 64ir^2z^2 \\ & - 48ir^4z^2 - 48ir^5z^2 - 24ir^4z^4 \\ & - 24ir^5z^4 + 24irz^4 + 48irz^2 \\ N_{126}(r, z) = & + 64ir^3z^2 \end{aligned} \quad (883)$$

$$D_{126}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (884)$$

$$R_{127}(r, z) = \frac{N_{127}(r, z)}{D_{127}(r, z)}. \quad (885)$$

$$\begin{aligned} & 96 - 528r^5 - 344r^3 - 344r^7 - 160z^2 - 80r \\ & - 80r^9 + 64z^4 + 96r^{10} + 444r^2 + 444r^8 \\ & + 852r^4 + 852r^6 - 4756r^4z^2 - 4756r^6z^2 \\ & - 3100r^4z^6 - 3100r^6z^6 - 1868r^2z^2 \\ & - 1868r^8z^2 - 1040r^5z^6 - 880r^5z^2 \\ & - 608r^5z^8 - 552r^3z^2 - 552r^7z^2 \\ & - 356r^2z^6 - 356r^8z^6 - 160rz^2 \\ & - 160r^9z^2 - 160r^{10}z^2 + 16rz^4 \\ & + 16r^9z^4 + 64r^{10}z^4 + 136r^3z^6 \\ & + 136r^7z^6 + 496r^4z^8 + 496r^6z^8 \\ & + 1080r^3z^4 + 1080r^7z^4 + 1492r^2z^4 \\ & + 1492r^8z^4 + 2864r^5z^4 + 6796r^4z^4 \\ N_{127}(r, z) = & + 6796r^6z^4 \end{aligned} \quad (886)$$

$$D_{127}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (887)$$

$$R_{128}(r, z) = \frac{N_{128}(r, z)}{D_{128}(r, z)}. \quad (888)$$

$$\begin{aligned} & -24ir^2 - 24ir^4 + 24ir^3 + 24ir^5 \\ & - 120ir^3z^4 - 72ir^2z^2 - 72ir^2z^4 \\ & - 72ir^3z^6 - 24ir^3z^2 - 24ir^2z^6 \\ & + 24ir^4z^2 + 24ir^5z^6 + 72ir^5z^2 \\ N_{128}(r, z) = & + 72ir^5z^4 + 72ir^4z^6 + 120ir^4z^4 \end{aligned} \quad (889)$$

$$D_{128}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (890)$$

$$R_{129}(r, z) = 16(z - 1)(z + 1) \left(r^2z^2 - 6r^2 - 2rz^2 + z^2 - 6\right) \quad (891)$$

$$R_{130}(r, z) = -60z^4 - 120z^2 - 52 \quad (892)$$

$$R_{131}(r, z) = \frac{N_{131}(r, z)}{D_{131}(r, z)}. \quad (893)$$

$$\begin{aligned} & 8i - 112ir^3 - 104ir - 8ir^5 + 24iz^4 \\ & + 48iz^2 + 104ir^4 + 112ir^2 - 432ir^2z^4 \\ & - 288ir^2z^6 - 240irz^2 - 120irz^4 \\ & - 64ir^3z^2 - 48ir^5z^2 - 24ir^5z^4 \\ & + 64ir^2z^2 + 120ir^4z^4 + 240ir^4z^2 \\ N_{131}(r, z) = & + 288ir^3z^6 + 432ir^3z^4 \end{aligned} \quad (894)$$

$$D_{131}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (895)$$

$$R_{132}(r, z) = \frac{N_{132}(r, z)}{D_{132}(r, z)}. \quad (896)$$

$$\begin{aligned} & -432r^5 - 232r^3 - 232r^7 - 48z^2 - 16r \\ & -16r^9 + 36r^2 + 36r^8 + 48z^4 + 108r^4 \\ & +108r^6 - 2928r^5z^6 - 2384r^5z^2 \\ & -1548r^4z^2 - 1548r^6z^2 - 1048r^3z^2 \\ & -1048r^7z^2 - 996r^4z^6 - 996r^6z^6 \\ & -672r^5z^8 - 324r^2z^2 - 324r^8z^2 \\ & -192rz^2 - 192r^9z^2 - 156r^2z^6 \\ & -156r^8z^6 - 48r^{10}z^2 + 48rz^4 \\ & +48r^9z^4 + 48r^{10}z^4 + 312r^3z^6 \\ & +312r^7z^6 + 528r^4z^8 + 528r^6z^8 \\ & +732r^2z^4 + 732r^8z^4 + 1620r^4z^4 \\ & +1620r^6z^4 + 2184r^3z^4 + 2184r^7z^4 \\ N_{132}(r, z) = & +4304r^5z^4 \end{aligned} \quad (897)$$

$$D_{132}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (898)$$

$$R_{133}(r, z) = 72z^4 + 144z^2 + 56 \quad (899)$$

$$R_{134}(r, z) = \frac{N_{134}(r, z)}{D_{134}(r, z)}. \quad (900)$$

$$\begin{aligned} & 32i - 112ir^4 - 80ir^2 - 32ir^5 + 80ir^3 \\ & +112ir - 432ir^3z^4 - 288ir^4z^2 \\ & -288ir^3z^6 - 144ir^4z^4 - 128ir^2z^2 \\ & +128ir^3z^2 + 144irz^4 + 288irz^2 \\ N_{134}(r, z) = & +288ir^2z^6 + 432ir^2z^4 \end{aligned} \quad (901)$$

$$D_{134}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (902)$$

$$R_{135}(r, z) = \frac{N_{135}(r, z)}{D_{135}(r, z)}. \quad (903)$$

$$\begin{aligned} & -216r^4 - 216r^6 - 112r^3 - 112r^7 - 96r^5 \\ & -72r^2 - 72r^8 - 64r - 64r^9 - 736r^5z^4 \\ & -720r^3z^4 - 720r^7z^4 - 432r^3z^6 \\ & -432r^7z^6 - 288r^4z^8 - 288r^6z^8 \\ & -216r^2z^2 - 216r^8z^2 - 216r^2z^4 \\ & -216r^8z^4 - 72r^2z^6 - 72r^8z^6 \\ & +72r^4z^6 + 72r^6z^6 + 216r^4z^2 \\ & +216r^6z^2 + 368r^3z^2 + 368r^7z^2 \\ & +576r^5z^8 + 792r^4z^4 + 792r^6z^4 \\ N_{135}(r, z) = & +864r^5z^6 + 1312r^5z^2 \end{aligned} \quad (904)$$

$$D_{135}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \quad (905)$$

$$R_{136}(r, z) = 8 \quad (906)$$

$$R_{137}(r, z) = -16i(r - 1) \quad (907)$$

$$R_{138}(r, z) = \frac{N_{138}(r, z)}{D_{138}(r, z)}. \quad (908)$$

$$\begin{aligned} & 32 - 96r - 96r^3 + 96z^2 + 128r^6 + 192r^2 \\ & +288r^4 - 800r^4z^2 - 416r^2z^2 \\ & -192r^3z^2 - 192r^4z^6 - 96r^6z^2 \\ & +96r^2z^4 + 96r^3z^4 + 96r^5z^4 \\ N_{138}(r, z) = & +96r^6z^4 + 192rz^2 + 576r^4z^4 \end{aligned} \quad (909)$$

$$D_{138}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (910)$$

$$R_{139}(r, z) = -48r^2z^4 + 48r^2z^2 - 128r^2 - 96rz^2 - 32r - 48z^2 - 80 \quad (911)$$

$$R_{140}(r, z) = -8 \quad (912)$$

$$R_{141}(r, z) = 16i(r - 1) \quad (913)$$

$$R_{142}(r, z) = \frac{N_{142}(r, z)}{D_{142}(r, z)}. \quad (914)$$

$$\begin{aligned} & -48 - 96r^2 - 64r - 48r^4 - 32r^3 + 32r^5 \\ & + 48z^2 - 512r^3z^2 - 240r^4z^2 \\ & - 144r^2z^2 - 96r^5z^2 - 48r^6z^2 \\ & + 48r^6z^4 + 96rz^2 + 96r^5z^4 \\ N_{142}(r, z) = & + 240r^2z^4 + 288r^4z^4 + 480r^3z^4 \end{aligned} \quad (915)$$

$$D_{142}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \quad (916)$$

$$R_{143}(r, z) = 16 \quad (917)$$

$$R_{144}(r, z) = -32i(r - 1) \quad (918)$$

$$R_{145}(r, z) = 64r \quad (919)$$

$$R_{146}(r, z) = -12z(z^2 + 1) \quad (920)$$

$$R_{147}(r, z) = -\frac{24iz(r - 1)(r + 1)^2(z^2 + 1)}{r^2 - 2rz + 1} \quad (921)$$

$$R_{148}(r, z) = \frac{N_{148}(r, z)}{D_{148}(r, z)}. \quad (922)$$

$$\begin{aligned} & -288r^3 - 288r^5 - 96r - 96r^7 + 32z + 32z^3 \\ & + 96r^2 + 96r^6 + 192r^4 - 1744r^4z^3 \\ & - 560r^3z^4 - 560r^5z^4 - 464r^3z^3 \\ & - 464r^5z^3 - 400r^4z^4 - 376r^2z^4 \\ & - 376r^6z^4 - 216r^2z^3 - 216r^6z^3 \\ & - 208rz^4 - 208r^7z^4 - 192r^4z^7 \\ & + 32r^8z + 32r^8z^3 + 72r^3z + 72r^5z \\ & + 80rz + 80r^7z + 80rz^2 + 80rz^3 \\ & + 80r^7z^2 + 80r^7z^3 + 104r^2z^2 \\ & + 104r^6z^2 + 196r^2z^5 + 196r^6z^5 \\ & + 232r^3z^5 + 232r^5z^5 + 232r^3z^6 \\ & + 232r^5z^6 + 256r^4z^2 + 304r^4z^6 \\ & + 356r^2z + 356r^6z + 488r^4z^5 + 648r^4z \\ N_{148}(r, z) = & + 840r^3z^2 + 840r^5z^2 \end{aligned} \quad (923)$$

$$D_{148}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \quad (924)$$

$$R_{149}(r, z) = \frac{24ir^2z(r - 1)(z + 1)^2(z^2 + 1)}{r^2 - 2rz + 1} \quad (925)$$

$$R_{150}(r, z) = 32r^2z^3 + 32r^2z - 48rz^4 + 32rz^3 + 32rz - 144r + 32z^3 + 32z \quad (926)$$

$$R_{151}(r, z) = 60z(z^2 + 1) \quad (927)$$

$$R_{152}(r, z) = -\frac{24iz(r - 1)(z^2 + 1)(r^2 - 6rz - 4r + 1)}{r^2 - 2rz + 1} \quad (928)$$

$$R_{153}(r, z) = \frac{N_{153}(r, z)}{D_{153}(r, z)}. \quad (929)$$

$$\begin{aligned}
& 48r + 48r^7 + 96r^2 + 96r^6 + 144r^3 + 144r^5 \\
& + 192r^4 - 480r^3z^4 - 480r^5z^4 \\
& - 456r^2z^4 - 456r^6z^4 - 432r^3z^3 \\
& - 432r^5z^3 - 360r^2z^3 - 360r^6z^3 \\
& - 360r^3z^6 - 360r^5z^6 - 264r^4z \\
& - 240r^4z^3 - 240r^4z^4 - 132r^2z \\
& - 132r^6z - 96rz^4 - 96r^7z^4 + 24r^2z^2 \\
& + 24r^6z^2 + 48rz + 48r^7z + 48rz^3 \\
& + 48r^7z^3 + 120r^3z + 120r^5z + 144rz^2 \\
& + 144r^7z^2 + 156r^2z^5 + 156r^6z^5 \\
& + 192r^4z^7 + 216r^3z^5 + 216r^5z^5 \\
& + 336r^4z^6 + 384r^4z^2 + 600r^3z^2 \\
N_{153}(r, z) = & + 600r^5z^2 + 984r^4z^5
\end{aligned} \tag{930}$$

$$D_{153}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right) \tag{931}$$

$$R_{154}(r, z) = -72z \left(z^2 + 1\right) \tag{932}$$

$$R_{155}(r, z) = -\frac{144irz(r-1)(z+1)(z^2+1)}{r^2 - 2rz + 1} \tag{933}$$

$$R_{156}(r, z) = \frac{72r^2z(r-1)^2(z+1)^2(z^2+1)}{(r^2 - 2rz + 1)^2} \tag{934}$$

$$R_{157}(r, z) = 48z^2 - 8 \tag{935}$$

$$R_{158}(r, z) = \frac{N_{158}(r, z)}{D_{158}(r, z)}. \tag{936}$$

$$\begin{aligned}
& 16i - 32ir^3 - 16ir - 16ir^5 + 16ir^4 \\
& + 32ir^2 + 96iz^2 - 96ir^4z^2 \\
& - 96ir^5z^2 - 64ir^2z^2 + 64ir^3z^2 \\
N_{158}(r, z) = & + 96irz^2
\end{aligned} \tag{937}$$

$$D_{158}(r, z) = \left(-r^2 + 2rz - 1\right) \left(r^2 + 2rz + 1\right) \tag{938}$$

$$R_{159}(r, z) = \frac{N_{159}(r, z)}{D_{159}(r, z)}. \tag{939}$$

$$\begin{aligned}
& 32 - 576r^5 - 384r^3 - 384r^7 - 96r - 96r^9 \\
& - 32z^2 + 32r^{10} + 160r^2 + 160r^8 + 320r^4 \\
& + 320r^6 - 2672r^4z^2 - 2672r^6z^2 \\
& - 2432r^5z^6 - 1088r^4z^6 - 1088r^6z^6 \\
& - 960r^5z^2 - 944r^2z^2 - 944r^8z^2 \\
& - 608r^3z^2 - 608r^7z^2 - 128rz^2 \\
& - 128r^9z^2 - 32r^{10}z^2 + 496r^2z^4 \\
& + 496r^8z^4 + 1312r^3z^4 + 1312r^7z^4 \\
N_{159}(r, z) = & + 3728r^4z^4 + 3728r^6z^4 + 3776r^5z^4
\end{aligned} \tag{940}$$

$$D_{159}(r, z) = \left(-r^2 + 2rz - 1\right)^2 \left(r^2 + 2rz + 1\right)^2 \tag{941}$$

$$R_{160}(r, z) = -\frac{96ir^2z^2(r-1)(r^2z^2 + r^2 + 4rz^2 + z^2 + 1)}{(r^2 - 2rz + 1)(r^2 + 2rz + 1)} \tag{942}$$

$$R_{161}(r, z) = -16(z-1)(z+1)(5r^2 + 2r + 5) \tag{943}$$

$$R_{162}(r, z) = -240z^2 + 8 \tag{944}$$

$$R_{163}(r, z) = \frac{N_{163}(r, z)}{D_{163}(r, z)}. \tag{945}$$

$$\begin{aligned}
& -16i - 32ir^2 - 16ir^4 + 16ir + 16ir^5 \\
& + 32ir^3 + 96iz^2 - 1152ir^2z^4 \\
& - 640ir^3z^2 - 480irz^2 - 96ir^5z^2 \\
N_{163}(r, z) = & + 480ir^4z^2 + 640ir^2z^2 + 1152ir^3z^4
\end{aligned} \tag{946}$$

$$D_{163}(r, z) = (-r^2 + 2rz - 1)(r^2 + 2rz + 1) \tag{947}$$

$$R_{164}(r, z) = \frac{N_{164}(r, z)}{D_{164}(r, z)}. \tag{948}$$

$$\begin{aligned}
& -48 - 480r^4 - 480r^6 - 384r^5 - 256r^3 \\
& - 256r^7 - 240r^2 - 240r^8 - 64r - 64r^9 \\
& - 48r^{10} + 48z^2 - 4224r^5z^6 - 2432r^5z^2 \\
& - 2064r^4z^4 - 2064r^6z^4 - 1312r^3z^2 \\
& - 1312r^7z^2 - 96rz^2 - 96r^9z^2 \\
& + 48r^{10}z^2 + 144r^2z^4 + 144r^8z^4 \\
& + 384r^2z^2 + 384r^8z^2 + 912r^4z^2 \\
& + 912r^6z^2 + 1344r^4z^6 + 1344r^6z^6 \\
N_{164}(r, z) = & + 2784r^3z^4 + 2784r^7z^4 + 4928r^5z^4
\end{aligned} \tag{949}$$

$$D_{164}(r, z) = (-r^2 + 2rz - 1)^2(r^2 + 2rz + 1)^2 \tag{950}$$

$$R_{165}(r, z) = 288z^2 - 16 \tag{951}$$

$$R_{166}(r, z) = \frac{N_{166}(r, z)}{D_{166}(r, z)}. \tag{952}$$

$$\begin{aligned}
& 32i - 64ir^3 - 32ir - 32ir^5 + 32ir^4 \\
& + 64ir^2 - 1152ir^3z^4 - 704ir^2z^2 \\
& - 576ir^4z^2 + 576irz^2 + 704ir^3z^2 \\
N_{166}(r, z) = & + 1152ir^2z^4
\end{aligned} \tag{953}$$

$$D_{166}(r, z) = (-r^2 + 2rz - 1)(r^2 + 2rz + 1) \tag{954}$$

$$R_{167}(r, z) = \frac{N_{167}(r, z)}{D_{167}(r, z)}. \tag{955}$$

$$\begin{aligned}
& -384r^5 - 256r^3 - 256r^7 - 64r - 64r^9 \\
& - 2176r^5z^4 - 1728r^3z^4 - 1728r^7z^4 \\
& - 1152r^4z^6 - 1152r^6z^6 - 864r^4z^2 \\
& - 864r^6z^2 - 288r^2z^2 - 288r^8z^2 \\
& - 288r^2z^4 - 288r^8z^4 + 1088r^3z^2 \\
& + 1088r^7z^2 + 2176r^5z^2 + 2304r^5z^6 \\
N_{167}(r, z) = & + 2592r^4z^4 + 2592r^6z^4
\end{aligned} \tag{956}$$

$$D_{167}(r, z) = (-r^2 + 2rz - 1)^2(r^2 + 2rz + 1)^2 \tag{957}$$

$$R_{168}(r, z) = 8\sqrt{3} \tag{958}$$

$$R_{169}(r, z) = 8i \tag{959}$$

$$R_{170}(r, z) = -\frac{8\sqrt{3}}{3} \tag{960}$$

$$R_{171}(r, z) = -\frac{16\sqrt{3}i}{3} \tag{961}$$

$$R_{172}(r, z) = \frac{4}{3} \tag{962}$$

$$R_{173}(r, z) = -8\sqrt{3} \tag{963}$$

$$R_{174}(r, z) = -8i \quad (964)$$

$$R_{175}(r, z) = 4i + 4\sqrt{3}i \quad (965)$$

$$R_{176}(r, z) = -16 \quad (966)$$

$$R_{177}(r, z) = -4\sqrt{3}i - 4i \quad (967)$$

$$R_{178}(r, z) = -16\sqrt{3} \quad (968)$$

$$R_{179}(r, z) = -4 \quad (969)$$

$$R_{180}(r, z) = 16\sqrt{3} \quad (970)$$

$$R_{181}(r, z) = 4 \quad (971)$$

$$R_{182}(r, z) = -\frac{4(3r^2z^2 + 7r^2 - 3rz^3 - 17rz + 3z^2 + 7)}{r^2 - 2rz + 1} \quad (972)$$

$$R_{183}(r, z) = \frac{N_{183}(r, z)}{D_{183}(r, z)}. \quad (973)$$

$$\begin{aligned} & -20\sqrt{3}ir^2 - 20\sqrt{3}ir^4 - 6\sqrt{3}iz \\ & + 20\sqrt{3}ir + 20\sqrt{3}ir^3 - 52\sqrt{3}ir^2z \\ & - 18\sqrt{3}ir^4z^2 - 12\sqrt{3}ir^2z^2 \\ & - 12\sqrt{3}ir^2z^3 + 6\sqrt{3}ir^5z \\ & + 12\sqrt{3}ir^3z^2 + 12\sqrt{3}ir^3z^3 \\ N_{183}(r, z) = & + 18\sqrt{3}irz^2 + 52\sqrt{3}ir^3z \end{aligned} \quad (974)$$

$$D_{183}(r, z) = 3r(-r^2 + 2rz - 1) \quad (975)$$

$$R_{184}(r, z) = 2r^2z + \frac{2r^2}{3} - 2rz^2 - \frac{2rz}{3} - 4r + 2z + \frac{2}{3} \quad (976)$$

$$R_{185}(r, z) = \frac{2\sqrt{3}ir(r-1)(z+1)}{9} \quad (977)$$

$$R_{186}(r, z) = -\frac{4\sqrt{3}(z-1)(z+1)(2r^2 - 3rz + 2)}{r^2 - 2rz + 1} \quad (978)$$

$$R_{187}(r, z) = \frac{2i(r-1)(z-1)(z+1)(3r^2 - 6rz - r + 3)}{r^2 - 2rz + 1} \quad (979)$$

$$R_{188}(r, z) = -\frac{2\sqrt{3}r(z-1)(z+1)(3r^2 - 6rz - 2r + 3)}{3(r^2 - 2rz + 1)} \quad (980)$$

$$R_{189}(r, z) = \frac{2(3r^4z - 3r^3z^2 + 7r^3 - 14r^2z - 3rz^2 + 7r + 3z)}{r(r^2 - 2rz + 1)} \quad (981)$$

$$R_{190}(r, z) = \frac{2\sqrt{3}i(r-1)(r^2z - r^2 - rz^2 + 3rz + z - 1)}{r^2 - 2rz + 1} \quad (982)$$

$$R_{191}(r, z) = -\frac{2r(z-1)}{3} \quad (983)$$

$$R_{192}(r, z) = \frac{2\sqrt{3}(r^2 + 1)(z-1)(z+1)}{r^2 - 2rz + 1} \quad (984)$$

$$R_{193}(r, z) = \frac{2ir(r-1)(z-1)(z+1)}{r^2 - 2rz + 1} \quad (985)$$

$$R_{194}(r, z) = -\frac{4\sqrt{3}(z-1)(z+1)(2r^2 - rz + 1)}{r^2 - 2rz + 1} \quad (986)$$

$$R_{195}(r, z) = \frac{N_{195}(r, z)}{D_{195}(r, z)}. \quad (987)$$

$$\begin{aligned}
& -2i - 8ir^2 + 2iz^2 - 4irz^3 + 2\sqrt{3}ir^3 \\
& + 4irz + 4\sqrt{3}ir + 8ir^2z^2 - 4\sqrt{3}ir^2z \\
& - 4\sqrt{3}irz^2 - 2\sqrt{3}ir^3z^2 \\
N_{195}(r, z) = & + 4\sqrt{3}ir^2z^3
\end{aligned} \tag{988}$$

$$D_{195}(r, z) = -r^2 + 2rz - 1 \tag{989}$$

$$R_{196}(r, z) = -2r(z-1)(z+1) + \frac{4\sqrt{3}(z-1)(z+1)}{3} \tag{990}$$

$$R_{197}(r, z) = -\frac{8(2r^2 - 4rz + z^2 + 1)}{r^2 - 2rz + 1} \tag{991}$$

$$R_{198}(r, z) = -\frac{2ir(3rz - 4)(2r^2 - 4rz + z^2 + 1)}{r^2 - 2rz + 1} - \frac{2\sqrt{3}i(2r^2 - 4rz + z^2 + 1)}{r^2 - 2rz + 1} \tag{992}$$

$$R_{199}(r, z) = \frac{N_{199}(r, z)}{D_{199}(r, z)}. \tag{993}$$

$$\begin{aligned}
& 2r^2 + 4r^4 - 12r^3z - 2rz - 2rz^3 + 2\sqrt{3}r \\
& + 4\sqrt{3}r^3 + 10r^2z^2 - 10\sqrt{3}r^2z \\
& - 4\sqrt{3}r^4z - 2\sqrt{3}r^2z^3 + 2\sqrt{3}rz^2 \\
N_{199}(r, z) = & + 8\sqrt{3}r^3z^2
\end{aligned} \tag{994}$$

$$D_{199}(r, z) = -r^2 + 2rz - 1 \tag{995}$$

$$R_{200}(r, z) = \frac{2\sqrt{3}(3r^2 + 1)(z-1)(z+1)}{r^2 - 2rz + 1} \tag{996}$$

$$R_{201}(r, z) = \frac{6ir^2(z-1)(z+1)}{r^2 - 2rz + 1} - \frac{2\sqrt{3}ir(z-1)(z+1)}{r^2 - 2rz + 1} \tag{997}$$

$$R_{202}(r, z) = \frac{2(3rz + 1)(2r^2 - 4rz + z^2 + 1)}{r^2 - 2rz + 1} \tag{998}$$

$$R_{203}(r, z) = \frac{2\sqrt{3}irz(2r^2 - 4rz + z^2 + 1)}{r^2 - 2rz + 1} - \frac{2ir(2r^2 - 4rz + z^2 + 1)}{r^2 - 2rz + 1} \tag{999}$$

$$R_{204}(r, z) = -16r + 16z \tag{1000}$$

$$R_{205}(r, z) = -4ir(r-z)(3rz-4) - 4\sqrt{3}i(r-z) \tag{1001}$$

$$R_{206}(r, z) = -4r(r-z)^2 + 4\sqrt{3}r(r-z)(rz-1) \tag{1002}$$

$$R_{207}(r, z) = 4(r-z)(3rz+1) \tag{1003}$$

$$R_{208}(r, z) = 4\sqrt{3}irz(r-z) - 4ir(r-z) \tag{1004}$$

$$R_{209}(r, z) = -\frac{4\sqrt{3}(z-1)(z+1)(r^2 - rz + 2)}{r^2 - 2rz + 1} \tag{1005}$$

$$R_{210}(r, z) = \frac{N_{210}(r, z)}{D_{210}(r, z)}. \tag{1006}$$

$$\begin{aligned}
& -2\sqrt{3}i + 2ir^3 + 8ir - 8irz^2 - 4ir^2z \\
& - 4\sqrt{3}ir^2 - 2ir^3z^2 + 2\sqrt{3}iz^2 \\
& + 4ir^2z^3 - 4\sqrt{3}irz^3 + 4\sqrt{3}irz \\
N_{210}(r, z) = & + 4\sqrt{3}ir^2z^2
\end{aligned} \tag{1007}$$

$$D_{210}(r, z) = -r^2 + 2rz - 1 \tag{1008}$$

$$R_{211}(r, z) = \frac{4\sqrt{3}r^2(z-1)(z+1)}{3} - 2r(z-1)(z+1) \tag{1009}$$

$$R_{212}(r, z) = -\frac{8(r^2z^2 + r^2 - 4rz + 2)}{r^2 - 2rz + 1} \tag{1010}$$

$$R_{213}(r, z) = \frac{N_{213}(r, z)}{D_{213}(r, z)}. \quad (1011)$$

$$\begin{aligned} & -12iz + 8ir^3 + 16ir - 38ir^2z \\ & -6ir^2z^3 - 4\sqrt{3}ir^2 - 2\sqrt{3}ir^4 \\ & + 8ir^3z^2 + 24irz^2 - 2\sqrt{3}ir^4z^2 \\ N_{213}(r, z) = & + 8\sqrt{3}ir^3z \end{aligned} \quad (1012)$$

$$D_{213}(r, z) = r(-r^2 + 2rz - 1) \quad (1013)$$

$$R_{214}(r, z) = \frac{N_{214}(r, z)}{D_{214}(r, z)}. \quad (1014)$$

$$\begin{aligned} & 4 + 2r^2 - 12rz - 4\sqrt{3}z - 2r^3z - 2r^3z^3 \\ & + 2\sqrt{3}r^3 + 4\sqrt{3}r + 10r^2z^2 - 10\sqrt{3}r^2z \\ N_{214}(r, z) = & -2\sqrt{3}r^2z^3 + 2\sqrt{3}r^3z^2 + 8\sqrt{3}r^2z^2 \end{aligned} \quad (1015)$$

$$D_{214}(r, z) = -r^2 + 2rz - 1 \quad (1016)$$

$$R_{215}(r, z) = \frac{2\sqrt{3}(r^2 + 3)(z - 1)(z + 1)}{r^2 - 2rz + 1} \quad (1017)$$

$$R_{216}(r, z) = \frac{2\sqrt{3}ir^2(z - 1)(z + 1)}{r^2 - 2rz + 1} - \frac{6ir(z - 1)(z + 1)}{r^2 - 2rz + 1} \quad (1018)$$

$$R_{217}(r, z) = \frac{2(r + 3z)(r^2z^2 + r^2 - 4rz + 2)}{r(r^2 - 2rz + 1)} \quad (1019)$$

$$R_{218}(r, z) = -\frac{2\sqrt{3}iz(r^2z^2 + r^2 - 4rz + 2)}{r^2 - 2rz + 1} + \frac{2i(r^2z^2 + r^2 - 4rz + 2)}{r^2 - 2rz + 1} \quad (1020)$$

$$R_{219}(r, z) = -8z^2 - 8 \quad (1021)$$

$$R_{220}(r, z) = \frac{2i(r - 1)(z^2 + 1)(r^2 - 5rz - 3r + 1)}{r^2 - 2rz + 1} \quad (1022)$$

$$R_{221}(r, z) = -2r(z + 1)(z^2 + 1) \quad (1023)$$

$$R_{222}(r, z) = \frac{4\sqrt{3}(z - 1)(z + 1)(2r^4 - 4r^3z + r^2z^2 + 5r^2 - 4rz + 2)}{r^2 - 2rz + 1} \quad (1024)$$

$$R_{223}(r, z) = 6z^2 + 6 \quad (1025)$$

$$R_{224}(r, z) = \frac{6ir(r - 1)(z + 1)(z^2 + 1)}{r^2 - 2rz + 1} \quad (1026)$$

$$R_{225}(r, z) = 8\sqrt{3}r(z - 1)(z + 1)(r^2 - rz + 2) \quad (1027)$$

$$R_{226}(r, z) = 16rz - 16 \quad (1028)$$

$$R_{227}(r, z) = -4\sqrt{3}ir(rz - 1) + \frac{4i(4r - 3z)(rz - 1)}{r} \quad (1029)$$

$$R_{228}(r, z) = 4\sqrt{3}(r - z)(rz - 1) - 4(rz - 1)^2 \quad (1030)$$

$$R_{229}(r, z) = -\frac{4(r + 3z)(rz - 1)}{r} \quad (1031)$$

$$R_{230}(r, z) = 4\sqrt{3}iz(rz - 1) - 4i(rz - 1) \quad (1032)$$

$$R_{231}(r, z) = 8\sqrt{3}(z - 1)(z + 1)(2r^2 - rz + 1) \quad (1033)$$

$$R_{232}(r, z) = 16z \quad (1034)$$

$$R_{233}(r, z) = -\frac{4iz(r - 1)(r^2 - 5rz - 3r + 1)}{r^2 - 2rz + 1} \quad (1035)$$

$$R_{234}(r, z) = 4rz(z+1) \quad (1036)$$

$$R_{235}(r, z) = -\frac{8\sqrt{3}r(z-1)(z+1)(2r^2-3rz+2)}{r^2-2rz+1} \quad (1037)$$

$$R_{236}(r, z) = -12z \quad (1038)$$

$$R_{237}(r, z) = -\frac{12irz(r-1)(z+1)}{r^2-2rz+1} \quad (1039)$$

$$R_{238}(r, z) = -\frac{4(3r^2z^2+7r^2+3rz^3+17rz+3z^2+7)}{r^2+2rz+1} \quad (1040)$$

$$R_{239}(r, z) = \frac{N_{239}(r, z)}{D_{239}(r, z)}. \quad (1041)$$

$$\begin{aligned} & -20\sqrt{3}ir - 20\sqrt{3}ir^3 - 6\sqrt{3}iz \\ & +20\sqrt{3}ir^2 + 20\sqrt{3}ir^4 - 52\sqrt{3}ir^2z \\ & -18\sqrt{3}irz^2 - 12\sqrt{3}ir^3z^2 \\ & -12\sqrt{3}ir^2z^3 + 6\sqrt{3}ir^5z \\ & +12\sqrt{3}ir^2z^2 + 12\sqrt{3}ir^3z^3 \\ N_{239}(r, z) = & +18\sqrt{3}ir^4z^2 + 52\sqrt{3}ir^3z \end{aligned} \quad (1042)$$

$$D_{239}(r, z) = 3r(r^2+2rz+1) \quad (1043)$$

$$R_{240}(r, z) = -2r^2z + \frac{2r^2}{3} - 2rz^2 + \frac{2rz}{3} - 4r - 2z + \frac{2}{3} \quad (1044)$$

$$R_{241}(r, z) = -\frac{2\sqrt{3}ir(r-1)(z-1)}{9} \quad (1045)$$

$$R_{242}(r, z) = -\frac{4\sqrt{3}(z-1)(z+1)(2r^2+3rz+2)}{r^2+2rz+1} \quad (1046)$$

$$R_{243}(r, z) = \frac{2i(r-1)(z-1)(z+1)(3r^2+6rz-r+3)}{r^2+2rz+1} \quad (1047)$$

$$R_{244}(r, z) = -\frac{2\sqrt{3}r(z-1)(z+1)(3r^2+6rz-2r+3)}{3(r^2+2rz+1)} \quad (1048)$$

$$R_{245}(r, z) = -\frac{2(3r^4z+3r^3z^2-7r^3-14r^2z+3rz^2-7r+3z)}{r(r^2+2rz+1)} \quad (1049)$$

$$R_{246}(r, z) = -\frac{2\sqrt{3}i(r-1)(r^2z+r^2+rz^2+3rz+z+1)}{r^2+2rz+1} \quad (1050)$$

$$R_{247}(r, z) = \frac{2r(z+1)}{3} \quad (1051)$$

$$R_{248}(r, z) = \frac{2\sqrt{3}(r^2+1)(z-1)(z+1)}{r^2+2rz+1} \quad (1052)$$

$$R_{249}(r, z) = \frac{2ir(r-1)(z-1)(z+1)}{r^2+2rz+1} \quad (1053)$$

$$R_{250}(r, z) = -\frac{4\sqrt{3}(z-1)(z+1)(2r^2+rz+1)}{r^2+2rz+1} \quad (1054)$$

$$R_{251}(r, z) = \frac{N_{251}(r, z)}{D_{251}(r, z)}. \quad (1055)$$

$$\begin{aligned} & 2i - 2iz^2 + 8ir^2 - 8ir^2z^2 - 4\sqrt{3}ir \\ & -4irz^3 - 2\sqrt{3}ir^3 + 4irz - 4\sqrt{3}ir^2z \\ & +2\sqrt{3}ir^3z^2 + 4\sqrt{3}irz^2 \\ N_{251}(r, z) = & +4\sqrt{3}ir^2z^3 \end{aligned} \quad (1056)$$

$$D_{251}(r, z) = r^2 + 2rz + 1 \quad (1057)$$

$$R_{252}(r, z) = -\frac{8(2r^2 + 4rz + z^2 + 1)}{r^2 + 2rz + 1} \quad (1058)$$

$$R_{253}(r, z) = \frac{2ir(3rz + 4)(2r^2 + 4rz + z^2 + 1)}{r^2 + 2rz + 1} - \frac{2\sqrt{3}i(2r^2 + 4rz + z^2 + 1)}{r^2 + 2rz + 1} \quad (1059)$$

$$R_{254}(r, z) = \frac{N_{254}(r, z)}{D_{254}(r, z)}. \quad (1060)$$

$$\begin{aligned} & -4r^4 - 2r^2 - 12r^3z - 10r^2z^2 \\ & -4\sqrt{3}r^3 - 2rz - 2\sqrt{3}r - 2rz^3 \\ & -10\sqrt{3}r^2z - 8\sqrt{3}r^3z^2 - 4\sqrt{3}r^4z \\ N_{254}(r, z) = & -2\sqrt{3}rz^2 - 2\sqrt{3}r^2z^3 \end{aligned} \quad (1061)$$

$$D_{254}(r, z) = r^2 + 2rz + 1 \quad (1062)$$

$$R_{255}(r, z) = \frac{2\sqrt{3}(3r^2 + 1)(z - 1)(z + 1)}{r^2 + 2rz + 1} \quad (1063)$$

$$R_{256}(r, z) = \frac{6ir^2(z - 1)(z + 1)}{r^2 + 2rz + 1} - \frac{2\sqrt{3}ir(z - 1)(z + 1)}{r^2 + 2rz + 1} \quad (1064)$$

$$R_{257}(r, z) = -\frac{2(3rz - 1)(2r^2 + 4rz + z^2 + 1)}{r^2 + 2rz + 1} \quad (1065)$$

$$R_{258}(r, z) = -\frac{2\sqrt{3}irz(2r^2 + 4rz + z^2 + 1)}{r^2 + 2rz + 1} - \frac{2ir(2r^2 + 4rz + z^2 + 1)}{r^2 + 2rz + 1} \quad (1066)$$

$$R_{259}(r, z) = -16r - 16z \quad (1067)$$

$$R_{260}(r, z) = 4ir(r + z)(3rz + 4) - 4\sqrt{3}i(r + z) \quad (1068)$$

$$R_{261}(r, z) = -4r(r + z)^2 - 4\sqrt{3}r(r + z)(rz + 1) \quad (1069)$$

$$R_{262}(r, z) = -4(r + z)(3rz - 1) \quad (1070)$$

$$R_{263}(r, z) = -4\sqrt{3}irz(r + z) - 4ir(r + z) \quad (1071)$$

$$R_{264}(r, z) = -\frac{4\sqrt{3}(z - 1)(z + 1)(r^2 + rz + 2)}{r^2 + 2rz + 1} \quad (1072)$$

$$R_{265}(r, z) = \frac{N_{265}(r, z)}{D_{265}(r, z)}. \quad (1073)$$

$$\begin{aligned} & -8ir - 2ir^3 + 2\sqrt{3}i - 4ir^2z - 2\sqrt{3}iz^2 \\ & + 2ir^3z^2 + 4\sqrt{3}ir^2 + 4ir^2z^3 \\ & + 8irz^2 - 4\sqrt{3}irz^3 - 4\sqrt{3}ir^2z^2 \\ N_{265}(r, z) = & +4\sqrt{3}irz \end{aligned} \quad (1074)$$

$$D_{265}(r, z) = r^2 + 2rz + 1 \quad (1075)$$

$$R_{266}(r, z) = -\frac{8(r^2z^2 + r^2 + 4rz + 2)}{r^2 + 2rz + 1} \quad (1076)$$

$$R_{267}(r, z) = \frac{N_{267}(r, z)}{D_{267}(r, z)}. \quad (1077)$$

$$\begin{aligned} & -16ir - 12iz - 8ir^3 - 38ir^2z - 24irz^2 \\ & -8ir^3z^2 - 6ir^2z^3 + 2\sqrt{3}ir^4 \\ N_{267}(r, z) = & +4\sqrt{3}ir^2 + 2\sqrt{3}ir^4z^2 + 8\sqrt{3}ir^3z \end{aligned} \quad (1078)$$

$$D_{267}(r, z) = r(r^2 + 2rz + 1) \quad (1079)$$

$$R_{268}(r, z) = \frac{N_{268}(r, z)}{D_{268}(r, z)}. \quad (1080)$$

$$\begin{aligned} & -4 - 2r^2 - 12rz - 10r^2z^2 - 4\sqrt{3}z - 4\sqrt{3}r \\ & -2r^3z - 2\sqrt{3}r^3 - 2r^3z^3 - 10\sqrt{3}r^2z \\ N_{268}(r, z) = & -8\sqrt{3}rz^2 - 2\sqrt{3}r^3z^2 - 2\sqrt{3}r^2z^3 \end{aligned} \quad (1081)$$

$$D_{268}(r, z) = r^2 + 2rz + 1 \quad (1082)$$

$$R_{269}(r, z) = \frac{2\sqrt{3}(r^2 + 3)(z - 1)(z + 1)}{r^2 + 2rz + 1} \quad (1083)$$

$$R_{270}(r, z) = \frac{2\sqrt{3}ir^2(z - 1)(z + 1)}{r^2 + 2rz + 1} - \frac{6ir(z - 1)(z + 1)}{r^2 + 2rz + 1} \quad (1084)$$

$$R_{271}(r, z) = \frac{2(r - 3z)(r^2z^2 + r^2 + 4rz + 2)}{r(r^2 + 2rz + 1)} \quad (1085)$$

$$R_{272}(r, z) = \frac{2\sqrt{3}iz(r^2z^2 + r^2 + 4rz + 2)}{r^2 + 2rz + 1} + \frac{2i(r^2z^2 + r^2 + 4rz + 2)}{r^2 + 2rz + 1} \quad (1086)$$

$$R_{273}(r, z) = \frac{2i(r - 1)(z^2 + 1)(r^2 + 5rz - 3r + 1)}{r^2 + 2rz + 1} \quad (1087)$$

$$R_{274}(r, z) = 2r(z - 1)(z^2 + 1) \quad (1088)$$

$$R_{275}(r, z) = \frac{4\sqrt{3}(z - 1)(z + 1)(2r^4 + 4r^3z + r^2z^2 + 5r^2 + 4rz + 2)}{r^2 + 2rz + 1} \quad (1089)$$

$$R_{276}(r, z) = -\frac{6ir(r - 1)(z - 1)(z^2 + 1)}{r^2 + 2rz + 1} \quad (1090)$$

$$R_{277}(r, z) = 8\sqrt{3}r(z - 1)(z + 1)(r^2 + rz + 2) \quad (1091)$$

$$R_{278}(r, z) = -16rz - 16 \quad (1092)$$

$$R_{279}(r, z) = 4\sqrt{3}ir(rz + 1) - \frac{4i(4r + 3z)(rz + 1)}{r} \quad (1093)$$

$$R_{280}(r, z) = -4\sqrt{3}(r + z)(rz + 1) - 4(rz + 1)^2 \quad (1094)$$

$$R_{281}(r, z) = \frac{4(r - 3z)(rz + 1)}{r} \quad (1095)$$

$$R_{282}(r, z) = 4\sqrt{3}iz(rz + 1) + 4i(rz + 1) \quad (1096)$$

$$R_{283}(r, z) = 8\sqrt{3}(z - 1)(z + 1)(2r^2 + rz + 1) \quad (1097)$$

$$R_{284}(r, z) = -16z \quad (1098)$$

$$R_{285}(r, z) = \frac{4iz(r - 1)(r^2 + 5rz - 3r + 1)}{r^2 + 2rz + 1} \quad (1099)$$

$$R_{286}(r, z) = 4rz(z - 1) \quad (1100)$$

$$R_{287}(r, z) = -\frac{8\sqrt{3}r(z - 1)(z + 1)(2r^2 + 3rz + 2)}{r^2 + 2rz + 1} \quad (1101)$$

$$R_{288}(r, z) = 12z \quad (1102)$$

$$R_{289}(r, z) = -\frac{12irz(r - 1)(z - 1)}{r^2 + 2rz + 1} \quad (1103)$$

$$R_{290}(r, z) = 4\sqrt{3}(z - 1)(z + 1) \quad (1104)$$

$$R_{291}(r, z) = -4ir(z - 1)(z + 1) \quad (1105)$$

$$R_{292}(r, z) = -\frac{4\sqrt{3}r^2(z - 1)(z + 1)}{3} \quad (1106)$$

$$R_{293}(r, z) = 4\sqrt{3} (3z^2 - 1) \quad (1107)$$

$$R_{294}(r, z) = -4ir (3z^2 - 1) \quad (1108)$$

$$R_{295}(r, z) = -\frac{4\sqrt{3}r^2 (3z^2 - 1)}{3} \quad (1109)$$

$$R_{296}(r, z) = -8\sqrt{3}z\sqrt{-z^2 + 1} \quad (1110)$$

$$R_{297}(r, z) = 8irz\sqrt{-z^2 + 1} \quad (1111)$$

$$R_{298}(r, z) = \frac{8\sqrt{3}r^2z\sqrt{-z^2 + 1}}{3} \quad (1112)$$

$$R_{299}(r, z) = \frac{16\sqrt{3}ir}{3} \quad (1113)$$

$$R_{300}(r, z) = \frac{4r^2}{3} \quad (1114)$$

$$R_{301}(r, z) = -4\sqrt{3} (z - 1) (z + 1) \quad (1115)$$

$$R_{302}(r, z) = 4ir (z - 1) (z + 1) \quad (1116)$$

$$R_{303}(r, z) = -4\sqrt{3} (3z^2 - 1) \quad (1117)$$

$$R_{304}(r, z) = 4ir (3z^2 - 1) \quad (1118)$$

$$R_{305}(r, z) = 8\sqrt{3}z\sqrt{-z^2 + 1} \quad (1119)$$

$$R_{306}(r, z) = -8irz\sqrt{-z^2 + 1} \quad (1120)$$

$$R_{307}(r, z) = 8z^2 + 8 \quad (1121)$$

$$R_{308}(r, z) = -2\sqrt{3}ir (z^2 + 1) - 2ir (z^2 + 1) \quad (1122)$$

$$R_{309}(r, z) = 24 (z - 1) (z + 1) \quad (1123)$$

$$R_{310}(r, z) = -6\sqrt{3}ir (z - 1) (z + 1) - 6ir (z - 1) (z + 1) \quad (1124)$$

$$R_{311}(r, z) = -16z\sqrt{-z^2 + 1} \quad (1125)$$

$$R_{312}(r, z) = \sqrt{-z^2 + 1} (4irz + 4\sqrt{3}irz) \quad (1126)$$

$$R_{313}(r, z) = 2ir (z^2 + 1) + 2\sqrt{3}ir (z^2 + 1) \quad (1127)$$

$$R_{314}(r, z) = -24 (z - 1) (z + 1) \quad (1128)$$

$$R_{315}(r, z) = 6ir (z - 1) (z + 1) + 6\sqrt{3}ir (z - 1) (z + 1) \quad (1129)$$

$$R_{316}(r, z) = 16z\sqrt{-z^2 + 1} \quad (1130)$$

$$R_{317}(r, z) = \sqrt{-z^2 + 1} (-4\sqrt{3}irz - 4irz) \quad (1131)$$

$$R_{318}(r, z) = -4\sqrt{3}irz - 4irz \quad (1132)$$

$$R_{319}(r, z) = -16\sqrt{-z^2 + 1} \quad (1133)$$

$$R_{320}(r, z) = \sqrt{-z^2 + 1} (4ir + 4\sqrt{3}ir) \quad (1134)$$

$$R_{321}(r, z) = 4irz + 4\sqrt{3}irz \quad (1135)$$

$$R_{322}(r, z) = 16\sqrt{-z^2 + 1} \quad (1136)$$

$$R_{323}(r, z) = \sqrt{-z^2 + 1} (-4\sqrt{3}ir - 4ir) \quad (1137)$$

$$R_{324}(r, z) = -8\sqrt{3}r^2 (z - 1) (z + 1) \quad (1138)$$

$$R_{325}(r, z) = -8\sqrt{3}r^2 (3z^2 - 1) \quad (1139)$$

$$R_{326}(r, z) = 16\sqrt{3}r^2 z \sqrt{-z^2 + 1} \quad (1140)$$

$$R_{327}(r, z) = 8ir \quad (1141)$$

$$R_{328}(r, z) = 8\sqrt{3}r^2 (z - 1)(z + 1) \quad (1142)$$

$$R_{329}(r, z) = 8\sqrt{3}r^2 (3z^2 - 1) \quad (1143)$$

$$R_{330}(r, z) = -16\sqrt{3}r^2 z \sqrt{-z^2 + 1} \quad (1144)$$

$$R_{331}(r, z) = -8ir \quad (1145)$$

References

- [1] Danylo Yerokhin. A minimal local covariant action for holographic dark energy: constraints, perturbations, and nonlinear dynamics. Unpublished manuscript, version V52, 27 September 2026, 2026. Main article with Appendices A–E. No arXiv identifier or journal reference has been assigned in this local package.